Shape Classification of Failure-Sets in Stationary Linear Filters

Critical values, gain families and information ordering

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Declaration

I declare that this thesis is my own work. Where it draws on the work of others, that work is cited, and the extent of the search for prior work is stated in §1.6.

Use of AI systems. This thesis was produced by directing AI systems towards a goal and helping them reach it. It began as a few paragraphs describing a theory about perception, which grew into a mathematical project: I met the standard equation presented here several times before asking whether it could be graphed, and the results of that question are what follows. Gemini helped me reach the research question. Claude, ChatGPT and Grok each checked the mathematics, and all of them carried out deep research; Muse also helped with that research and found many of the missing DOIs behind §1.6 and Appendix C. NotebookLM gave me summaries and audio overviews along the way. The simulations were run with Claude Code, a separate AI agent made the independent referee pass described in §8.5, and Claude compiled and revised this text. At every stage I asked for each prompt and output to be explained and picked apart before writing a fresh prompt and starting the cycle again. Because of the scale of the material, my role in this final paper is mostly that of editor: I read over and checked every output, and any mistakes that remain are ones I let through.

Abstract

A filter specification fixes a tolerance R^2 on the error variance and asks under which operating conditions the error exceeds it. As one parameter varies (observation precision, noise correlation time, sampling interval, process noise or a design parameter), the answer is a set, the failure set, and its shape matters: a threshold, a bounded window, two rays, or a mixture. This thesis classifies those shapes for scalar stationary linear filters of an Ornstein–Uhlenbeck signal.

For any continuous variance curve with finitely many turning points, the class of the failure set at every tolerance is fixed by a finite signature: the endpoint limits, the turning-point values and two approach directions. For the white-noise relay, every stationary variance lies on one surface in two dimensionless coordinates, on which the floor crossing is the line $q = 2$ and the optimal filter is the lower envelope. Along precision, a continuous variance curve is reached by some continuous gain if and only if it never falls below the optimum. At any tolerance tighter than the floor every failure set therefore contains a lower ray, and every further restriction comes from the gain family: natural single-stage gains have at most one window component, and a Hill-type gain produces a mixed class, a lower ray with a window.

When one observation process is a causal garbling of another, the optimal filtering error is monotone between them. Along such a sweep the optimal filter never windows, so a window, two rays or a mixed class certifies a miscalibrated filter. For additive stationary Gaussian noise the ordering is decided by the noise spectra, and within a power-normalised family of coloured noises the condition is sharp. Without the ordering the optimal filter can window, as it does along correlation time at matched variance.

The results were checked, mostly under pre-registered criteria. Of 182 recorded checks, 37 failed, and each failure was diagnosed; 23 of 24 re-checks under corrected criteria passed. The class sets of sampled relays and of two-stage gains rest on numerical evidence, not proof.

Acknowledgements

The computations used Python with NumPy, SciPy, SymPy, mpmath and Matplotlib (Appendix E).

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Notation

Symbol	Meaning	Defined
θ	the swept parameter	§2.1
$\sigma^2(\theta)$	actual stationary error variance	§2.1
R^2	tolerance; R^2/F is the relative tolerance	§1.1
$\mathcal{F}(R^2)$	failure set $\{\theta : \sigma^2(\theta) > R^2\}$	§2.1
(ℓ, h, k)	class: lower ray, upper ray, number of window components	§2.1
L_-, L_+	endpoint limits of σ^2	§2.1
$\mathcal{C}$	the set of critical tolerances	§2.1
ϑ	stability margin of a class: inf of $\lvert\sigma^2 - R^2\rvert$ away from the crossings	§2.8
λ	mean-reversion rate of the signal	§3.1
Σ	half the diffusion intensity of the signal	§1.7, §3.1
$F = \Sigma/\lambda$	the floor: the error of not filtering	§1.7
Π	observation precision	§3.1
G	filter gain	§3.1
$\lambda_G = \lambda + G$	closed-loop rate of a constant-gain relay	§3.1 (proof of Prop. 3.1), §6.1
$G\star$, $\sigma\star^2$	optimal (Kalman–Bucy) gain and error	§3.4
H	high-precision limit $\Sigma/(\lambda + G_\infty)$	§3.1
$\nu = 2F\Pi/\lambda$	normalised precision	§3.2
$q = G/(F\Pi)$	normalised gain (this meaning only)	§3.2
$\mathcal{V}(\nu, q)$	normal-form surface, σ^2/F	§3.2
$q\star(\nu)$	optimal envelope	§3.4
$q\pm$	roots of the tolerance boundary $\mathcal{V} = R^2/F$	§3.5
π_R	forced-lower-ray precision, $2\lambda(F - R^2)/R^4$	§3.5, §5.1
$\varepsilon = \Pi G'/G$	elasticity of the gain in Π	§3.7
Π_0, c	prior precision and coupling of the Bayesian weight	§1.2, chapter 4
$\kappa = c/(F\Pi_0)$	over-weighting ratio	§1.2, §5.3
M, m	maximum and minimum turning-point values	§5.3–5.4
λ', Σ', $F' = \Sigma'/\lambda'$	design parameters of a misspecified Kalman gain	§5.3
$\chi = \Sigma/\Sigma'$	process-noise ratio, true over design	§5.3 (Prop. 5.9)
η	sensory (measurement) noise process	§6.1
τ	noise correlation time	§6.1
$t = \lambda\tau$	normalised correlation time	§3.8, §6.1
H_G, V_w	relay limits along τ: Σ/λ_G and $(\Sigma + G^2/(2\Pi))/\lambda_G$	§6.1
P	optimal stationary error along τ	§6.1–6.2

Symbol	Meaning	Defined
v	noise variance at matched variance	§6.2
Δt	sampling interval	§6.3
$a = e^{-\lambda \Delta t}$	per-sample decay factor (§6.3 onward only)	§6.3
$s = 1 - a^2$	one minus the squared decay factor	§6.3
Fs	process-noise variance per sample, Var w	§6.3
$v_s = 1/(\Pi \Delta t)$	per-sample sampling variance	§6.3
L	per-sample gain of a discrete filter	§6.3
P^+, P^-, $\bar{P}$	posterior, prior and averaged errors	§6.3
$\varphi = F\Pi\Delta t$	sampling ratio	§6.3
ρ	over-trust factor	§6.3, §6.6
β	exponent of the power-normalised family	§7.3

1. Introduction

A filter that must keep its error within a fixed specification can fail on a set of operating conditions with a definite shape: a threshold, a bounded window, two rays, or a mixture. This thesis classifies those shapes for scalar stationary linear filters, for each parameter sweep defined in §1.3, and shows which shapes an optimal filter can never produce.

1.1 The question

A filter is usually judged at one operating point: how large is its error, and how far is it from the optimum? A specification asks something different. It fixes a tolerance R^2 and asks for which operating conditions the actual error variance σ^2 exceeds it. As a condition θ varies (observation precision, noise correlation time, sampling interval, process noise, or a design parameter), the answer is a set,

$$\mathcal{F}(R^2) = \{\theta : \sigma^2(\theta) > R^2\}$$

and its shape matters in practice. A threshold means that the filter fails below some precision. A window means that it fails only at intermediate precision, while both worse and better observations are acceptable. This thesis asks:

> Under what conditions does the failure set of a stationary linear filter change shape as an environment or design parameter varies, and which shapes are impossible for an optimal filter?

1.2 Starting point: a bounded failure window

This research began with a single relay: a filter for an Ornstein–Uhlenbeck state whose gain at each precision is the one-step Bayesian weight $c\Pi/(\Pi + \Pi_0)$, used in place of the steady-state Kalman–Bucy gain [1]. The weight is a precision-weighting rule of the kind studied in perception and neuroscience [2, 3]. The starting study verified three findings that set up the thesis (Appendix D):

- **The excess identity.** The relay is worse than no filtering exactly when its gain exceeds $2F\Pi$, where $F = \Sigma/\lambda$ is the floor (defined in §1.7). The optimal gain never exceeds $F\Pi$ (Proposition 3.5).
- **The window theorem.** For gains with natural limits, some tolerance fails on a bounded set of precisions away from zero if and only if $\sup G/\Pi > 2F$ (Theorem 3.4). For the Bayesian-weight gain this is $\kappa > 2$, where $\kappa = c/(F\Pi_0)$ (Proposition 5.8). In the worked example ($\lambda = c = 1$, $\Sigma = 0.01$, $\Pi_0 = 1$, so $F = 0.01$ and $\kappa = 100$), at $R^2 = 0.05$ the relay fails exactly for $\Pi \in (1/9, 4)$.
- **The optimal filter never windows** along precision, because its error decreases strictly (Theorem 3.4).

That study answered the question for one gain family along one sweep. This thesis extends

the answer to each sweep in Table 1.5, to the natural gain families used with those sweeps, and to the optimal filter.

1.3 The object of study

A sweep is a one-parameter family of filters, indexed by a parameter θ. For each θ the filter has a stationary error variance $\sigma^2(\theta)$, and the failure set $\mathcal{F}(R^2)$ is the set of θ at which σ^2 exceeds R^2. That set is open, so it is a union of disjoint open intervals. Its **class** records three things: whether the set contains a ray at the lower end (ℓ), a ray at the upper end (h), and how many components k it has away from both ends. A **window** is a failure set with no ray and $k \geq 1$; a **multi-window** is the case $k \geq 2$. The named classes are Empty, All, lower ray, upper ray, two rays, window, multi-window, and mixed (a ray plus window components).

Two kinds of sweep are studied:

- **Environment sweeps** change the observations or the signal: precision Π, correlation time τ, sampling interval Δt, process noise Σ. The optimal filter changes along them.
- **Design sweeps** change the filter only: prior precision Π_0 and coupling c. They classify miscalibration.

A classification is **complete** when the class is determined at every parameter value and every tolerance, equality cases included, and every class claimed impossible has a proof.

1.4 Approach: two pillars and a bridge

The thesis rests on two organising results joined by one theorem.

- **The bridge: the critical-value signature (Theorem 2.2).** For any piecewise-monotone variance curve, the class at every tolerance is fixed by a finite list: the endpoint limits, the values at turning points, and two approach directions. Classifying a model reduces to computing that signature (chapter 2).
- **Pillar 1: the normal form (Theorem 3.2).** For the white-noise relay, every stationary variance is one surface evaluated along a path: $\sigma^2/F = \mathcal{V}(\nu, q)$, with $\nu = 2F\Pi/\lambda$ and $q = G/(F\Pi)$. The floor crossing is the line $q = 2$, the optimal filter is the lower envelope, and each sweep or gain family is a path (chapter 3).
- **Pillar 2: causal information ordering (Theorem 7.2).** If one observation process can be built from another causally, using only its past and present together with independent randomness, the optimal filtering error is monotone between them. Along such a sweep the optimal filter cannot window (chapter 7).

The resulting picture: a variance curve determines its signature, which determines the class. A gain family restricts which signatures are reachable. Optimality and information ordering restrict them further.

1.5 Main results

Every continuous gain schedule fails at tight tolerances. For any continuous gain schedule and any $R^2 < F$, the failure set contains all precisions below $\pi_R = 2\lambda(F - R^2)/R^4$

(Corollary 5.2). So a window class needs a tolerance at least as loose as the uncorrected error, for every gain, with no regularity assumptions.

Along precision, the model forbids only one thing. A variance curve is reachable by some continuous gain if and only if it never dips below the optimal variance (Theorem 5.1). Every class is then available to some gain, except that below the floor every failure set contains a lower ray (Corollaries 5.2–5.3). Smooth gains with the natural limits of Theorem 3.4 can have any number of window components (Corollary 5.4). The restrictions come from the gain family:

- natural single-stage gains have at most one window component (Propositions 5.8–5.11);
- the sigmoidal Hill gain, with zero initial slope, produces the mixed class lower ray plus window (Theorem 5.12, Proposition 5.13), and a mixed class of this kind needs G/Π to be non-monotone (Proposition 5.14);
- sums of two natural stages produce multi-window classes (Numerical result 5.15).

Each sweep has its own repertoire.

Sweep	Relay classes (natural gains)	Optimal-filter classes	Labels
Precision Π	All, lower ray, window, Empty; mixed (1, 0, 1) for Hill	lower ray, Empty	3.4, 5.8–5.14
Prior Π_0, coupling c	All, lower ray, upper ray, two rays, Empty; never a window	none (design sweeps)	4.1–4.5
Correlation time τ, matched power	All, lower ray, Empty	lower ray, Empty	6.3, 6.6–6.7
Correlation time τ, matched variance	All, upper ray, Empty	window, Empty	6.10–6.12
Sampling interval Δt	adds two rays, (0, 1, 1) and (1, 0, 1)	posterior: lower ray, Empty; prior and time-averaged: All, upper ray, Empty	6.13–6.21
Process noise Σ	absolute tolerance: All, upper ray; relative tolerance: lower ray, Empty, or a window for the over-trusting gain when $\rho > 2$	absolute: upper ray; relative: lower ray, Empty	6.23–6.28

Where the optimal filter can and cannot window. Along any causally ordered sweep, the optimal filter's failure set is Empty, All, or a single ray at the less informative end (Corollary 7.3). Within a one-parameter family of coloured noises, $S(\omega) = K\tau^\beta/(1 + \tau^2\omega^2)$, the condition is sharp: the sweep is ordered, the optimal error is monotone, and optimal windows are absent, exactly when $\beta \leq 0$ or $\beta \geq 2$ (Theorem 7.8). Correlation time at matched variance is $\beta = 1$, and there the optimal filter does window (Proposition 6.10, Counterexample 6.11). Ordering must be causal. Sensor delay gives experiments that differ only by a time shift, yet the optimal filter windows (Proposition 7.11). For additive stationary Gaussian noise the test is exact: causal ordering, pathwise ordering, and dominance of the noise spectra almost everywhere coincide (Theorem 7.6).

A diagnostic. Along a causally ordered sweep, any window, two rays, or mixed class certifies that the filter is not optimal (Corollary 7.4). The bounded window of §1.2 is therefore a

signature of miscalibration, not of the environment.

1.6 Contributions and relation to prior work

The novelty statements in this section describe the searches reported here. Searches on 11 and 12 October 2026 covered the English-language scholarly web, arXiv, publisher pages, bibliographic catalogues and Google Patents. Searches on 14 October 2026 added the zbMATH abstract and review corpus, through its public interface, and repeated the Google Patents search; neither found relevant work. MathSciNet, the full text of IEEE and SIAM journals (checked by title and abstract only), zbMATH's formula search and Espacenet were not searched. A statement that no prior work was found means that none was found in these sources. It is not a claim that none exists.

1. **The classification of failure-set topology** for stationary linear filters, including the forced lower ray and each sweep's complete class table. In these searches, no prior work was found that defines the set $\{\theta : \sigma^2(\theta) > \mathrm{R}^2\}$ for a stationary linear filter and classifies its shape. The nearest precedents differ in kind. Sinopoli et al. [54] and Wu et al. [55] classify boundedness under random observation loss, by a single critical arrival rate and by a critical curve respectively. Jiang et al. [56] and Bosov et al. [57] describe error rising with measurement quality in nonlinear filters, without a sweep classification. Gotovos et al. [58] estimate a superlevel set of an unknown field, not of a known error function. Tenne and Singh [59] give closed-form steady-state variances for design, without thresholding them. The neighbouring literature computes actual error under mismatch at given operating points [4–9], or designs filters to meet a specification. Related work studies filtering under misspecified models [10, 11] and when feedback helps at all [12, 13].
2. **Expressivity of gain families.** Realisability is the scalar filtering analogue of covariance assignment, developed by Hotz and Skelton for matrix state feedback in multivariable control [14, 15]. The Hill mixed class appears to be new. The Hill-type gain form is the kind used in divisive-normalisation models [16, 17].
3. **Optimality.** The power-family threshold (Theorem 7.8), the matched-variance optimal window (Counterexample 6.11), and the latency counterexample (Counterexample 7.12) appear to be new. Theorem 7.2 is presented as a filtering instance of known results. Monotonicity of optimal performance under garbling orders goes back to Blackwell [18, 19], extends to sequential experiments [20], and, in stochastic control where observation channels are themselves design variables, appears in [21]. It is also studied for the comparison of information structures in control and teams [22, 23]. What is specific here is the causal garbling of observation processes and its consequence for filtering error. Theorem 7.6 extends the static characterisation for linear normal experiments [24] to stationary Gaussian processes ordered by their spectra. No process version was found.
4. **Smaller contributions.** The normal form, where dimensionless indices are precedented by Kalata's tracking index [25]. The explicit statement of the excess threshold $\mathrm{G} = 2\mathrm{F}\Pi$. This is an elementary consequence of the stationary variance formula. It is recorded because later results use it, not as a result in its own right. And the observation that, under matched-precision sampling, the posterior error falls with

Δt while the prior and time-averaged errors rise.

The variance of a fixed-gain filter, the Kalman–Bucy steady state [1], the discrete variance factor $\alpha/(2-\alpha)$, which Saho and Masugi [60] give for α–β–γ filters with acceleration measurements, and which for exponentially weighted averages goes back to Roberts [26], the coloured-noise methods [27, 28], and the delay prediction covariance [29] are background. The framework of chapter 2 is the fixed-level reading of zero-dimensional persistence [30, 31].

1.7 Scope and conventions

- **Model.** Scalar, linear, stationary, Gaussian. The gain is constant in time at each parameter value but may vary across parameter values.
- **Error.** Always the actual stationary error variance, never the variance a filter reports from its own model.
- **Noise convention.** Σ is half the diffusion intensity: without correction the state obeys $dx = -\lambda x\, dt + \sqrt{(2\Sigma)}\, dW$. The floor is therefore $F = \Sigma/\lambda$, the stationary variance of the unobserved state. It is the error of not filtering at all.
- **Sampling.** Each sample has noise variance $v_s = 1/(\Pi\Delta t)$. The time-averaged error is the primary metric. Posterior and prior errors are reported alongside, because they move in opposite directions.
- **Moving floor.** For the Σ sweep, both absolute (R^2 fixed) and relative (R^2/F fixed) tolerances are reported, because they give opposite rays.
- **Excluded.** Multivariate states, nonlinear dynamics, time-varying gains, and applied relevance studies. Chapter 9 lists these as future work.

1.8 How results were established

Every result has one type (Definition, Theorem, Proposition, Lemma, Corollary, Numerical result, Conjecture, or Counterexample) and one number by chapter. Theorem 5.12 is therefore the twelfth numbered result of chapter 5. Each result also has a permanent key in the project repository. Appendix A lists the keys and the checks that verified each result.

Analytic results were checked symbolically and numerically, under criteria written and committed before each run. A failed check stays recorded as a failure, with a diagnosis. Appendix D gives the full record. Of 182 recorded checks with a verdict, most of them pre-registered, 37 failed, and each failure was diagnosed. Of 24 re-checks made under corrected criteria in the final verification task, 23 passed (chapter 8). Blind searches of 10^6 and 4.4×10^6 classifications found no class outside the predicted sets (Numerical result 8.2).

1.9 Outline

- **Chapter 2** sets out the framework: conventions, the critical-tolerance theorem, the signature, the events at which classes change, reparameterisation, and stability.
- **Chapter 3** derives the normal form of the white-noise relay.
- **Chapter 4** treats the design sweeps as horizontal paths on that surface.
- **Chapter 5** studies the precision sweep and the expressivity of gain families.

- **Chapter 6** leaves white noise: correlation time, sampling interval, and process noise.
- **Chapter 7** proves the optimality results and their limits.
- **Chapter 8** reports verification, including every recorded failure.
- **Chapter 9** sets out the discussion: what the thesis shows, how large the effects are, the limits of scope, and the open problems.
- **Chapter 10** states the answer to the central question at the scope the proofs support, lists the contributions, and closes the thesis.

2. Framework: The Critical-Value Signature

The topology of a failure set is fixed by a finite list of numbers: the endpoint limits and turning-point values of the variance curve, together with two approach directions (Theorem 2.2). This chapter states that theorem and organises its consequences for the rest of the thesis. Corollary 2.8 and Proposition 2.9 extend the theorem to reparameterisation and to stability. The other results apply it.

2.1 Setting and conventions

A sweep varies one parameter θ over an open interval and records the actual stationary error variance $\sigma^2(\theta)$ of a filter. Variances are always actual, never the variance a filter reports from its own model.

- **H1 (domain).** θ ranges over $D = (\theta_-, \theta_+)$, with $-\infty \le \theta_- < \theta_+ \le \infty$. Endpoints never belong to D. Behaviour there enters only through limits.
- **H2 (regularity).** σ^2 is continuous on D, with finitely many turning points $t_1 < \ldots < t_n$ and strictly monotone pieces between them. The endpoint limits $L\pm = \lim \sigma^2$ as $\theta \to \theta\pm$ exist in $[0, \infty]$.
- **H2′ (plateaus).** As H2, but finitely many pieces may be constant. Used only in §2.5.

The following are definitions, not hypotheses. The tolerance R^2 is always positive.

- **Failure set.** For $R^2 > 0$ the failure set is open:

$$\mathcal{F}(R^2) = \{\theta \in D : \sigma^2(\theta) > R^2\}$$

- **Approach direction.** σ^2 approaches a finite L_- from above if $\sigma^2 > L_-$ on the first piece, so that the first piece increases away from it. Otherwise it approaches from below. At θ_+, σ^2 approaches from above if the last piece decreases towards L_+.
- **Class.** $\mathcal{F}(R^2)$ is a disjoint union of open intervals. Its class is (ℓ, h, k). $\ell = 1$ if a component contains a neighbourhood of θ_- (a lower ray). $h = 1$ likewise at θ_+ (an upper ray). k counts the components bounded away from both ends (window components).
- **Critical tolerances.** $\mathcal{C}$ is the set of finite values among L_-, $\sigma^2(t_1)$, …, $\sigma^2(t_n)$, L_+.

Named class	(ℓ, h, k)
Empty	(0, 0, 0)
All	$\mathcal{F} = D$
Lower ray	(1, 0, 0)
Upper ray	(0, 1, 0)
Two rays	(1, 1, 0), $\mathcal{F} \neq D$
Window	(0, 0, k), $k \ge 1$
Multi-window	(0, 0, k), $k \ge 2$
Mixed	$k \ge 1$ and $\ell + h \ge 1$

A failure set bounded away from both ends is often called a window. This thesis uses that meaning for the window class (0, 0, k), $k \geq 1$, and calls each bounded component a window component. The two terms are not interchangeable. A mixed class contains window components but is not a window class.

2.2 Topology alone excludes nothing

Proposition 2.1 (realisability). For every ℓ, $h \in \{0, 1\}$ and every finite $k \geq 0$ there is a continuous, piecewise strictly monotone function on $(0, \infty)$ and a tolerance whose failure set has class (ℓ, h, k). All is realised too. Classes with $k = \infty$ need infinitely many turning points and are excluded by H2.

Proof. Take a piecewise-linear zigzag with k interior peaks of height 2, separated by troughs of height 0. It starts at height 2 if $\ell = 1$ and at height 0 otherwise, and ends likewise for h. Set $R^2 = 1$. For All, take any function bounded below by 2. □

Every restriction in this thesis therefore comes from the filtering model, never from topology.

2.3 The critical-tolerance theorem

Theorem 2.2. Assume H1 and H2.

1. **Local constancy.** The class of $\mathcal{F}(R^2)$ is constant on each connected component of $(0, \infty) \setminus \mathcal{C}$. It can change only when R^2 crosses or equals a critical tolerance.
2. **Rays.** $\ell = 1$ if $R^2 < L_-$, and $\ell = 0$ if $R^2 > L_-$. If $R^2 = L_-$, then $\ell = 1$ if and only if σ^2 approaches L_- from above. The same holds for h with L_+.
3. **Window components.** Each window component contains an interior local maximum t_j with $\sigma^2(t_j) > R^2$, and distinct components contain distinct maxima. Hence k is at most the number of interior maxima above R^2, with equality if and only if each such maximum lies in its own component touching neither end.
4. **Determining data.** The class of $\mathcal{F}(R^2)$, for every R^2, is determined by the ordered sequence $(L_-, \sigma^2(t_1), \ldots, \sigma^2(t_n), L_+)$ and the two approach directions. The positions t_j are needed only to locate the components.

Proof. (1) On each piece, σ^2 is a strictly monotone bijection onto the open interval between its end values, all of which lie in $\mathcal{C}$ or are infinite. For $R^2 \notin \mathcal{C}$, the piece's contribution to $\mathcal{F}$ is empty, the whole piece, or a one-sided subinterval at its larger end, decided by where R^2 sits relative to the end values. Adjacent contributions join exactly when the shared turning-point value exceeds R^2. All of this is constant between critical tolerances. (2) Near θ_-, σ^2 is close to L_-. At equality the first piece is strictly monotone, so it lies entirely above or entirely below L_-. (3) A window component (a, b) has $a, b \in D$ with $\sigma^2(a) = \sigma^2(b) = R^2$ and $\sigma^2 > R^2$ between, so σ^2 attains an interior maximum there, which is a turning point. Disjoint components contain distinct maxima. (4) The construction in (1) uses only the end values of each piece, and (2) adds the approach directions. At a tolerance equal to an interior turning-point value the same holds: a minimum with value R^2 is excluded from $\mathcal{F}$ and splits the component around it, and a maximum with value R^2 contributes nothing. Under H2 the approach directions are themselves determined by the sequence. The first piece rises if and only if $\sigma^2(t_1) > L_-$, or, with no turning point, if and only if $L_+ > L_-$. □

Applied to the Bayesian-weight gain with $\kappa > 2$ (Proposition 5.8), the sequence (F, M, H), with both ends approached from above, returns its four regimes, equality cases included. Here M is the peak value and H the high-precision limit, both defined in Proposition 5.8.

Theorem 2.2 is the one-dimensional case of a known picture. The superlevel sets of a function on an interval, followed as the threshold moves, form a zero-dimensional persistence module, equivalently a merge tree. Its changes occur only at endpoint values and critical values. See [32] for join trees of superlevel sets, [33] for the Morse-theoretic backbone, and [30, 34, 35] for persistence and merge trees. What this thesis adds is the reading at a fixed tolerance, with equality cases and approach directions. That is the reading a specification needs.

2.4 The critical-value signature

Definition 2.3. The *critical-value signature* of a curve satisfying H2 is

$$S(\sigma^2) = (L_-,\ \sigma^2(t_1),\ \ldots,\ \sigma^2(t_n),\ L_+;\ d_-,\ d_+)$$

where d± ∈ {above, below} are the approach directions. Its **critical-tolerance diagram** is the (θ, R^2) half-plane cut by horizontal lines at the values in $\mathcal{C}$. Each band between consecutive lines carries one class, and each line carries its own equality class.

By Theorem 2.2(4) the class is a function of S and R^2 alone, so classifying a model means computing its signature. The precision sweep of the Bayesian-weight relay (Proposition 5.8) shows the reading.

Band of R^2	Class ($\kappa > 2$)
$R^2 \leq H$	All
$H < R^2 \leq F$	Lower ray
$F < R^2 < M$	Window
$R^2 \geq M$	Empty

Figure B.1 (Appendix B): a schematic curve (not a filter's variance) with its critical tolerances, and the class in each band between them.

A second level sits above the diagram. A family parameter (κ, Π_0, ρ, β) changes the signature itself. It can add or remove turning points, or reorder their values. The values of that parameter where the signature changes length are the **onset conditions** of later chapters, such as $\kappa = 2$ (Proposition 5.8) and the Hill pair birth (Theorem 5.12).

2.5 Where the class changes

For a fixed curve the class changes only at the first four events in the table below. Pair birth changes the signature as a family parameter varies. The last row changes nothing. Each row is a consequence of Theorem 2.2 or of its plateau extension.

Event	Occurs when	Effect on the class	Label
Boundary	$R^2 = L_-$ or L_+	a ray appears or disappears; at equality the approach direction decides	Thm 2.2(2)
Extremum	$R^2 = \sigma^2(t_j)$	a component is born or dies (maximum), or two components merge or split (minimum)	Thm 2.2(3)
Equal height	two maxima separated by a lower minimum share a value M_2, each in a window just below it	k drops by 2 at the single tolerance M_2	Cor. 2.5
Plateau	R^2 equals a plateau value (H2′)	the plateau leaves the failure set and can split a component	Thm 2.4
Pair birth	a family parameter creates a minimum–maximum pair through a nondegenerate fold	a new band of tolerances opens from zero width; values separate like $\delta^{3/2}$, positions like $\delta^{1/2}$	Prop. 2.7
Non-event	a stationary point that is not a turning point	none	Cor. 2.6

The first four are events in R^2 for one curve. Pair birth is an event in the family parameter. It changes the signature, and so the diagram itself.

The four results behind the table are these.

Theorem 2.4 (plateaus). Under H2′ instead of H2, Theorem 2.2 holds with three changes. Plateau values join the critical tolerances. At R^2 equal to a plateau value, the plateau is excluded from $\mathcal{F}$, which can split or remove components. An end adjacent to a plateau at value L carries a ray if and only if $R^2 < L$.

Proof. As for Theorem 2.2(1)–(2). A constant piece of value p contributes the whole piece to $\mathcal{F}$ if $R^2 < p$, and nothing otherwise. That is constant on each interval of tolerances that avoids p. □

Corollary 2.5 (equal-height maxima). Suppose two interior maxima share the value M_2, are separated by a lower minimum, and just below M_2 each lies in a component touching neither end (the equality condition of Theorem 2.2(3)). Then just below M_2 both count in k, and at and above M_2 both windows are gone. So k drops by 2 at one tolerance. The class is otherwise unaffected.

Proof. Theorem 2.2(3)–(4). □

Families with at most one maximum cannot produce this event.

Corollary 2.6 (degenerate stationary points). A stationary point that is not a turning point never changes the class. A flat-topped maximum acts as a maximum.

Proof. By Theorem 2.2(4) the class depends only on the ordered turning-point values, the endpoint limits and the approach directions. □

Proposition 2.7 (pair birth). Suppose a minimum–maximum pair is born as a family parameter passes $\delta = 0$ through a nondegenerate fold. Near the birth point, in a local coordinate x, $d\sigma^2/d\theta \approx a(x^2 - b\delta)$, with $a, b \neq 0$. Then for $b\delta > 0$ the turning-point

positions separate like $\delta^{1/2}$, and the values like $M - m \sim \delta^{3/2}$. So the band of tolerances that sees the pair opens continuously from zero width.

Proof. The roots of $a(x^2 - b\delta)$ are $\pm\sqrt{(b\delta)}$. The change in value between them is $(4/3)|a|(b\delta)^{3/2}$. □

Verification status: Theorem 2.4 and Corollaries 2.5 and 2.6 were checked in Task 10 (T10.5): the plateau case; k = 2 just below an equal-height pair and Empty at and above it, by both classification routes; and 1,000 tolerances with identical classes with and without an inflection point. For Proposition 2.7, T10.5 fitted the exponents 1.50007 (values) and 0.50002 (positions) for the Hill gain with n = 2. The pre-registered pair-birth check failed on its criterion (Appendix D, Task 10). The result rests on the proof. A re-check under a corrected criterion passed in Task 19 (H-10.5).

2.6 From shape to topology

Non-monotonicity, extrema and window components are three different properties, and only the last is topological. The chain runs in one direction.

1. **Monotone curve.** The class is Empty, All or a single ray (lower if decreasing, upper if increasing). No other class is possible.
2. **Non-monotone curve.** It has at least one turning point. A curve whose only turning point is a minimum gives at most two rays, never a window component. This is the mechanism of Theorem 4.1.
3. **Interior maximum above R^2.** It lies in some component of $\mathcal{F}(R^2)$.
4. **Window component.** That component is a window component only if, on each side of the maximum, the curve falls to R^2 or below before reaching the end. That happens through a minimum at or below R^2, or through an endpoint limit below R^2 (or equal to it and approached from below).

So a non-monotone curve can fail to window at a given tolerance, and one curve can show a window at one tolerance and a ray at another. Proposition 5.13 shows the Hill gain reaching All, a lower ray, a mixed class, a window and Empty within one family.

2.7 Reparameterisation

Corollary 2.8. Let $\varphi : D' \to D$ be a continuous strictly monotone bijection between open intervals, and set $\tilde{\sigma}^2 = \sigma^2 \circ \varphi$. For every R^2, the failure set $\tilde{\mathcal{F}}(R^2) = \varphi^{-1}(\mathcal{F}(R^2))$ has the same k. If φ is increasing, it has the same (ℓ, h). If φ is decreasing, ℓ and h are exchanged. The signature is preserved, reversed when φ is decreasing.

Proof. φ is a homeomorphism, so it maps components to components and preserves being bounded away from both ends. An increasing φ maps neighbourhoods of θ'_- onto neighbourhoods of θ_-. A decreasing one maps them onto neighbourhoods of θ_+. □

Classes are therefore properties of the failure set, not of the scale on which a sweep is plotted. Π, $\log \Pi$ and Π^3 give the same classes. Orientation still matters. Sweeping $1/\Pi$ instead of Π, or the sampling rate $1/\Delta t$ instead of Δt, turns every lower ray into an upper ray. Window widths and positions are not invariant. The starting study's window (1/9, 4) spans a factor of 36 in Π, but a different length in any other coordinate.

2.8 Structural stability

Away from critical tolerances, the class survives small errors in the variance curve, provided the error does not add wiggles where the curve crosses the tolerance.

Proposition 2.9. Let σ^2 satisfy H2, and let $R^2 \notin \mathcal{C}$. Then the crossing set $\{\theta : \sigma^2(\theta) = R^2\}$ is finite, say $\{x_1, \ldots, x_m\}$. Choose disjoint compact intervals $N_i \subset D$, each inside one monotone piece, with x_i in the interior of N_i. Set

$$\vartheta = \inf\{\, |\sigma^2(\theta) - R^2| : \theta \in D \setminus \textstyle\bigcup_i \operatorname{int} N_i \}$$

Then $\vartheta > 0$, and every continuous $\tilde{\sigma}^2$ on D with

1. sup over D of $|\tilde{\sigma}^2 - \sigma^2| < \vartheta$, and
2. $\tilde{\sigma}^2$ strictly monotone on each N_i,

has failure set $\tilde{\mathcal{F}}(R^2)$ of the same class as $\mathcal{F}(R^2)$.

Proof.

- **Finitely many crossings.** Since R^2 is not a turning-point value, crossings lie inside pieces, at most one per strictly monotone piece.
- $\vartheta > 0$**.** The function $|\sigma^2 - R^2|$ is continuous and positive on the closed set $K = D \setminus \cup \operatorname{int} N_i$, which contains no crossing. Near $\theta\pm$ it tends to $|L\pm - R^2|$, which is positive because $L\pm \in \mathcal{C}$ when finite (or to ∞). So it is bounded below on the tails, and on the compact remainder.
- **Signs off the crossings.** On K, condition 1 gives $\tilde{\sigma}^2 - R^2$ the same sign as $\sigma^2 - R^2$.
- **One crossing per** N_i**.** The ends of N_i lie in K, where $\sigma^2 - R^2$ takes opposite signs. So does $\tilde{\sigma}^2 - R^2$. By condition 2, $\tilde{\sigma}^2$ crosses R^2 exactly once in N_i, in the same direction.
- **Conclusion.** $\tilde{\sigma}^2 - R^2$ and $\sigma^2 - R^2$ have the same ordered pattern of positive and negative intervals along D, ends included. Components correspond one to one, and so do the components touching each end. □

Sufficient condition. If σ^2 is C^1 with non-zero slope on each N_i, condition 2 holds for any C^1 perturbation whose slope differs from $\sigma^{2\prime}$ by less than $\min |\sigma^{2\prime}|$ on each N_i.

Why condition 2 is needed. Closeness in sup norm alone does not suffice. A wiggle of height below ϑ placed at a crossing can cross R^2 several times, and so create extra, arbitrarily short components.

Direction in condition 2. Condition 2 names no direction, and none is needed. By condition 1, $\tilde{\sigma}^2 - R^2$ has the same sign as $\sigma^2 - R^2$ at both ends of N_i, and those signs are opposite. So a strictly monotone $\tilde{\sigma}^2$ on N_i is monotone in the same direction as σ^2.

Reading. The margin ϑ is the operational meaning of 'thin'. When a class occupies a narrow band of tolerances, ϑ is small, and modelling errors larger than ϑ can change the class. The Hill mixed band has median width 5×10^{-4} F (Proposition 5.13). The sampled relay's prior dips below the floor shrink like $e^{-2\lambda\Delta t}$ with the position of the minimum, from about 95% of F when it lies at $\lambda\Delta t < 1$ to below 10^{-300} for $\lambda\Delta t$ above about 380 (§6.5, Proposition 6.20). The far-out ones are thinner than any modelling error. At a critical tolerance there is no margin at all. Raising or lowering a maximum by any amount at R^2 equal to its value creates or removes a component.

Relation to persistence stability. Persistence diagrams are stable in sup norm. The bottleneck distance between the diagrams of two functions is at most the sup-norm distance between them [31]. The class at one fixed tolerance is not stable in that sense. A small perturbation can push an extremum across R^2, or add a short component near a crossing. Proposition 2.9 is the fixed-level counterpart: it buys stability with a margin ϑ and a monotonicity condition near the crossings. The contrast is between a summary over all levels at once, which is stable, and a single slice, which is not.

Verification status: Corollary 2.8 and Proposition 2.9 were checked numerically in Task 15 under pre-registered criteria. For 2.8: 1,000 curves, each with increasing and decreasing, smooth and piecewise-linear reparameterisations, at 50 or more tolerances per curve including every critical value; 200,000 classifications, no class or signature mismatch. For 2.9: 1,000 perturbations on 500 curves satisfying both conditions; no class change. A perturbation that breaks only condition 2 (a wiggle of height 0.9ϑ inside one N_i) changed the class from an upper ray to (0, 1, 4), as the remark above predicts.

2.9 What the framework supplies

Theorem 2.2 reduces every classification in this thesis to two tasks: compute the signature of a curve, or constrain which signatures a family can reach. The remaining chapters divide that work.

- **Chapter 3** gives the tool for computing signatures of the white-noise relay: a normal form in which every curve is a path across one fixed surface.
- **Chapters 4–6** compute signatures sweep by sweep.
- **Chapter 5** also measures *expressivity*: which signatures a gain family can reach, from every allowed class (arbitrary continuous gains) down to at most one window component (natural single-stage gains).
- **Chapter 7** constrains the optimal filter. Along a causal information ordering its curve is monotone, so its signature has no turning points.

The spine is therefore: curve $\to$ signature $\to$ class; gain family $\to$ reachable signatures; optimality and information ordering $\to$ further restriction.

3. The Normal Form of the White-Noise Relay

For the scalar continuous-time relay with white sensory noise, every stationary variance is one fixed surface evaluated along a path: $\sigma^2(\theta) = \mathrm{F} \cdot \mathcal{V}(\nu(\theta),\ \mathrm{q}(\theta))$ (Theorem 3.2). The surface carries the floor crossing, the optimal filter, and every tolerance boundary. A sweep or a gain family only chooses the path. This is the first of the thesis's two pillars. The normal form covers white sensory noise, and matched-power coloured noise in §3.8. It does not reach sampled observations, correlation time at matched variance, or sensor delay. Chapters 6 and 7 treat those.

Verification status: Theorem 3.2 was checked by hand and then verified symbolically in Task 14 (Part G), in the reading $\nu = 2\Sigma\Pi/\lambda^2$, $q = G/(F\Pi)$.

3.1 The model

A scalar error x is corrected towards a noisy signal with a gain G that depends on the observation precision Π:

$$\dot{x} = -(\lambda + G)\, x + G\, \xi_s + \xi_{\mathrm{sim}}, \qquad \sigma^2 = \frac{\Sigma + G^2/(2\Pi)}{\lambda + G}$$

Proposition 3.1 (stationary variance). Here ξ_{s} is white noise of intensity $1/\Pi$, and ξ_{sim} is white noise of intensity 2Σ. The stationary variance is the formula above. If $\mathrm{G} \to 0$ and $\mathrm{G}^2/\Pi \to 0$ as $\Pi \to 0$, σ^2 tends to the floor $\mathrm{F} = \Sigma/\lambda$. If $\mathrm{G} \to \mathrm{G}_\infty \in (0, \infty)$ as $\Pi \to \infty$, it tends to the high-precision limit $\mathrm{H} = \Sigma/(\lambda + \mathrm{G}_\infty) < \mathrm{F}$.

So Σ is half the diffusion intensity: without correction the state obeys $\mathrm{dx} = -\lambda \mathrm{x}\ \mathrm{dt} + \sqrt{(2\Sigma)}\ \mathrm{dW}$, whose stationary variance is the floor $\mathrm{F} = \Sigma/\lambda$. Under the other common convention, $\mathrm{dx} = -\lambda \mathrm{x}\ \mathrm{dt} + \sigma\ \mathrm{dW}$, the same floor is $\sigma^2/(2\lambda)$. The equation is exactly the error equation of a constant-gain filter for a hidden Ornstein–Uhlenbeck state. So σ^2 is an actual error variance, as in the sensitivity analysis of mismatched filters [4–9, 36–38].

Proof. The state x is an Ornstein–Uhlenbeck process with rate $\lambda_{\mathrm{G}} = \lambda + \mathrm{G}$ and noise intensity $\mathrm{D} = \mathrm{G}^2/\Pi + 2\Sigma$. Its stationary variance is $\mathrm{D}/(2\lambda_{\mathrm{G}})$, which is the formula above. The two limits follow by substitution. □

3.2 Coordinates and the surface

Two dimensionless numbers carry all the dependence on λ, Σ, Π and G:

$$\nu = \frac{2F\Pi}{\lambda} = \frac{2\Sigma\Pi}{\lambda^2}, \qquad q = \frac{G}{F\Pi}$$

ν compares the observation precision with the floor, over one relaxation time $1/\lambda$. The gain q is measured against the floor precision FΠ. In these coordinates the variance takes one form.

Theorem 3.2 (normal form).

$$\frac{\sigma^2}{F} = \mathcal{V}(\nu, q) = \frac{1 + \nu q^2/4}{1 + \nu q/2}$$

Proof. Divide σ^2 by $F = \Sigma/\lambda$: $\sigma^2/F = (1 + G^2/(2\Pi\Sigma))/(1 + G/\lambda)$. With $G = qF\Pi$, $G/\lambda = \nu q/2$ and $G^2/(2\Pi\Sigma) = q^2F\Pi/(2\lambda) = \nu q^2/4$. □

Written with $g = G/\lambda = \nu q/2$, the same identity is $(1 + g^2/\nu)/(1 + g)$. Lemma 6.23 uses it for the process-noise sweep. The coordinate q is preferred here because the floor crossing and the optimal filter become simple in it (§3.3–3.4). Reducing filter performance to dimensionless indices is not new. Kalata's tracking index [25] does it with one parameter for α–β trackers. The contribution here is this two-parameter form for the continuous-time relay, and its use as a map of failure regions. Two partial derivatives are used throughout:

$$\frac{\partial \mathcal{V}}{\partial q} = \frac{\nu\,(\nu q^2 + 4q - 4)}{2(\nu q + 2)^2}, \qquad \frac{\partial \mathcal{V}}{\partial \nu} = \frac{q\,(q - 2)}{(\nu q + 2)^2}$$

3.3 The floor crossing is the line q = 2

Subtracting the floor gives

$$\mathcal{V}(\nu, q) - 1 = \frac{\nu\, q\,(q - 2)}{4 + 2\nu q}$$

so $\sigma^2 > F$ if and only if $q > 2$, whatever λ, Σ and Π are. This is **Corollary 3.3 (excess identity)**. In dimensional form it reads $G > 2F\Pi$. The numerator of $\mathcal{V}$ adds the noise the correction injects, $\nu q^2/4$ in units of F. The denominator divides by the variance it removes, $1 + \nu q/2$. Injection beats removal exactly when $q > 2$.

Theorem 3.4 (window theorem). Let G be continuous along the precision sweep, with $G \to 0$ and $G^2/\Pi \to 0$ as $\Pi \to 0$, and $G \to G_\infty \in (0, \infty)$. Some tolerance yields a window class if and only if $\sup q > 2$ along the path. Every R^2 between F and $\sup \sigma^2$ then gives class (0, 0, k) with $k \geq 1$. The optimal filter never windows along precision, because its error decreases strictly (Proposition 3.5). Each family's condition is that supremum.

Gain family	q along Π	sup q	Window condition	Label
Bayesian weight $c\Pi/(\Pi + \Pi_0)$	$c/(F(\Pi + \Pi_0))$, decreasing	$\kappa = c/(F\Pi_0)$, at $\Pi \to 0$	$\kappa > 2$	Prop. 5.8
Capped min(gΠ, c)	g/F, then $c/(F\Pi)$	g/F	$g > 2F$	Prop. 5.10
Hill n = 2	$c\Pi/(F(\Pi^2 + {\Pi_0}^2))$, peaked at Π_0	$c/(2F\Pi_0)$	$\Pi_0 < c\lambda/(4\Sigma)$	Prop. 5.13
Misspecified Kalman	decreasing from F'/F	F'/F	$F' > 2F$	Prop. 5.9
Precision-blind G_0	$G_0/(F\Pi)$, unbounded at $\Pi \to 0$	∞	none: $G \to 0$ fails, curve starts at ∞	Prop. 5.11

The over-weighting ratio κ is therefore not a special Bayesian quantity. It is the value of q at the low-precision end, and it equals sup q only for gains whose G/Π decreases. For the Hill gain, q starts at 0 and peaks inside the domain. For $n = 2$ its condition is $c/(F\Pi_0) > 4$, not > 2. For general n it is $c/(F\Pi_0) > 2/h(n)$, where $h(n) = (n-1)^{(n-1)/n}/n$ is the maximum of $t^{n-1}/(1+t^n)$. For example, $2/h(n) \approx 3.78$ at $n = 1.5$ and $n = 3$, and ≈ 3.51 at $n = 4$.

The condition sup $q > 2$ governs the **window class**, not window components. In the worked example ($\lambda = c = 1$, $\Sigma = 0.01$), the Hill gain with $25 \le \Pi_0 < 31.475$ has sup $q \le 2$. Yet it produces a window component inside the mixed class (1, 0, 1) (Proposition 5.13).

3.4 The optimal filter is the lower envelope

At fixed ν, $\partial\mathcal{V}/\partial q$ vanishes where $\nu q^2 + 4q - 4 = 0$. That equation has one positive root:

$$q^\star(\nu) = \frac{2}{1+\sqrt{1+\nu}}, \qquad \mathcal{V}_\star(\nu) = \mathcal{V}(\nu, q^\star) = q^\star(\nu)$$

The identity $\mathcal{V}\star = q\star$ holds because $\nu q\star^2/4 = 1 - q\star$ at the root. So $\mathcal{V}\star = (2 - q\star)/(1 + \nu q\star/2) = q\star$. In dimensional form it is $G\star = \sigma\star^2\Pi$, the steady-state Kalman–Bucy gain [1, 39, 40] (Proposition 3.5, Lemma 3.6).

Proposition 3.5 (optimal gain). For this plant the steady-state Riccati equation $0 = 2\Sigma - 2\lambda p - \Pi p^2$ gives

$$G^\star(\Pi) = p\,\Pi = -\lambda + \sqrt{\lambda^2 + 2\Sigma\Pi}, \qquad \sigma^2_\star(\Pi) = p = \frac{2\Sigma}{\lambda + \sqrt{\lambda^2 + 2\Sigma\Pi}}$$

$G\star$ also minimises σ^2 over constant gains, so $\sigma^2(\Pi) \ge \sigma\star^2(\Pi)$ for every gain. $\sigma\star^2$ decreases strictly from F as Π grows, so $G\star = \sigma\star^2\Pi < F\Pi$ for every $\Pi > 0$. In the coordinates of §3.2, $q\star = G\star/(F\Pi)$ and $\sigma\star^2/F = \mathcal{V}\star = q\star$.

Lemma 3.6 (single minimum in the gain). At fixed Π, σ^2 is strictly decreasing in G on $(0, G\star)$, and strictly increasing on $(G\star, \infty)$.

Proof of Proposition 3.5 and Lemma 3.6. The Riccati equation and its positive root give the steady-state Kalman–Bucy filter [1, 39, 40]. Differentiating the formula of Proposition 3.1 in G gives $\partial\sigma^2/\partial G = (G^2 + 2\lambda G - 2\Sigma\Pi)/(2\Pi(\lambda + G)^2)$, whose only positive zero is $G\star$. This is Lemma 3.6, and it gives the minimality in Proposition 3.5. $\sigma\star^2$ decreases strictly in Π, because $\sqrt{(\lambda^2 + 2\Sigma\Pi)}$ increases. It tends to $2\Sigma/(2\lambda) = F$ as $\Pi \to 0$. Dividing $G\star$ by $F\Pi$ with $\nu = 2\Sigma\Pi/\lambda^2$ gives $q\star = 2(\sqrt{(1+\nu)} - 1)/\nu = 2/(1 + \sqrt{(1+\nu)})$. □

Verification status: Proposition 3.5 in Tasks 1 and 7; Lemma 3.6 symbolically in Task 8 (T8.1).

Three consequences follow at once.

- $\mathcal{V}$ is strictly decreasing in q below $q\star$, and strictly increasing above it, at every ν (Lemma 3.6).
- $q\star < 1$, so the optimal filter sits at least a factor of 2 below the floor-crossing line $q = 2$.
- $\mathcal{V}\star$ decreases strictly from 1 ($\nu \to 0$) to 0 ($\nu \to \infty$). Along the precision sweep the optimal filter's classes are therefore lower ray or Empty, as Chapter 7 obtains from

information ordering (Proposition 7.7(a)).

From $\partial\mathcal{V}/\partial\nu$, a path at constant q rises with precision when $q > 2$, and falls when $q < 2$. An over-gained relay gets worse as observations improve.

3.5 Tolerance boundaries

The boundary $\mathcal{V} = R^2/F$ is the quadratic $q^2 - 2(R^2/F)\,q + 4(1 - R^2/F)/\nu = 0$, with roots

$$q_{\pm}(\nu;\, R^2/F) = \frac{R^2}{F} \pm \sqrt{\Big(\frac{R^2}{F}\Big)^2 + \frac{4\,(R^2/F - 1)}{\nu}}$$

Since $\mathcal{V} - R^2/F$ has the sign of $(q - q_-)(q - q_+)$, the failure region at fixed ν is $q > q_+$ **or** $q < q_-$. Three regimes follow.

Tolerance	Failure region in the (ν, q) plane	Consequence
$R^2/F > 1$ (looser than the floor)	$q > q_+$ only ($q_- < 0$). q_+ falls from ∞ at $\nu \to 0$ to $2R^2/F$ at $\nu \to \infty$	failure needs over-gain; windows live here
$R^2/F = 1$ (the floor)	$q > 2$	the excess identity
$R^2/F < 1$ (tighter than the floor)	everything for $\nu < \nu_R = 4(1 - R^2/F)/(R^2/F)^2$; below the envelope, $q < q_-$ or $q > q_+$	every gain fails at low precision

In dimensional form, $\nu < \nu_R$ is $\Pi < \pi_R = 2\lambda(F - R^2)/R^4$. Below that precision the whole surface lies above the tolerance. This is the forced lower ray of Corollary 5.2. The two branches meet on the envelope, at $q = R^2/F$ and $\nu = \nu_R$. Above the floor only the upper side matters. Below it, under-gain fails too.

Figure B.2 (Appendix B): the (ν, q) plane with the line $q = 2$, the envelope $q_\star(\nu)$, and the boundaries for $R^2/F = 0.5$, 1 and 2.

3.6 The inverse map: realisability

Solving $\mathcal{V} = v/F$ for q inverts the surface:

$$q_{\pm} = \mathcal{V} \pm \sqrt{\mathcal{V}^2 + \frac{4(\mathcal{V} - 1)}{\nu}}, \qquad G_+ = F\Pi\, q_+ = \Pi v + \sqrt{\Pi\,(\Pi v^2 + 2\lambda v - 2\Sigma)}$$

G_+ is exactly the realising gain of Theorem 5.1. The root is real if and only if $\nu\mathcal{V}^2 + 4\mathcal{V} - 4 \geq 0$. That is, $\mathcal{V} \geq \mathcal{V}_\star$: the same quadratic as the envelope. So the surface is two-sheeted over its envelope. This is the scalar filtering analogue of covariance assignment [14, 15, 41–44], which characterises the state covariances a feedback gain can assign in multivariable control. In the scalar filter the attainable set is the whole half-line above the optimum, with an explicit gain. The upper sheet $q_+ \geq q_\star$ holds the over-gained relays. The lower sheet $q_- \leq q_\star$ holds the under-gained ones.

Two results of Chapter 5 read directly off this picture.

- **Realisability (Theorem 5.1).** Any continuous curve on or above the envelope is

reached by a continuous gain on the upper sheet. Nothing below the envelope is reachable.

- **The branch switch (Corollary 5.4).** On the upper sheet $G_+ \geq \Pi v$, so $G_+/\Pi \to 2H$ whenever $v \to H > 0$. A gain with a finite limit must cross from the upper sheet to the lower one, and can do so only where its path touches the envelope.

3.7 Sweeps as paths

A sweep is a path $\theta \mapsto \gamma(\theta) = (\nu(\theta), q(\theta))$. Along it, $\sigma^2(\theta) = F(\theta) \cdot \mathcal{V}(\gamma(\theta))$. Along a path parameterised by ν,

$$\frac{d\mathcal{V}}{d\nu} = \frac{q\,(q-2) + \frac{\nu}{2}\,(\nu q^2 + 4q - 4)\,q'(\nu)}{(\nu q + 2)^2}$$

which is the elasticity lemma (Lemma 3.7) in normal form. Its turning points are the turning points of the variance curve, so a gain family's signature is read from its path.

Lemma 3.7 (elasticity form). For any differentiable gain $G(\Pi) > 0$ with elasticity $\varepsilon = \Pi G'/G$,

$$\frac{d\sigma^2}{d\Pi} = \frac{G\,N}{2\Pi^2(\lambda + G)^2}, \qquad N = \varepsilon\,(G^2 + 2\lambda G - 2\Sigma\Pi) - G(\lambda + G)$$

The turning points of $\sigma^2(\Pi)$ are the sign changes of N. When G is strictly monotone, N is a function of G alone.

Proof. $d\sigma^2/d\Pi = \partial\sigma^2/\partial G \cdot G' + \partial\sigma^2/\partial\Pi = (G^2 + 2\lambda G - 2\Sigma\Pi)G'/(2\Pi(\lambda + G)^2) - G^2/(2\Pi^2(\lambda + G))$. Then use $\Pi G' = \varepsilon G$. □

Verification status: symbolically in Task 10 (T10.1(a)).

The failure set is the preimage of the failure region. With a fixed floor, $\mathcal{F}(R^2) = \{\theta : \mathcal{V}(\gamma(\theta)) > R^2/F\}$. With a moving floor, the threshold moves too: $\mathcal{F}(R^2) = \{\theta : \mathcal{V}(\gamma(\theta)) > R^2/F(\theta)\}$.

Sweep	ν	q	Floor	Path on the surface	Chapter
Prior precision Π_0, coupling c	fixed	moves with G	fixed	horizontal	4
Precision Π	$\propto \Pi$	$G(\Pi)/(F\Pi)$	fixed	a curve set by the gain family	5
Process noise Σ	$\propto \Sigma$	$G/(F\Pi)$	moves ($F \propto \Sigma$)	relative tolerance: fixed boundary; absolute: boundary R^2/F moves along the path	6

A horizontal path is a section of the surface, $\mathcal{V}(\nu, \cdot)$. Each section is unimodal, with its minimum at $q^\star(\nu)$. So a monotone gain sweep at fixed precision has at most one turning point, a minimum, and never a window component (Theorem 4.1). Under a relative tolerance the threshold is constant again. That is why the Σ sweep then classifies like the precision

sweep (Lemma 6.23).

3.8 Scope

The normal form covers the scalar, stationary, continuous-time relay with white sensory noise, and a gain that is constant in time at each parameter value. Within that scope the white-noise results of Chapters 3–5 read as objects on one surface.

Result	On the surface
Excess identity (Corollary 3.3)	the line q = 2
Optimal gain (Proposition 3.5, Lemma 3.6)	the envelope q = q⋆(ν)
Window theorem (Theorem 3.4)	sup q > 2 along the path
Monotone gain sweeps (Theorem 4.1)	horizontal paths through a unimodal section
Forced lower ray (Corollary 5.2)	for $\nu < \nu_{\mathrm{R}}$ the surface lies above the tolerance
Realisability (Theorem 5.1)	the inverse map, upper sheet
Branch switch (Corollary 5.4)	crossing sheets at an envelope touch
Elasticity (Lemma 3.7)	the path derivative
Process-noise sweep, relative tolerance (Lemma 6.23)	a path against a fixed boundary

One extension is immediate. Under matched-power coloured noise, handled by measurement differencing [27, 28, 45], Lemma 6.1 becomes a three-coordinate surface with t = λτ:

$$\mathcal{V}_\tau(\nu, q, t) = \frac{1 + \dfrac{\nu q^2}{4\,(1 + t\,(1 + \nu q/2))}}{1 + \nu q/2},$$

$$\frac{P}{F} = \frac{2}{1 + \nu t + \sqrt{(1+\nu t)^2 + \nu(1-t)^2}}$$

The second formula is the optimal filter along τ (Corollary 6.5). At t = 0 both reduce to the white-noise surface and its envelope. Chapter 6 uses them.

The normal form does not cover sampled observations (6.13–6.22). It does not cover correlation time at matched variance or in the power family (Proposition 6.10, Theorem 7.8), or sensor delay (Proposition 7.11). There the optimal filter's behaviour is decided by the information structure of the observations, not by the shape of $\mathcal{V}$. Chapter 7 takes over. Theorem 2.2 applies on both sides.

4. Design Sweeps as Horizontal Paths

A design sweep changes the filter but not what it observes. Its path on the surface $\mathcal{V}(\nu, q)$ therefore runs horizontally, at fixed ν. Each horizontal section is unimodal, with its minimum at $q\star$, so every design sweep has at most one turning point, a minimum. It can never produce a window component or a mixed class (Theorem 4.1). The two design parameters of the Bayesian-weight relay, prior precision and coupling, are classified completely here (Propositions 4.2–4.5). Both reach their best value exactly at the optimal filter.

4.1 Why design sweeps are horizontal

A design sweep varies a parameter θ of the gain only. Precision Π, relaxation rate λ and process noise Σ stay fixed. So $\nu = 2F\Pi/\lambda$ and the floor F are fixed, and only $q(\theta) = G(\theta)/(F\Pi)$ moves. By Lemma 3.6 the section $q \mapsto \mathcal{V}(\nu, q)$:

- equals 1 at $q = 0$, since no correction gives the floor;
- decreases strictly to its minimum $\mathcal{V}\star(\nu) = q\star(\nu)$ at $q = q\star$;
- increases strictly after it, passing 1 again at $q = 2$, and growing without bound.

Theorem 4.1 (monotone gain sweeps cannot window). Fix Π. Let $\theta \mapsto G(\theta)$ be continuous and strictly monotone on D. Then $\sigma^2(G(\theta), \Pi)$ is strictly monotone, or strictly decreasing and then strictly increasing, on D. Every failure set is therefore Empty, All, a lower ray, an upper ray or two rays. No window component and no mixed class can occur.

Proof. Composing the section of Lemma 3.6, which decreases and then increases, with a strictly monotone map gives a function with at most one interior turning point, a minimum. Such a function is strictly quasiconvex, so its sublevel sets $\{\sigma^2 \le R^2\}$ are intervals, possibly a point or empty. The complement of an interval in an open interval is at most two rays. □

The theorem covers every design rule whose gain is strictly monotone in its design parameter, not only the Bayesian weight. The verification record tests this on several rules.

Task 8 ran 20,000 random parameter sets, spread over six decades per parameter, through five monotone gain sweeps each, with ten tolerances per sweep: 10^6 classifications in all. The classifications were computed by root-finding on monotone pieces, not by checking grid points. No window, mixed class or interior maximum appeared. The Task 13 audit repeated the check for $c \cdot \tanh(\Pi/\Pi_0)$, $c(1 - e^{-\Pi/\Pi_0})$ and $c\Pi^2/(\Pi^2 + {\Pi_0}^2)$, each swept in Π_0. The audit found no class outside the stated sets. Route agreement failed in one row (row 12), described below.

Row 12 of the Task 13 audit concerns the Π_0 sweep of monotone gains. At tolerances set to the stored limit L_- of increasing tanh-shaped and exp-shaped curves, route (ii)'s single sample point lay within its equality tolerance. The gap there was at most 2.49×10^{-41} relative, inside the 10^{-40} tolerance, so route (ii) read Empty where route (i) read All. Task 13 recorded 122 such cases, in 23 parameter sets, and attributed them to code. In Task 19 the tolerance was set to 10^{-300}. Then 45 cases in 8 sets persisted, and they were attributed to the criterion, since the stored limit is accurate only to about 10^{-25}. At the exact limit the

class is All, by Theorem 2.2(2). Route (ii) was not wrong at the level of its own precision; it was asked to resolve a difference below that precision.

Two consequences shape the rest of the chapter.

- **The optimal filter is fixed along a design sweep**, because the observations do not change. Design sweeps therefore classify miscalibration only. There is no optimal-filter column.
- **A sweep whose gain passes through G⋆ has minimum value exactly $\sigma\star^2(\Pi)$.** The best setting of any design parameter reproduces the steady-state Kalman–Bucy filter. Both directions of misweighting are worse.

4.2 Prior precision

Fix $\Pi > 0$ and $c > 0$. The Bayesian-weight gain $G(\Pi_0) = c\Pi/(\Pi + \Pi_0)$ decreases strictly from c, as $\Pi_0 \to 0$, to 0, as $\Pi_0 \to \infty$. So q moves left along the section, from $q_c = c/(F\Pi)$ to 0. Write

$$V_c = \sigma^2(c, \Pi) = \frac{\Sigma + c^2/(2\Pi)}{\lambda + c}, \qquad \Pi_A = \frac{c(c + 2\lambda)}{2\Sigma}, \qquad \Pi_B = \frac{c}{2F} = \frac{c\lambda}{2\Sigma}$$

so that $\Pi_B < \Pi_A$. In normal form the three cases below are simply where the starting point q_c sits on the section. $\Pi \geq \Pi_A$ means $q_c \leq q\star$. $\Pi_B \leq \Pi < \Pi_A$ means $q\star < q_c \leq 2$. $\Pi < \Pi_B$ means $q_c > 2$.

Proposition 4.2 (prior precision: endpoints and extrema).

- $L_+ = F$, approached from below. $L_- = V_c$. Its approach direction is stated in the next two bullets.
- If $\Pi \geq \Pi_A$ ($c \leq G\star$), σ^2 increases strictly from V_c, approached from above, to F. There is no interior extremum.
- If $\Pi < \Pi_A$ ($c > G\star$), there is exactly one interior extremum, a global minimum at $\Pi_0\star = \Pi(c/G\star - 1)$, with value $\sigma\star^2(\Pi)$. V_c is then approached from below.
- $V_c > F$ if and only if $\Pi < \Pi_B$. $V_c = F$ if and only if $\Pi = \Pi_B$.

Proof. Compose Lemma 3.6 with the strictly decreasing map $\Pi_0 \mapsto G$. The gain passes through G⋆ if and only if $c > G\star$, which is $c^2 + 2\lambda c > 2\Sigma\Pi$, that is $\Pi < \Pi_A$. Solving $c\Pi/(\Pi + \Pi_0) = G\star$ gives $\Pi_0\star$. Near $\Pi_0 = \infty$, $G \to 0^+$, where σ^2 decreases in G, so σ^2 increases in Π_0 and F is approached from below. Near $\Pi_0 = 0$, $G \to c^-$. If $c > G\star$, σ^2 increases in G there, so it decreases in Π_0 and V_c is approached from below. If $c \leq G\star$ the reverse holds. V_c against F is Corollary 3.3 at $G = c$. □

Proposition 4.3 (prior precision: classification at every tolerance).

Case P1, $\Pi \geq \Pi_A$ (sequence V_c, F):

Tolerance	Class
$R^2 \leq V_c$	All
$V_c < R^2 < F$	Upper ray
$R^2 \geq F$	Empty

Case P2, $\Pi_B \le \Pi < \Pi_A$ (sequence V_c, $\sigma\star^2$, F, with $\sigma\star^2 < V_c \le F$):

Tolerance	Class
$R^2 < \sigma\star^2$	All
$R^2 = \sigma\star^2$	Two rays (D minus the single point $\Pi_0\star$)
$\sigma\star^2 < R^2 < V_c$	Two rays
$V_c \le R^2 < F$	Upper ray (empty band when $\Pi = \Pi_B$)
$R^2 \ge F$	Empty

Case P3, $\Pi < \Pi_B$ (sequence V_c, $\sigma\star^2$, F, with $\sigma\star^2 < F < V_c$):

Tolerance	Class
$R^2 < \sigma\star^2$	All
$\sigma\star^2 \le R^2 < F$	Two rays (at equality, D minus $\Pi_0\star$)
$F \le R^2 < V_c$	Lower ray
$R^2 \ge V_c$	Empty

Proof. Theorem 2.2 applied to Proposition 4.2. F is approached from below, so $R^2 = F$ never gives an upper ray. V_c is approached from above in P1, so $R^2 = V_c$ gives All there. It is approached from below in P2 and P3, so $R^2 = V_c$ gives no lower ray. The minimum value $\sigma\star^2$ is attained only at $\Pi_0\star$, which is the single point removed at $R^2 = \sigma\star^2$. □

Worked example ($\lambda = c = 1$, $\Sigma = 0.01$, so $F = 0.01$, $\Pi_B = 50$, $\Pi_A = 150$):

Π	Case	$\Pi_0\star$	$\sigma\star^2$	V_c
1	P3	99.498	0.0099505	0.255
60	P2	64.162	0.0080540	0.0091667
500	P1	none	—	0.0055

Reading. The best fixed prior at a given precision is the one that makes the Bayesian weight equal the Kalman–Bucy gain. Both a prior that is too weak and one that is too strong are worse. At low precision the two-ray band is extremely thin: at $\Pi = 1$, $\sigma\star^2$ is only 0.5% below F. At intermediate precision it is wide: at $\Pi = 60$ it spans 0.0080540 to 0.0091667, about 11% of F. The prior sweep never windows, whatever the precision.

4.3 Coupling

Fix Π and Π_0, and write $K = \Pi/(\Pi + \Pi_0) \in (0, 1)$. The gain $G = cK$ increases strictly from 0 to ∞ as c does. So q moves right along the whole section, from 0 to ∞. It therefore always passes $q\star$, and always crosses the floor line $q = 2$.

Proposition 4.4 (coupling: endpoints and extrema). $L_- = F$, approached from below, and $L_+ = \infty$. There is exactly one interior extremum, a global minimum at

$$c^\star = \frac{G^\star(\Pi)}{K} = \frac{(\Pi + \Pi_0)(\sqrt{\lambda^2 + 2\Sigma\Pi} - \lambda)}{\Pi}$$

with value $\sigma\star^2(\Pi)$. The variance returns to F at $c_F = 2F\Pi/K = 2F(\Pi + \Pi_0)$, and $\sigma^2 > F$ if and only if $c > c_F$.

Proof. Lemma 3.6 composed with the strictly increasing map $c \mapsto cK$, which passes through $G\star$ and through $2F\Pi$. □

Proposition 4.5 (coupling: classification at every tolerance). There is no case split.

Tolerance	Class
$R^2 < \sigma\star^2$	All
$R^2 = \sigma\star^2$	Two rays (D minus the single point $c\star$)
$\sigma\star^2 < R^2 < F$	Two rays
$R^2 \geq F$	Upper ray (c_R, ∞), where $c_R > c\star$ solves $\sigma^2 = R^2$. At $R^2 = F$, $c_R = c_F$.

The class is never Empty, because $\sigma^2 \to \infty$ as $c \to \infty$. It is never a window.

Proof. Theorem 2.2 with the sequence $(F, \sigma\star^2, \infty)$, and F approached from below. □

Worked example ($\lambda = 1$, $\Sigma = 0.01$, $\Pi = \Pi_0 = 1$): $c\star = 0.019901$, $\sigma\star^2 = 0.0099505$, $c_F = 0.04$.

Reading. Coupling behaves like a volume control on the correction. Too little coupling wastes the observations. Too much admits more measurement noise than the damping is worth. Beyond c_F the relay is worse than no filtering at all, at every tolerance at or above the floor.

4.4 Summary

Sweep	Path on the surface	Classes	Never
Any monotone gain at fixed Π	horizontal, monotone in q	Empty, All, lower ray, upper ray, two rays	window, mixed (Theorem 4.1)
Prior precision Π_0	horizontal, from q_c down to 0	P1: All, upper ray, Empty. P2: adds two rays. P3: All, two rays, lower ray, Empty	window (Proposition 4.3)
Coupling c	horizontal, from 0 up to ∞	All, two rays, upper ray	Empty, window (Proposition 4.5)

Verification. Lemma 3.6 and Propositions 4.2 and 4.4 were checked symbolically, every identity reducing to zero (Task 8, checks T8.1–T8.2). Propositions 4.3 and 4.5 were checked at every row, including each critical tolerance exactly, at $\pm 10^{-6}$ relative, and the boundary cases $\Pi = \Pi_A$ and $\Pi = \Pi_B$: 62 of 62 classes were correct, computed by root-finding on monotone pieces (T8.3). Theorem 4.1 was checked on 10^6 classifications (T8.4). Task 13 re-audited all three rows on 1,000 fresh parameter sets each, with no class outside the stated

sets. Row 12 of that audit, described in §4.1, failed its pre-registered route-agreement rule. Its re-check in Task 19 also failed, on its criterion (§8.2).

Figure B.3 (Appendix B): σ^2 along the prior sweep at $\Pi = 1$ (case P3, U-shaped) and $\Pi = 500$ (case P1, monotone), and along the coupling sweep at $\Pi = \Pi_0 = 1$, with $c\star$ and c_F marked.

5. The Precision Sweep: Expressivity of Gain Families

Along precision, the gain family chooses the path through the (ν, q) plane. ν grows in proportion to Π, and q = G(Π)/(FΠ) is whatever the design rule makes it. This chapter asks two questions. What does the model itself allow along precision, whatever the gain? And what do the gain families used in practice actually produce? The first has a short answer: the model rules out exactly one thing, a variance below the optimum (§5.1). Every further restriction belongs to a gain family (§5.2–5.5).

5.1 What the model allows

Theorem 5.1 (gain realisability). Fix λ, $\Sigma > 0$. A function v : $(0, \infty) \to (0, \infty)$ is the variance curve σ^2(G(Π), Π) of some continuous gain schedule G ≥ 0 if and only if v is continuous and v $\geq \sigma\star^2$ everywhere. One such gain is the upper-branch inverse

$$G_+(v, \Pi) = \Pi v + \sqrt{\Pi\,(\Pi v^2 + 2\lambda v - 2\Sigma)}$$

Proof. Necessity is Proposition 3.5. For sufficiency, σ^2(G, Π) = v is the quadratic $G^2 - 2\Pi vG + 2\Pi(\Sigma - \lambda v) = 0$, with roots G$\pm$ = Πv $\pm$ $\sqrt{}$(Π(Πv^2 + 2λv − 2Σ)). The radicand is non-negative if and only if $\Pi v^2 + 2\lambda v - 2\Sigma \geq 0$. The positive root of that quadratic in v is G$\star$/Π = $\sigma\star^2$ (Proposition 3.5), so the condition is v $\geq \sigma\star^2$. Then G$_+$(v(Π), Π) is continuous, and G$_+$ $\geq$ Πv > 0. □

In the coordinates of Chapter 3 this is the inverse map of §3.6. A curve on or above the envelope is reached on the upper sheet. Nothing below the envelope is reached at all. The result is the scalar filtering analogue of the covariance assignment theory of Hotz and Skelton and its successors [14, 15, 41–44]. That theory concerns matrix state feedback. The contribution here is the scalar filtering form and its explicit gain.

Corollary 5.2 (forced lower ray). For every continuous gain schedule and every $R^2 <$ F, the failure set contains (0, π_R), with $\pi_R = 2\lambda(F - R^2)/R^4$. Hence $\ell = 1$. At a tolerance tighter than the floor, no gain gives Empty, a pure window, an upper ray alone, or any other class with $\ell = 0$.

Proof. $\sigma\star^2$ decreases strictly from F (Proposition 3.5). It equals R^2 where $\Pi R^4 + 2\lambda R^2 - 2\Sigma = 0$, that is, at $\Pi = \pi_R$. For $\Pi < \pi_R$, $\sigma^2 \geq \sigma\star^2 > R^2$. □

So ‘windows need loose tolerances’ holds for every gain schedule, with no regularity assumption. In the (ν, q) plane, π_R is the precision ν_R of §3.5, below which the whole surface lies above the tolerance.

Corollary 5.3 (exact class set along precision). For continuous, piecewise strictly monotone variance curves: for $R^2 <$ F the achievable classes are exactly those with $\ell = 1$ (any h, any finite k, including All). For $R^2 \geq$ F every class with finite k is achievable.

Proof. Impossibility is Corollary 5.2. For achievability, by Theorem 5.1 it is enough to build

a continuous, piecewise strictly monotone $v \geq \sigma\star^2$ with the target class. For $R^2 < F$, follow $\sigma\star^2$ on $(0, \pi_R]$ and then a zigzag in $\log \Pi$ with troughs at R^2 and peaks above it. For $R^2 \geq F$, use troughs at F and peaks above R^2, starting above R^2 (for $\ell = 1$) or on $\sigma\star^2$ (for $\ell = 0$). For All, take $v = \sigma\star^2 + R^2$. □

The constructions above may use gains that diverge or oscillate at the ends of the sweep. Under the regularity of Theorem 3.4 the endpoint limits are pinned, but the number of windows is not.

Corollary 5.4 (unbounded windows for regular gains). For every $k \geq 1$ there is a C^∞ gain schedule with the regularity of Theorem 3.4 ($G \to 0$ and $G^2/\Pi \to 0$ as $\Pi \to 0$; $G \to G_\infty \in (0, \infty)$) whose failure set at $R^2 = 1.25F$ has exactly k windows.

Proof. Take the Bayesian-weight gain G_B with $1 < \kappa < 2$ (Proposition 5.8). Its variance $v_B = \sigma^2(G_B, \cdot)$ stays below F, and is smooth and strictly decreasing. Near $\Pi = 0$, $G_B > G\star$, because its initial slope is $\kappa F > F$. For large Π, $G_B < G\star$. So G_B first meets $G\star$ at some Π_c, where v_B touches $\sigma\star^2$, and $G_B = G_+(v_B, \cdot)$ on $(0, \Pi_c)$. Choose k disjoint closed intervals inside $(0, \Pi_c)$. On each, set $v = v_B + b$, with $b \geq 0$ smooth and equal to 0 near the interval's ends, such that v rises strictly to a peak value 1.5F and then falls strictly. On the rising flank take $b' > |v_B'|$. On the falling flank take $b' \leq 0$. Set $v = v_B$ elsewhere, $G = G_+(v, \cdot)$ on $(0, \Pi_c)$, and $G = G_B$ beyond. Outside the bump intervals $G = G_B$, so G is smooth and keeps the regularity of G_B at both ends. Each bump rises strictly and then falls strictly, so it crosses 1.25F exactly twice. And $v_B < F$ elsewhere. □

The construction has to leave the upper sheet before the end of the sweep. On that sheet $G_+ \geq \Pi v$, so $G_+/\Pi \to 2H$ whenever $v \to H > 0$: the gain grows like $2H\Pi$ and has no finite limit. A gain with a finite limit can cross to the lower sheet only where its path touches the envelope (§3.6). In the proof that crossing is the point Π_c.

Figure B.4 (Appendix B): a realised curve with three windows at $R^2 = 1.25F$ on a Bayesian-weight base curve, and the forced lower ray at $R^2 = 0.6F$.

Corollary 5.5 (what regularity pins). Under the regularity of Theorem 3.4, $L_- = F$ and $L_+ = H = \Sigma/(\lambda + G_\infty)$. So ℓ is fixed by R^2 against F (with the approach direction when $R^2 = F$), and h by R^2 against H. The number k is unbounded (Corollary 5.4). Since $H < F$, there is never an upper ray when $R^2 \geq F$.

Proof. The limits are those of Proposition 3.1. The rest is Theorem 2.2. □

Along precision, then, the model rules out exactly one thing: being better than the optimal filter. Every further restriction is a property of the gain family.

Verification. Theorem 5.1 was checked symbolically (Task 10, T10.1) and by 101 of 101 target constructions (T10.2). Corollaries 5.2 and 5.3 were checked by the same 101 constructions and by 24 of 24 forbidden targets, each blocked (T10.2). Two symbolic items failed in the pre-registered SymPy route, which could not simplify square roots of squares. The identities hold by the factor-radicand route (Appendix D, Task 10), and both passed a re-check with a denesting step in Task 19 (H-10.1a, H-10.1b). For Corollary 5.4, the construction first pre-registered failed its check (Appendix D, Task 10). The construction given in the proof passed for k = 1 to 5, with the gain converging ($G(10^{12})/G(10^{11}) - 1 = 6 \times 10^{-10}$).

5.2 Tools for gain families

Three lemmas do most of the work for particular families. Lemma 3.7 locates turning points through the elasticity $\varepsilon = \Pi G'/G$. The next two fix the excess set and the high-precision end.

Lemma 5.6 (excess components). $\{\sigma^2 > F\} = \{G/\Pi > 2F\}$ pointwise. Hence at $R^2 = F$ the failure set is the excess set. For $R^2 \geq F$ every failure component lies in an excess component. If G/Π is unimodal or decreasing, there is at most one window at $R^2 = F$.

Proof. Corollary 3.3, applied at each Π: $\sigma^2 > F$ if and only if $q = G/(F\Pi) > 2$. For $R^2 \geq F$, $\{\sigma^2 > R^2\} \subset \{\sigma^2 > F\}$. If G/Π is unimodal or decreasing, $\{G/\Pi > 2F\}$ is an interval. □

Lemma 5.7 (bounded gains approach H from above). If $G < G_\infty$ everywhere, then $\sigma^2 \geq \Sigma/(\lambda + G) > H$, so H is approached from above and never attained.

Proof. $\sigma^2 \geq \Sigma/(\lambda + G)$, because $G^2/(2\Pi) \geq 0$. And $\Sigma/(\lambda + G) > \Sigma/(\lambda + G_\infty) = H$. □

5.3 Monotone and saturating families

The starting study's gain is the one-step Bayesian weight, a precision-weighting rule of the kind studied in [2, 3]. The gain is standard. The classification of its failure sets along precision is the contribution.

Proposition 5.8 (Bayesian-weight gain along precision). Let $G = c\Pi/(\Pi + \Pi_0)$, with over-weighting ratio $\kappa = (c/\Pi_0)/F$. Then σ^2 has one turning point, a maximum M, if and only if $\kappa > 2$. Otherwise it is strictly decreasing. The endpoint limits are F (approached from above if and only if $\kappa > 2$) and $H = \Sigma/(\lambda + c)$ (approached from above). The classes are:

- $\kappa > 2$: All ($R^2 \leq H$), lower ray ($H < R^2 \leq F$), window ($F < R^2 < M$), Empty ($R^2 \geq M$);
- $\kappa \leq 2$: All ($R^2 \leq H$), lower ray ($H < R^2 < F$), Empty ($R^2 \geq F$).

Proof. Here $\varepsilon = \Pi_0/(\Pi + \Pi_0) = 1 - G/c$. Eliminating $\Pi = \Pi_0 G/(c - G)$ in Lemma 3.7 gives $N = -(G/c)(G^2 + 2\lambda G + 2\Sigma\Pi_0 - \lambda c)$. As Π runs over $(0, \infty)$, G runs once over $(0, c)$. The quadratic has the positive root $G = -\lambda + \sqrt{(\lambda^2 + \lambda c - 2\Sigma\Pi_0)}$, which lies below c if and only if $\lambda c > 2\Sigma\Pi_0$, that is, $\kappa > 2$. Near $\Pi = 0$ the quadratic is close to $2\Sigma\Pi_0 - \lambda c$, so $N > 0$ there when $\kappa > 2$, and the turning point is a maximum. $G/\Pi \to \kappa F$, so F is approached from above if and only if $\kappa > 2$ (Lemma 5.6). H is approached from above by Lemma 5.7. The classes follow from Theorem 2.2. □

The over-weighting ratio decides the window, and the peak is unique.

Proposition 5.9 (misspecified Kalman gain). Let $G = -\lambda' + \sqrt{(\lambda'^2 + 2\Sigma'\Pi)}$, the steady-state gain designed for wrong parameters λ' and Σ', with $F' = \Sigma'/\lambda'$ and $\chi = \Sigma/\Sigma'$. If $F' > 2F$, σ^2 has exactly one turning point, a maximum, at the gain

$$G_M = \frac{-2\chi\lambda' + \sqrt{4\chi^2\lambda'^2 + 2(1+\chi)\lambda'(\lambda - 2\chi\lambda')}}{1+\chi}, \qquad \Pi_M = \frac{G_M^2 + 2\lambda' G_M}{2\Sigma'}$$

If $F' \leq 2F$, σ^2 is strictly decreasing. $L_- = F$, approached from above if and only if $F' > 2F$, and $L_+ = 0$. The classes are lower ray ($0 < R^2 \leq F$), window ($F < R^2 < M$) and Empty

($R^2 \geq M$) when $F' > 2F$. Otherwise they are lower ray ($R^2 < F$) and Empty ($R^2 \geq F$). All never occurs.

Proof. $\Pi(G) = (G^2 + 2\lambda' G)/(2\Sigma')$, and $\varepsilon = (G + 2\lambda')/(2(G + \lambda'))$. Lemma 3.7 gives $N = G \cdot p(G)/(2(G + \lambda'))$, with $p(G) = -(1 + \chi)G^2 - 4\chi\lambda' G + 2\lambda'(\lambda - 2\chi\lambda')$. The function p is strictly decreasing on $G > 0$, and $p(0) > 0$ if and only if $F' > 2F$. So there is at most one sign change, from + to −. As $\Pi \to \infty$, $G^2/(2\Pi) \to \Sigma'$ while $G \to \infty$, so $\sigma^2 \to 0$. The direction at $\Pi \to 0$ follows from Lemma 5.6, since $G/\Pi \to F'$. □

The onset matches the Bayesian-weight onset with κ replaced by F'/F, but the peak is higher. At over-weighting ratio 100 ($\kappa = F'/F = 100$, with $\lambda = \lambda' = c = 1$, $\Sigma = 0.01$), the peak is $M/F = 17.575$ for the misspecified Kalman gain, against 9.2875 for the Bayesian weight. The ratio of the two peaks is 1.892.

Figure B.5 (Appendix B): peak variance M/F against the initial over-weighting for the two gains.

Proposition 5.10 (capped gain). Let $G = \min(g\Pi, c)$. On the linear branch $d\sigma^2/d\Pi = g(g\lambda/2 - \Sigma)/(\lambda + g\Pi)^2$, whose sign is that of $g - 2F$. On the capped branch σ^2 is strictly decreasing. So:

- $g > 2F$: one maximum, at the kink $\Pi = c/g$;
- $g < 2F$: σ^2 is strictly decreasing;
- $g = 2F$: a plateau $\sigma^2 \equiv F$ on $(0, c/g]$, then strictly decreasing. By Theorem 2.4 the classes are All ($R^2 \leq H$), lower ray ($H < R^2 < F$) and Empty ($R^2 \geq F$).

The classes that occur are All, lower ray, window and Empty.

Proposition 5.11 (precision-blind gain). For a constant gain $G = G_0$, σ^2 decreases strictly from ∞ to $H = \Sigma/(\lambda + G_0)$. The classes are All ($R^2 \leq H$) and lower ray ($R^2 > H$).

A precision-blind gain over-gains at low precision without bound ($q = G_0/(F\Pi) \to \infty$). Its curve starts at ∞ and never returns to F. It fails the regularity of Theorem 3.4 at the low-precision end, and it never windows.

5.4 The Hill gain and the mixed class

The Hill gain $G = c\Pi^n/(\Pi^n + {\Pi_0}^n)$ has the Hill form that appears in divisive-normalisation models of neural gain control [16, 17]. It is used here as an estimator gain, and the neural interpretation is not claimed. Unlike the families of §5.3, its ratio G/Π rises and then falls. So q starts at 0, peaks inside the sweep, and decays.

Theorem 5.12 (Hill gain, every real $n > 1$). Let $G = c\Pi^n/(\Pi^n + {\Pi_0}^n)$ with $n > 1$. Let $t = \Pi/\Pi_0$ and $T = t^n$, and set

$$P_n(T) = -(\lambda + c)\,T^2 + (n-1)(c + 2\lambda)\,T + (2n-1)\lambda,$$

$$\varphi_n(t) = \frac{t^{\,n-1} P_n(t^n)}{(1 + t^n)^2}$$

(a) σ^2 has either no turning point or exactly two, a minimum followed by a maximum. The pair exists if and only if $\Pi_0 < {\Pi_0}^{\mathrm{pair}}(n) = c \cdot \varphi_n(t\star)/(2n\Sigma)$, where $t\star$ is the unique maximiser of φ_n. At equality there is a single degenerate stationary point.

(b) For $n = 2$, $t\star^2 = 2\sqrt{3} - 3$ and

$$\Pi_0^{\text{pair}} = \frac{c\,\varphi_*}{4\Sigma}, \qquad \varphi_* = \frac{t^*(\lambda(16\sqrt{3} - 24) + c(14\sqrt{3} - 24))}{16 - 8\sqrt{3}}$$

Proof. With $\varepsilon = n/(1 + T)$, Lemma 3.7 gives $N(1 + T)^3 = c \cdot T \cdot P_n(T) - 2n\Sigma\Pi_0 \cdot t \cdot (1 + T)^2$. So sign $N = \text{sign}(c \cdot \varphi_n(t)/(2n) - \Sigma\Pi_0)$. As a function of T, $\varphi_n = T^a\, P_n(T)/(1 + T)^2$, with $a = (n - 1)/n \in (0, 1)$. Its derivative is $d\varphi_n/dT = T^{a-1}\, R(T)/(1 + T)^3$, with $R(T) = a \cdot P_n \cdot (1 + T) + T(1 + T)P_n{}' - 2T \cdot P_n$. Write $P_n = p_2T^2 + p_1T + p_0$, with $p_2 = -(\lambda + c) < 0$, $p_1 = (n - 1)(c + 2\lambda) > 0$ and $p_0 = (2n - 1)\lambda > 0$. The cubic R has coefficients $a \cdot p_0 > 0$, $a(p_0 + p_1) + p_1 - 2p_0$, $(a - 1)p_1 + (a + 2)p_2 < 0$ and $a \cdot p_2 < 0$. Whatever the sign of the linear coefficient, there is exactly one sign change, so by Descartes' rule R has exactly one positive root. Hence φ_n rises from $\varphi_n(0^+) = 0$ to its maximum at $t\star$, and then decreases to $-\infty$. Near $t = 0$, $N(1 + T)^3/t = c \cdot t^{n-1}P_n - 2n\Sigma\Pi_0(1 + T)^2 \to -2n\Sigma\Pi_0 < 0$. So N is negative throughout, or changes sign $-$, $+$, $-$: a minimum, then a maximum. For $n = 2$, $R(t^2) = -\tfrac{1}{2}(t^4 + 6t^2 - 3)((\lambda + c)t^2 + \lambda)$, whose positive root is $t\star^2 = 2\sqrt{3} - 3$. □

At $\lambda = c = 1$, $\Sigma = 0.01$, the pair threshold is $\Pi_0{}^{\text{pair}} = 30.243$, 31.475, 37.820, 44.180 and 62.711 for $n = 1.5$, 2, 3, 4 and 8.

Proposition 5.13 (Hill gain: classification and the mixed class). Let $H = \Sigma/(\lambda + c)$. σ^2 tends to F from below as $\Pi \to 0$, since $G/\Pi \to 0 < 2F$. It tends to H from above as $\Pi \to \infty$ (Lemma 5.7). $H < m < F$ when the pair exists, so two rays never occur. With the pair and $M > F$ (for $n = 2$, $\Pi_0 < c\lambda/(4\Sigma)$, the window condition of Theorem 3.4):

Tolerance	Class
$R^2 \le H$	All
$H < R^2 < m$	Lower ray
$m \le R^2 < F$	Lower ray + window (1, 0, 1)
$F \le R^2 < M$	Window
$R^2 \ge M$	Empty

With the pair and $M \le F$: All ($R^2 \le H$), lower ray ($H < R^2 < m$), (1, 0, 1) ($m \le R^2 < M$), lower ray ($M \le R^2 < F$), Empty ($R^2 \ge F$). Without the pair: All ($R^2 \le H$), lower ray ($H < R^2 < F$), Empty ($R^2 \ge F$).

Proof. Theorem 2.2 applied to the signature of Theorem 5.12, with the endpoint limits and directions above. □

The mixed class is narrow in practice. Across the census of Task 10, the band width $(\min(M, F) - m)/F$ has median 5×10^{-4}. Its range is 4×10^{-22} to 0.96. In 16% of parameter sets the mixed band exists with no window above F. In the worked example ($\lambda = c = 1$, $\Sigma = 0.01$, $n = 2$), the Hill gain with $25 \le \Pi_0 < 31.475$ has sup $q \le 2$, yet it produces a window component inside the mixed class.

Figure B.6 (Appendix B): for the Hill gain with n = 2, the turning-point values over Π_0, the mixed band and the window band, with the pair birth and the window onset marked.

The families of §5.3 cannot produce this class. The next proposition gives the reason. A

gain that increases while G/Π strictly decreases never has a minimum at or below the floor.

Proposition 5.14 (no mixed class when G/Π decreases). In the setting of Theorem 3.4, suppose moreover that G is increasing and $G(\Pi)/\Pi$ is strictly decreasing. This holds for any sum of Bayesian stages $c_i\Pi/(\Pi + \Pi_{0,i})$. Then σ^2 is strictly decreasing wherever $\sigma^2 \leq F$. So every interior minimum lies above F, and no failure set combines a lower ray with a window. The classes (1, 0, k), $k \geq 1$, do not occur.

Proof. $2\Pi^2(\lambda + G)^2 \cdot d\sigma^2/d\Pi = \Pi G'(2\lambda G + G^2 - 2\Sigma\Pi) - G^2(\lambda + G)$. And $\sigma^2 \leq F$ if and only if $2\Sigma\Pi \geq \lambda G$. There the bracket is at most $G(\lambda + G)$. So, since $G' \geq 0$, the right side is at most $G(\lambda + G)(\Pi G' - G)$. This is negative, because G/Π is strictly decreasing. For a Bayesian stage, $\Pi G_i{}' - G_i = -c_i\Pi^2/(\Pi + \Pi_{0,i})^2 < 0$. □

5.5 Two-stage gains

Sums of natural stages behave differently once the stages are widely separated in Π_0.

Numerical result 5.15 (two-stage gains). Sums of two widely separated natural stages produce more than one maximum. Two-channel Bayesian gains show (max, min, max) in 7 of 2,000 parameter sets, giving multi-window failure sets. Two-channel Hill gains show (min, max, min, max) in 63 of 2,000 sets, giving mixed multi-window failure sets. For the Bayesian sums the windows always lie above F and never join a lower ray, as Proposition 5.14 requires. No bound on the number of turning points of such sums is known (Chapter 9).

5.6 Summary

Gain family	Turning points	Classes along Π	Window condition	Result
Any continuous gain	any	for $R^2 < F$, every class with $\ell = 1$. For $R^2 \geq F$, every class.	none	Theorem 5.1, Corollaries 5.2–5.3
Regular gains (Theorem 3.4)	any	ℓ fixed by F, h by H. k unbounded.	sup q > 2	Corollaries 5.4–5.5
Bayesian weight	at most 1 (maximum)	All, lower ray, window, Empty	$\kappa > 2$	Proposition 5.8
Misspecified Kalman	at most 1 (maximum)	lower ray, window, Empty; never All	$F' > 2F$	Proposition 5.9
Capped	at most 1 (maximum at the kink), or a plateau	All, lower ray, window, Empty	$g > 2F$	Proposition 5.10
Precision-blind	none	All, lower ray	never	Proposition 5.11
Hill, $n > 1$	none, or a minimum then a maximum	All, lower ray, (1, 0, 1), window, Empty	$M > F$ ($n = 2$: $\Pi_0 < c\lambda/(4\Sigma)$)	Theorem 5.12, Proposition 5.13

Gain family	Turning points	Classes along Π	Window condition	Result
Increasing G with G/Π decreasing	any	no (1, 0, k)	none	Proposition 5.14
Two-stage sums	up to 4 observed	multiple windows; mixed multi-window (Hill sums)	none	Numerical result 5.15

Verification.

- Proposition 5.8 was proved in the starting study and checked there (Tasks 1, 3 and 7). Its reduction was re-checked in Task 10 (T10.3, turning-point position error 6×10^{-31}). Its classes were checked in T10.7 and in the Task 13 class audit (row 2).
- Proposition 5.9: symbolic by the factor-radicand route (T10.1), 2,000 of 2,000 parameter sets (T10.3), and Task 13 audit row 3.
- Propositions 5.10 and 5.11: T10.3 (and T10.5 for the plateau), and Task 13 audit rows 4 and 5.
- Theorem 5.12(b) was checked symbolically (T10.1(e)) and on all 2,000 random sets with n = 2 (961 with the pair, 1,039 without), with no disagreement in the number of roots. Part (a) was checked in Task 13 (A5): symbolic identities, a regression of ${\Pi_0}^{\text{pair}}(n)$ against Task 10's values for n = 1.5, 2, 3, 4 and 8, and 5,000 random sets with real n in [1.01, 10]. Task 10's census covered 10,000 sets (T10.3).
- Proposition 5.13: 4,868 sets with two turning points, 77,888 tolerances, no mismatch (T10.4). Task 13 audit rows 6 and 7.
- Proposition 5.14 was proved after a failure. Row 8 of the Task 13 class audit stated a class (1, 0, 1) for sums of Bayesian stages that cannot occur. The class set was corrected, and the proof's steps were checked in SymPy (Appendix D, Task 13).
- Numerical result 5.15 comes from an exploratory census (T10.3) and Task 13 audit rows 8 and 9.

6. Beyond White Noise

Chapters 3–5 kept the sensory noise white and varied the gain or the precision. This chapter keeps the gain rule fixed and changes the noise or the clock instead. The sweeps are the noise's correlation time τ (§6.1–6.2), the sampling interval Δt (§6.3–6.5), and the process noise Σ, which moves the floor itself (§6.6). Each sweep is classified by Theorem 2.2. Two results matter most for Chapter 7. Along matched-variance τ the optimal filter windows (Counterexample 6.11). Along Δt its error metrics move in opposite directions (Theorem 6.19).

6.1 Correlation time at matched zero-frequency power

Let the sensory noise η be an Ornstein–Uhlenbeck process with correlation time $\tau > 0$. Normalise it so that its zero-frequency power stays $1/\Pi$:

$$C_\eta(\varsigma) = \frac{e^{-|\varsigma|/\tau}}{2\Pi\tau}, \qquad S_\eta(\omega) = \frac{1/\Pi}{1+\tau^2\omega^2}$$

As $\tau \to 0$ this tends to white noise of intensity $1/\Pi$, the model of Chapter 3. The relay's gain is set by precision alone, so along τ it is a constant $G > 0$. ($G = 0$ gives $\sigma^2 \equiv F$.) Write $\lambda_G = \lambda + G$.

Lemma 6.1 (relay variance).

$$\sigma^2(\tau) = \frac{1}{\lambda_G}\left(\Sigma + \frac{G^2}{2\Pi(1+\lambda_G\tau)}\right), \qquad \sigma^2 > F \iff \frac{G}{\Pi} > 2F(1+\lambda_G\tau)$$

Proof. The process-noise part contributes Σ/λ_G. A first-order filter with rate λ_G, driven by Ornstein–Uhlenbeck input of variance v and rate $1/\tau$, has output variance $v/(\lambda_G(\lambda_G + 1/\tau))$. With $v = G^2/(2\Pi\tau)$ this is $G^2/(2\Pi\lambda_G(1 + \lambda_G\tau))$. The excess form follows by subtracting F. □

In the coordinates of Chapter 3, with $t = \lambda\tau$, this is the three-coordinate surface $\mathcal{V}_\tau(\nu, q, t)$ of §3.8. Colouring divides the injected noise $\nu q^2/4$ by $1 + t(1 + \nu q/2)$. So the excess threshold rises from $q > 2$ to $q > 2(1 + \lambda_G\tau)$.

Proposition 6.2 (relay: endpoints and monotonicity). σ^2 is strictly decreasing in τ, from $V_w = (\Sigma + G^2/(2\Pi))/(\lambda + G)$, approached from below, to $H_G = \Sigma/(\lambda + G)$, approached from above. Always $H_G < F$. $V_w > F$ if and only if $G > 2F\Pi$.

Proof. The only τ-dependence is the factor $1/(1 + \lambda_G\tau)$. The approach directions follow for a decreasing function. V_w against F is Corollary 3.3. □

Proposition 6.3 (relay: classification). The classes are All ($R^2 \le H_G$), lower ray $(0, \tau_R)$ ($H_G < R^2 < V_w$), and Empty ($R^2 \ge V_w$), with

$$\tau_R = \frac{1}{\lambda_G}\left(\frac{G^2}{2\Pi(\lambda_G R^2 - \Sigma)} - 1\right)$$

At $R^2 = F$ the class is the lower ray $(0, \tau_F)$, with $\tau_F = (G/(2F\Pi) - 1)/\lambda_G$, if $G > 2F\Pi$. Otherwise it is Empty.

Proof. Theorem 2.2 with the signature (V_w, H_G). Solve Lemma 6.1 for τ. □

In the worked example ($\lambda = c = 1$, $\Sigma = 0.01$, $\Pi = \Pi_0 = 1$, so $G = 0.5$): $V_w = 0.09$, $H_G = 0.0066667$, and $\tau_F = 16$. Also $\sigma^2(1) = 0.04$, $\sigma^2(10) = 0.011875$, and $\sigma^2(16) = 0.01$. At matched zero-frequency power, colouring the noise helps a fixed relay, and failure concentrates near the white limit.

The optimal filter along τ is built from standard tools. Measurement differencing and the steady-state Riccati equation with correlated noise are in [27, 28, 45]. The contribution here is the closed form of Proposition 6.4 and its use along τ.

Proposition 6.4 (optimal stationary error). With measurement differencing [27, 28, 45], $z = .\{y\} + y/\tau$. The optimal stationary error is the positive root of

$$(1-\lambda\tau)^2P^2 + \Big(4\Sigma\tau + \frac{2\lambda}{\Pi}\Big)P - \frac{2\Sigma}{\Pi} = 0, \qquad P(\tau) = \frac{4\Sigma}{\Pi\,(b + \sqrt{D})}$$

with $b = 4\Sigma\tau + 2\lambda/\Pi$ and $D = b^2 + 8\Sigma(1-\lambda\tau)^2/\Pi$. The rationalised form has no singularity at $\lambda\tau = 1$.

Proof. $z = hs + v$, with $h = 1/\tau - \lambda$ and $v = w_s + w_\eta$. The measurement intensity is $R_v = 2\Sigma + 1/(\Pi\tau^2)$, and the cross-intensity is 2Σ. The steady-state Riccati equation with correlated noise is $0 = -2\lambda P + 2\Sigma - (hP + 2\Sigma)^2/R_v$. Multiplying by $-\tau^2 R_v$ gives the quadratic. Its constant term is negative and its leading coefficient non-negative, so it has exactly one positive root. □

Corollary 6.5 (normalised optimal error). With $\nu = 2\Sigma\Pi/\lambda^2$ and $t = \lambda\tau$,

$$\frac{P}{F} = \frac{2}{1 + \nu t + \sqrt{(1+\nu t)^2 + \nu\,(1-t)^2}}$$

Proof. Multiply Proposition 6.4 through by Π: $\Pi b = 2\lambda(1 + \nu t)$ and $\Pi\sqrt{D} = 2\lambda\sqrt{((1+\nu t)^2 + \nu(1-t)^2)}$. Then $P/F = (\lambda/\Sigma)\cdot 4\Sigma/(\Pi(b + \sqrt{D}))$ gives the formula. □

At $t = 0$ the formula reduces to the envelope $q\star(\nu)$ of Proposition 3.5.

Theorem 6.6 (optimal error decreases in τ). P is strictly decreasing on $(0, \infty)$, with $P(0^+) = \sigma\star^2(\Pi)$ and $P(\tau) \sim 1/(\Pi\tau(1 + \sqrt{(1 + \lambda^2/(2\Sigma\Pi))})) \to 0$.

Proof. b increases in τ, and $dD/d\tau = 32\Sigma^2\tau + 16\Sigma\lambda^2\tau/\Pi > 0$, so $b + \sqrt{D}$ increases. At $\tau = 0$, $P = 2\Sigma/(\lambda + \sqrt{(\lambda^2 + 2\Sigma\Pi)}) = \sigma\star^2$. For large τ, $b \sim 4\Sigma\tau$ and $\sqrt{D} \sim 4\Sigma\tau\sqrt{(1 + \lambda^2/(2\Sigma\Pi))}$. □

Theorem 6.6 also follows from information ordering. At matched zero-frequency power the observation processes are ordered by causal garbling along τ (Proposition 7.7(b)), and Theorem 7.2 applies.

Proposition 6.7 (optimal: classification). The failure set is the lower ray $(0, \tau\star_R)$ for $0 < R^2 < \sigma\star^2$, and Empty for $R^2 \geq \sigma\star^2$, with

$$\tau_R^\star = \frac{2\lambda R^2 - 4\Sigma + \sqrt{\Delta}}{2\lambda^2 R^2}, \qquad \Delta = 16\Sigma^2 - 16\lambda\Sigma R^2 + \frac{8\lambda^2(\Sigma - \lambda R^2)}{\Pi}$$

The optimal filter never produces All, a window, an upper ray or two rays along τ.

Proof. Theorem 2.2 with Theorem 6.6. Setting $P = R^2$ in Proposition 6.4 gives a quadratic in τ with leading coefficient $\lambda^2 R^4 > 0$, and constant term $R^4 + 2\lambda R^2/\Pi - 2\Sigma/\Pi$. That constant term is negative if and only if $R^2 < \sigma\star^2$. □

Proposition 6.8 (best constant-gain relay). Let $V(\tau) = \inf$ over $G \geq 0$ of $\sigma^2(G, \tau)$. Then $P \leq V$, V is strictly decreasing, $V(0^+) = \sigma\star^2$, and $V(\infty) = 0$.

Proof. A constant-gain relay is a causal linear estimator, so $P \leq V$. $G = 0$ is never the infimiser, since $\partial\sigma^2/\partial G = -\Sigma/\lambda^2 < 0$ there. Take $\tau_1 < \tau_2$. If the infimum at τ_1 is attained at $G_1 > 0$, then $V(\tau_2) \leq \sigma^2(G_1, \tau_2) < V(\tau_1)$. If it is approached as $G \to \infty$, its value is $1/(2\Pi\tau_1) > 1/(2\Pi\tau_2) \geq V(\tau_2)$. As $\tau \to \infty$, $\sigma^2(G, \tau) \to \Sigma/(\lambda + G)$, whose infimum is 0. □

Proposition 6.9 (strict gap). $P(\tau) < V(\tau)$ for every $\tau > 0$. No constant-gain relay is optimal under coloured noise.

Proof.

1. If a relay with gain G attained P, its estimate would be the conditional mean (by uniqueness of the L^2 projection). Its error $e = s - \hat{s}$ would then be orthogonal to every past observation.
2. For the relay, $\dot{e} = -\lambda_G e - G\eta + w_s$. For $\varsigma \geq 0$, w_s after $t - \varsigma$ is independent of $y(t - \varsigma)$, so

$$c(\varsigma) = \mathbb{E}[e(t)\, y(t-\varsigma)] = e^{-\lambda_G \varsigma} c(0) - \frac{G}{2\Pi\tau} \frac{e^{-\varsigma/\tau} - e^{-\lambda_G \varsigma}}{\lambda_G - 1/\tau} \quad (\lambda_G \neq 1/\tau)$$

   with $-(G/(2\Pi\tau)) \cdot \varsigma \cdot e^{-\lambda_G \varsigma}$ in place of the last term when $\lambda_G = 1/\tau$. For $G > 0$ the $e^{-\varsigma/\tau}$ term (or the $\varsigma e^{-\lambda_G \varsigma}$ term) is non-zero, so $c \not\equiv 0$. For $G = 0$, $c(\varsigma) = Fe^{-\lambda\varsigma} \not\equiv 0$. So no relay with finite gain is optimal.
3. As $G \to \infty$, $\sigma^2 \to \text{Var}\ \eta = 1/(2\Pi\tau)$, the error of the estimator $\hat{s} = y$. That estimator is not optimal either, since $E[(s - y)y] = -\text{Var}\ \eta \neq 0$.
4. $\sigma^2(G, \tau)$ is continuous on the compactified range $[0, \infty]$, so its minimum $V(\tau)$ is attained there and exceeds $P(\tau)$. □

In the worked example ($\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$): $\sigma\star^2 = 0.0099505$. $P(1) = 0.0098039$, the case $\tau = 1$ (so $\lambda\tau = 1$), against $V(1) = 0.0099010$. $P(10) = 0.0067813$. The fixed relay with $G = 0.5$ gives 0.04 and 0.011875, and tends to $H_G = 0.0066667$ as $\tau \to \infty$, while $P \to 0$.

6.2 Correlation time at matched variance

The same physical change, narrowing the noise bandwidth, can be normalised differently. Fix the noise variance v instead of its zero-frequency power:

$$C_\eta(\varsigma) = v\, e^{-|\varsigma|/\tau}, \qquad S_\eta(\omega) = \frac{2v\tau}{1 + \tau^2\omega^2}$$

Proposition 6.10 (optimal filter at matched variance). The optimal stationary error is the positive root of

$$A(\tau)P^2 + BP - C = 0, \qquad A = \frac{(1-\lambda\tau)^2}{\tau}, \quad B = 4(\Sigma + \lambda v), \quad C = 4\Sigma v$$

that is, P = 2C/(B + √(B² + 4AC)). Consequently:

1. P is strictly decreasing in A, so it is unimodal in τ, with its maximum at τ = 1/λ and $P_{max} = \Sigma v/(\Sigma + \lambda v)$.
2. A(τ) = A(1/(λ²τ)), so P is symmetric in log τ about τ = 1/λ.
3. P ∼ 2√(Σvτ) as τ → 0, and P ∼ (2/λ)√(Σv/τ) as τ → ∞.
4. For $0 < R^2 < P_{max}$ the failure set is the window (τ_-, τ_+). For $R^2 \geq P_{max}$ it is Empty, where

$$\tau_\pm = \frac{(2\lambda + A_R) \pm \sqrt{A_R\,(4\lambda + A_R)}}{2\lambda^2}, \qquad A_R = \frac{C - BR^2}{R^4}, \qquad \tau_-\tau_+ = \frac{1}{\lambda^2}$$

Proof. The correlated-noise Riccati equation of Proposition 6.4 holds with w_η of intensity 2v/τ, so $R_v = 2\Sigma + 2v/\tau$. Multiplying by $-\tau R_v$ gives the quadratic. Its constant term is negative and B > 0, so there is one positive root, and it decreases in A ≥ 0. Items 1–3 follow from the shape of A. For item 4, set $P = R^2$, solve for A, then solve $\lambda^2\tau^2 - (2\lambda + A_R)\tau + 1 = 0$. □

Filtering under coloured measurement noise is standard, by state augmentation or measurement differencing [27, 28, 45, 46]. The effect of mis-specified time-correlated model error on Kalman smoothing has been studied [47]. That work concerns model error, not measurement noise. In the searches reported in §1.6, no comparison of the two normalisations of the same noise was found.

Counterexample 6.11 (the optimal filter can window). The statement 'the optimal filter never windows' holds along precision (Theorem 3.4) and along matched-zero-frequency τ (Theorem 6.6). It fails along matched-variance τ. By Proposition 6.10 the optimal failure set is a window for every $0 < R^2 < P_{max}$.

Proposition 6.12 (relay at matched variance). For fixed G > 0,

$$\sigma^2(\tau) = \frac{1}{\lambda_G}\left(\Sigma + \frac{G^2 v\,\tau}{1 + \lambda_G\tau}\right)$$

This is strictly increasing, from H_G to $(\Sigma + G^2v/\lambda_G)/\lambda_G$, and at least P throughout. The classes are All ($R^2 \leq H_G$), upper ray ($H_G < R^2$ below the top value), and Empty (R^2 at or above the top value).

The same change of bandwidth gives a monotone optimal filter at matched zero-frequency power, and a windowed one at matched variance. The normalisation is a modelling choice, and it changes the topology. Chapter 7 explains the difference. At matched zero-frequency power the observations are ordered by causal garbling along τ (Proposition 7.7). At matched variance they cannot be ordered that way, since an ordering would rule out the optimal window (Corollary 7.3).

Figure B.7 (Appendix B): stationary error against τ for the relay and the optimal filter, at matched zero-frequency power (left) and matched variance (right).

Verification for §6.1 and §6.2.

- Lemma 6.1 was checked by an augmented-state Lyapunov route (Task 9, T9.1(a), (b)).
- Propositions 6.2 and 6.3: T9.1, T9.4 (80,000 classifications); Task 13 audit row 13.
- Proposition 6.4 was checked against the closed-form Wiener filter to 2×10^{-17} (T9.2). The pre-registered route through a regularised algebraic Riccati equation failed its tolerance, because the regularisation converges as $\sqrt{\varepsilon}$ with a large prefactor. An extended-precision Riccati solution agrees (Appendix D, Task 9). A re-check at 100 digits passed in Task 19 (H-9.2).
- Corollary 6.5 was verified symbolically in Task 14 (Part G).
- Theorem 6.6 and Propositions 6.7 and 6.8: T9.1, T9.4, T9.5; Task 13 audit row 14. T9.5's monotonicity check failed at double precision: 285 and 13 flags, none real at higher precision (Appendix D, Task 9). A re-check at 50 digits on 400,000 samples passed in Task 19 (H-9.5).
- Proposition 6.9 was proved in Task 13 (A7): a symbolic covariance identity, and 10,000 parameter sets at 60 digits, with no failure. Task 9 had recorded strictness only as a numerical result.
- Proposition 6.10: three routes agree to 1.0×10^{-14}, and the shape is exact to 10^{-50} (T10.6); Task 13 audit row 16. The pre-registered check of its shape near $\tau = 0$ failed as specified (attribution: criterion; Appendix D, Task 10). A re-check under a corrected criterion passed in Task 19 (H-10.6).
- Counterexample 6.11 and Proposition 6.12: T10.6; Task 13 audit row 15.

6.3 Sampling: models and excess sets

Sampling replaces the continuous observation by samples every Δt. In the point-sampled model with matched floor (M-P), write $s = 1 - a^2$ with $a = e^{-\lambda \Delta t}$. The state and observations are

$$x_{k+1} = a\, x_k + w_k, \quad a = e^{-\lambda \Delta t}, \quad \operatorname{Var} w = Fs = F(1 - a^2);$$

$$z_k = x_k + v_k, \quad \operatorname{Var} v = v_s = \frac{1}{\Pi \Delta t}$$

The per-sample noise variance v_s is the variance of white noise of intensity $1/\Pi$ averaged over the interval. So every sample becomes more precise as the interval grows. Write $\varphi = F\Pi\Delta t$.

Between samples the error evolves continuously, and it is periodic in time with period Δt. The curve of the error over one period is the **periodic error curve**. Three metrics summarise it:

- the posterior error at sample times, P^+ (**posterior**);
- the prior error just before a sample, $P^- = a^2 P^+ + Fs$ (**prior**);
- the time average over the period, $\bar{P} = F - (F - P^+) \cdot s/(2\lambda\Delta t)$ (**averaged**).

The posterior and prior errors are the minimum and the supremum of the periodic error curve. The averaged error is its mean. The averaged error is the primary metric of this thesis, because it is the error a user experiences across the period. Posterior and prior errors are reported alongside it.

Four filters are compared.

- The optimal filter (steady-state Kalman).
- **D1**, the sampled relay, with $\mathrm{L} = 1 - \mathrm{e}^{-\mathrm{G}\Delta\mathrm{t}}$.
- **D2**, the discrete Bayesian weight, with $\mathrm{L} = \mathrm{c}\Pi\Delta\mathrm{t}/(\Pi\Delta\mathrm{t} + \Pi_0)$ and $0 < \mathrm{c} < 2$.
- **D3**, the discrete Kalman filter over-trusting precision by a factor ρ.

For a fixed gain L,

$$P^{+} = \frac{(1-L)^2 F s + L^2 v_s}{1-(1-L)^2 a^2}$$

The integrate-and-dump model (M-I), in which each sample is the average of the observation over its interval, is treated in Chapter 7, where it is ordered by causal garbling (Corollary 7.14, Theorem 7.15).

Proposition 6.13 (discrete excess condition). For a fixed gain $0 \le \mathrm{L} < 2$, the prior error is

$$P^{-} = \frac{a^2 L^2 v_s + F s}{1 - a^2 (1-L)^2}, \qquad P^{+} - F = \frac{P^{-} - F}{a^2}$$

and $\mathrm{P}^- > \mathrm{F}$ if and only if $\mathrm{P}^+ > \mathrm{F}$ if and only if $\mathrm{L}/(2-\mathrm{L}) > \varphi$, where $\varphi = \mathrm{F}/\mathrm{v_s} = \mathrm{F}\Pi\Delta\mathrm{t}$. The steady-state Kalman gain $\mathrm{L}\star$ satisfies $\mathrm{L}\star/(2-\mathrm{L}\star) < \varphi/2$ for every $\Delta\mathrm{t} > 0$.

Proof. With Var w = Fs and $\mathrm{s} = 1 - \mathrm{a}^2$, the numerator of $\mathrm{P}^- - \mathrm{F}$ is $\mathrm{a}^2\mathrm{L}^2\mathrm{v_s} + \mathrm{Fs} - \mathrm{F} + \mathrm{Fa}^2(1-\mathrm{L})^2 = \mathrm{a}^2\mathrm{L}(\mathrm{Lv_s} - \mathrm{F}(2-\mathrm{L}))$. The denominator is positive for $0 \le \mathrm{L} < 2$. So $\mathrm{P}^- > \mathrm{F}$ if and only if $\mathrm{Lv_s} > \mathrm{F}(2-\mathrm{L})$, that is, $\mathrm{L}/(2-\mathrm{L}) > \mathrm{F}/\mathrm{v_s} = \varphi$. $\mathrm{P}^- = \mathrm{a}^2\mathrm{P}^+ + \mathrm{Fs}$ gives $\mathrm{P}^+ - \mathrm{F} = (\mathrm{P}^- - \mathrm{F})/\mathrm{a}^2$. For the optimal filter, $\mathrm{L}\star = \mathrm{P}^-/(\mathrm{P}^- + \mathrm{v_s})$ with $\mathrm{P}^- < \mathrm{F}$, so $\mathrm{L}\star/(2 - \mathrm{L}\star) = \mathrm{P}^-/(\mathrm{P}^- + 2\mathrm{v_s})$. This is below $\mathrm{F}/(2\mathrm{v_s})$ if and only if $\mathrm{P}^-(2\mathrm{v_s} - \mathrm{F}) < 2\mathrm{v_s}\mathrm{F}$, which holds because $\mathrm{P}^- < \mathrm{F}$. The optimal prior is the positive root of $\mathrm{f}(\mathrm{M}) = \mathrm{M}^2 + \mathrm{s}(\mathrm{v_s} - \mathrm{F})\mathrm{M} - \mathrm{sFv_s}$, and $\mathrm{f}(\mathrm{F}) = \mathrm{F}^2\mathrm{a}^2 > 0$. □

The ratio $\mathrm{L}/(2-\mathrm{L})$ is the discrete-time variance factor of an exponentially weighted average. For white noise of variance v, the average $\mathrm{x_k} = (1-\alpha)\mathrm{x_{k-1}} + \alpha\mathrm{y_k}$ has stationary variance $\alpha^2\mathrm{v}/(1-(1-\alpha)^2) = \alpha\mathrm{v}/(2-\alpha)$. This factor is standard, and it appears in Saho and Masugi [60] for α–β–γ filters with acceleration measurements. Its use for exponentially weighted averages goes back to Roberts [26]. The excess inequality in Proposition 6.13 is derived here. As in continuous time, the optimal gain sits at most halfway to the excess threshold.

Lemma 6.14 (shared excess set). For any filter on M-P, $\mathrm{P}^- - \mathrm{F} = \mathrm{a}^2(\mathrm{P}^+ - \mathrm{F})$ and $\bar{\mathrm{P}} - \mathrm{F} = (\mathrm{P}^+ - \mathrm{F}) \cdot \mathrm{s}/(2\lambda\Delta\mathrm{t})$. Hence all three metrics exceed F on the same set, $\{\mathrm{L}/(2-\mathrm{L}) > \varphi\}$.

Proof. The first identity is Proposition 6.13. The second is the definition of $\bar{\mathrm{P}}$. □

Proposition 6.15 (D1: the tanh lemma). $\mathrm{L}/(2-\mathrm{L}) = \tanh(\mathrm{G}\Delta\mathrm{t}/2)$. So the excess set is $\{\mathrm{G} \cdot \mathrm{h}(\mathrm{G}\Delta\mathrm{t}) > \mathrm{F}\Pi\}$ with $\mathrm{h}(\mathrm{x}) = \tanh(\mathrm{x}/2)/\mathrm{x}$ strictly decreasing from 1/2. The excess set is the lower ray $(0, \Delta\mathrm{t_E})$ if and only if $\mathrm{G} > 2\mathrm{F}\Pi$, which is the continuous-time condition. Otherwise it is empty. Here $\Delta\mathrm{t_E}$ solves $\tanh(\mathrm{G}\Delta\mathrm{t}/2) = \mathrm{F}\Pi\Delta\mathrm{t}$.

Proof. $\mathrm{h}'(\mathrm{x}) = (\mathrm{x} - \sinh\mathrm{x})/(2\mathrm{x}^2\cosh^2(\mathrm{x}/2)) < 0$, since $\sinh\mathrm{x} > \mathrm{x}$, and $\mathrm{h} \to 1/2$ as $\mathrm{x} \to 0$. □

Sampling never creates excess that continuous time lacks, and longer intervals remove it.

Proposition 6.16 (D2). The excess set is the lower ray $\Delta t < \Delta t_E = (c - 2F\Pi_0)/((2 - c)F\Pi)$ if and only if $\kappa = c/(F\Pi_0) > 2$. Otherwise it is empty.

The model requires $0 < c < 2$, so that $0 < L < 2$. For $c > 2$ the gain tends to c, $L/(2 - L) > \varphi$ no longer characterises excess, and the excess set can be all of $(0, \infty)$. For example, with $\lambda = 1$, $\Sigma = 0.01$, $\Pi = \Pi_0 = 1$ and $c = 3$, $P^+ > F$ at every Δt, and $P^+ \to 4F$.

Proposition 6.17 (D3). The excess set is $\{\varphi < 1 \text{ and } s(1 - \varphi)(\rho(1 + \varphi) - 2) > 4\varphi\}$. Its small-$\Delta t$ behaviour gives the onset. With $\nu = 2\Sigma\Pi/\lambda^2$, there is excess near $\Delta t = 0$ if and only if $\rho > 2 + \nu$. This is the continuous-time excess condition for $G = -\lambda + \sqrt{(\lambda^2 + 2\rho\Sigma\Pi)}$, since $\sqrt{(1 + \rho\nu)} > 1 + \nu$ if and only if $\rho > 2 + \nu$.

Proof. D3's gain is $L = m/(1 + m)$, with m the positive root of $Q(m) = m^2 + s(1 - \rho\varphi)m - s\rho\varphi$. By Lemma 6.14, excess holds if and only if $L/(2 - L) > \varphi$, that is, if and only if $m(1 - \varphi) > 2\varphi$. If $\varphi \geq 1$ this fails. If $\varphi < 1$ it says $m > T = 2\varphi/(1 - \varphi)$, that is, $Q(T) < 0$, since Q has one positive root and is increasing there. Also $Q(T)(1 - \varphi)^2/\varphi = 4\varphi - s(1 - \varphi)(\rho(1 + \varphi) - 2)$. Near $\Delta t = 0$, with $x = \lambda\Delta t$ and $\varphi = \nu x/2$, the difference $s(1 - \varphi)(\rho(1 + \varphi) - 2) - 4\varphi$ is $2x(\rho - 2 - \nu) + O(x^2)$. At $\rho = 2 + \nu$ the first two orders cancel, and the difference equals $-\nu(3\nu^2 + 6\nu + 4)x^3/6 + O(x^4) < 0$. So there is no excess near 0 in that case either. □

The onset $\rho > 2$ is the condition for excess at some precision (Proposition 6.25). At fixed precision it is not enough. In 178 of 1,000 census sets, $2 < \rho \leq 2 + \nu$, and there is no excess near $\Delta t = 0$.

Proposition 6.18 (endpoint limits). With $G_\rho = -\lambda + \sqrt{(\lambda^2 + 2\rho\Sigma\Pi)}$:

Filter	$\Delta t \to 0$ (all metrics)	$\Delta t \to \infty$
Optimal	$\sigma\star^2$	posterior $\to 0$; prior and averaged $\to$ F from below
D1	$\sigma^2(G, \Pi)$	posterior $\to 0$; prior $\to$ F from below (gap $\propto e^{-2\lambda\Delta t}$); averaged $\to$ F from below, like $-F/(2\lambda\Delta t)$
D2	$\sigma^2(c\Pi/\Pi_0, \Pi)$	posterior $\to (1 - c)^2F$, with $\Delta t \cdot (P^+ - (1 - c)^2F) \to c[c + 2(1 - c)F\Pi_0]/\Pi$; prior and averaged $\to$ F from below
D3	$\sigma^2(G_\rho, \Pi)$	posterior $\to 0$; prior and averaged $\to$ F from below

Verification for §6.3 and Proposition 6.18. Symbolic checks of the limits (Task 12, A5). The pre-registered route for the D2 coefficient failed, because SymPy's limit routine could not evaluate it. The series route confirms it. The failure is attributed to code (Appendix D, Task 12). A re-check by the series route passed in Task 19 (H-12.A5).

6.4 The optimal filter along the sampling interval

Theorem 6.19 (optimal M-P error along Δt).

(a) The posterior error P^+ is strictly decreasing, from $\sigma\star^2$ to 0.

(b) The prior error P^- is strictly increasing, from $\sigma\star^2$ to F.

(c) The averaged error $\bar{P}$ is strictly increasing, from $\sigma\star^2$ to F.

Proof. Write $x = \lambda\Delta t$, $s = 1 - e^{-2x}$, $s' = 2e^{-2x}$, and $v_s = \lambda/(\Pi x)$.

(a) $P^+ = F/(1+\mu)$, where $\mu > 0$ solves $\mu^2 + (1-\varphi)\mu - \varphi/s = 0$. The root μ is strictly increasing in φ and in $u := \varphi/s$, and both increase in x, since $e^y > 1 + y$.

(b) $M = P^-$ is the positive root of $f(M) = M^2 + s(v_s - F)M - sFv_s$. Its derivative $f'(M) = 2M + s(v_s - F)$ is positive there, because the product of the roots, $-sFv_s$, is negative. Implicit differentiation, with $v_s{}' = -v_s/x$, gives $\operatorname{sign}(dM/dx) = \operatorname{sign} B$, where $B = FMs' - (F-M)v_s(s/x - s')$. On $f = 0$, $sv_s(F-M) = M(M - sF)$, so $B = M[Fs' - (M - sF)(s/x - s')/s]$. Now $f(F) = F^2(1-s) > 0$ and $f(sF) = -sFv_s(1-s) < 0$. These give $sF < M < F$, hence $0 < M - sF < F(1-s)$. Also $s/x > s'$, since $s - xs' = e^{-2x}(e^{2x} - 1 - 2x) > 0$. So $B > (MF/s)[ss' - (1-s)(s/x - s')] = (MF/s)(e^{-2x}/x)(2x - 1 + e^{-2x}) > 0$. The last factor is positive, because $y \mapsto e^{-y} - 1 + y$ vanishes at 0 and has derivative $1 - e^{-y} > 0$.

(c) By Lemma 6.14 and (a), $\bar{P} = F - (F/2)Q$ with $Q = \mu s/((1+\mu)x)$. So it suffices that Q decreases. With $\varphi = \varpi x$, where $\varpi := F\Pi/\lambda = \nu/2$ is fixed along Δt, and $E = 1/x - s'/s > 0$, implicit differentiation of the μ equation and $\varphi/s = \mu(\mu + 1 - \varphi)$ give $(2\mu + 1 - \varphi)\mu' = \varphi\mu/x + \mu(\mu + 1 - \varphi)E$, where $2\mu + 1 - \varphi > 0$ and $\mu + 1 - \varphi > 0$. Then $d \ln Q/dx = \mu'/(\mu(1+\mu)) - E$. Also $(1+\mu)(2\mu + 1 - \varphi)E - [\varpi + (\mu + 1 - \varphi)E] = E\mu(2 + 2\mu - \varphi) - \varpi$, with $\mu(2 + 2\mu - \varphi) = \varphi(2/s + \mu)$. Hence $d \ln Q/dx < 0$ if and only if $1 < xE(2/s + \mu) = (1 - xs'/s)(2/s + \mu)$. This holds, since $\mu > 0$ and

$$\Big(1 - \frac{xs'}{s}\Big)\frac{2}{s} = 1 + \frac{2e^{-2x}\,(\sinh 2x - 2x)}{s^2} \geq 1$$

The limits are those of Proposition 6.18. □

So the optimal filter's classes along Δt depend on the metric. For the posterior error they are: lower ray ($R^2 < \sigma\star^2$) or Empty ($R^2 \geq \sigma\star^2$). For the prior and averaged errors they are: All ($R^2 \leq \sigma\star^2$, since $\sigma\star^2$ is approached from above), upper ray ($\sigma\star^2 < R^2 < F$), and Empty ($R^2 \geq F$).

Each sample carries the signal with noise variance $v_s = 1/(\Pi\Delta t)$, so longer intervals make every sample more precise. But the samples are sparser, so the error grows more between them. The posterior error sees only the first effect. The prior and averaged errors see both, and the second dominates. The ingredients are standard sampled-data filtering [40]. A trade-off in the sampling rate, with an optimum once delays are priced in, is known [48]. Theorem 6.19 adds the direction of each metric under this matched-precision scaling. The statement depends on the scaling $v_s = 1/(\Pi\Delta t)$. A fixed per-sample variance is a different question, and it is not treated here. Because the posterior error falls while the prior and averaged errors rise, the M-P family is not ordered by causal garbling in either direction (Proposition 7.16).

Figure B.8 (Appendix B): optimal error against Δt, with the posterior error falling towards 0 and the prior and averaged errors rising to F.

6.5 Sampled relays

Proposition 6.20 (two rays from an over-weighted sampled relay). Let D1 have $G > 2F\Pi$, and assume H2 for these curves (finitely many turning points; Conjecture 6.22 would

give it). Then the prior and averaged errors start at $\sigma^2(G, \Pi) > F$ and end at F, approached from below. So each has an interior minimum below F. For R^2 between the lowest minimum and F, the failure set contains a lower and an upper ray: two rays, or (1, 1, k) if there are several minima.

Proof. Continuity and Proposition 6.18. □

The dips differ greatly in depth. For the averaged error, $\bar{P} - F \sim -F/(2\lambda\Delta t)$, a modest dip. For the prior error, the gap is multiplied by $e^{-2\lambda\Delta t}$. By Lemma 6.14, $P^- - F = a^2(P^+ - F)$ with $a^2 = e^{-2\lambda\Delta t}$, so the prior dip's relative depth $(F - P^-)/F$ shrinks roughly like $e^{-2\lambda\Delta t}$ at the minimum. Across the 907 D1 census sets with $G > 2F\Pi$, it ranges from about 0.95 when the minimum lies at $\lambda\Delta t < 1$, through about 10^{-17} near $\lambda\Delta t = 17$, to below 10^{-300} for $\lambda\Delta t$ above about 380. The smallest is $10^{-32,880,317}$, at $\lambda\Delta t \approx 3.8 \times 10^7$. These depths were computed in mpmath at 80 significant digits (more for $\lambda\Delta t < 1$), evaluating the factor $e^{-2\lambda\Delta t}$ and the posterior gap separately, since a direct difference $P^- - F$ would need millions of digits (Task 12, re-checked in Task 19). When the minimum lies at small $\lambda\Delta t$, the prior's two rays are substantial. When it lies far out, they exist mathematically but carry no practical weight: in the worked example the minimum is at $\lambda\Delta t \approx 100$ and the dip is about 10^{-90}. The averaged error's two rays are real but narrow.

In the worked example (D1, $G = 0.5$, $\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$), the posterior error rises from 0.09 to a single maximum 0.20457 at $\Delta t = 2.4645$, and then falls to 0. So it has a window for $0.09 < R^2 < 0.20457$. The averaged error crosses F at $\Delta t_E = 100$ (to 20 digits), dips to 0.0099875 at $\Delta t = 200$ (a dip of 0.125%), and returns to F from below.

Figure B.9 (Appendix B): the sampled relay D1 along Δt. Posterior error P^+ (left) and averaged error $\bar{P}$ (right).

Numerical result 6.21 (census of sampled relays). On 2,000 parameter sets per family, every class agreed between the two classification routes:

Family	Metric	Turning-point shapes	Classes found
D1	posterior	none, max	Empty, lower ray, window
D1	prior	none, min, max–min	All, Empty, lower ray, upper ray, two rays, (0, 1, 1)
D1	averaged	none, min, max–min	All, Empty, lower ray, upper ray, two rays, (0, 1, 1)
D2	posterior	none, max, min, min–max	All, Empty, lower ray, upper ray, two rays, window, (1, 0, 1)
D2	prior, averaged	none, min	All, Empty, lower ray, upper ray, two rays
D3	posterior	none, max	Empty, lower ray, window
D3	prior	none, min, max–min	All, Empty, lower ray, upper ray, two rays, (0, 1, 1)
D3	averaged	none, min	All, Empty, lower ray, upper ray, two rays

The D3 excess set at $R^2 = F$ (posterior) was a lower ray or Empty in 1,000 of 1,000 sets. Two rays in D1 averaged error also occur with $G \leq 2F\Pi$. For example, $G = 0.019$, $\lambda = 1$,

$\Sigma = 0.01$, $\Pi = 1$ gives a maximum, then a minimum below the starting value. So $G > 2F\Pi$ is sufficient for two rays (Proposition 6.20), but not necessary.

Natural gains along Δt give two mixed classes. The upper ray plus window (0, 1, 1) comes from a max–min shape: a hump before the slow return to F. The lower ray plus window (1, 0, 1) comes from the min–max shape of D2 posterior.

Conjecture 6.22 (at most two turning points). Along Δt, D1–D3 have at most two turning points in every metric.

No case with more than two was found in the census or in the Task 13 audit, and no proof is known (Chapter 9).

Verification for §6.5.

- Proposition 6.20 rests on Proposition 6.18. Task 12's pre-registered check of a minimum below F failed, because its zero tolerance was too coarse for the prior's dip (attribution: criterion; Appendix D, Task 12). A re-check with the tolerance scaled to the dip passed on 907 of 907 sets in Task 19 (H-12.A7).
- Numerical result 6.21: Task 12 (A8, Part C), 6,000 sets × 3 metrics; Task 13 audit rows 24–32. A qualitative regression against quick-check numbers in the Task 12 prompt failed. The error was in those predicted numbers (attribution: prediction; Appendix D, Task 12). Against independently recomputed values, the regression passed in Task 19 (H-12.A8).
- Lemma 6.14 and Propositions 6.15–6.17 were checked symbolically, with the census counts given in the text (Task 12, A1–A4; Task 13 for the Proposition 6.17 equality case).
- Theorem 6.19: (a) symbolic in Task 12 (A6). (b) and (c) in Task 13 (A1, A2): every step symbolic, and 10,000 parameter sets at 100 digits with no failure in any link. Two identities passed only by a post-hoc rewrite route, recorded as failures of the pre-registered route (Appendix D, Task 13). Both passed a corrected re-check in Task 19 (H-13.A-sym). Part (c) had been a numerical result in Task 12 (400,000 points).

6.6 Process noise: a moving floor

Fix λ and Π, and sweep Σ. The floor $F = \Sigma/\lambda$ now moves with the parameter, so the tolerance convention matters. Two are used. **Absolute** tolerance fixes R^2. **Relative** tolerance fixes the ratio R^2/F. In the coordinates of Chapter 3, $\nu = 2\Sigma\Pi/\lambda^2$ is the normalised precision of §3.2, and $g = G/\lambda = \nu q/2$.

Lemma 6.23 (Σ scaling). $\sigma^2/F = (1 + g^2/\nu)/(1 + g)$. For a gain that depends on (Σ, Π) only through ν, the relative-tolerance Σ sweep and the precision sweep classify identically in ν.

Proof. Divide σ^2 by $F = \Sigma/\lambda$, as in Theorem 3.2. Under a relative tolerance the ratio R^2/F is fixed, so the failure set is $\{\nu : \mathcal{V} > R^2/F\}$ along the same path in either sweep. □

Proposition 6.24 (Σ-independent gain). Absolute: σ^2 is affine and increasing in Σ, from $G^2/(2\Pi(\lambda + G))$ (approached from above) to ∞. The classes are All and upper ray. Relative: σ^2/F decreases from ∞ to $1/(1 + g)$. The classes are All and lower ray. The excess set ($R^2/F = 1$) is $\Sigma < \lambda G/(2\Pi)$.

Proposition 6.25 (Σ-tracking Kalman gain, over-trust ρ). Let $G = -\lambda + \sqrt{(\lambda^2 + 2\rho\Sigma\Pi)}$ and $w = \sqrt{(1 + \rho\nu)}$. Absolute: σ^2 is strictly increasing from 0, and the class is upper ray. Relative:

$$\frac{\sigma^2}{F} = \frac{(1+\rho)\,w + (1-\rho)}{w\,(w+1)}$$

This starts at 1 and tends to 0, with a single maximum at $w_M = [(\rho - 1) + \sqrt{(2\rho(\rho - 1))}]/(1 + \rho)$ if and only if $\rho > 2$. It is strictly decreasing otherwise. The classes are:

- $\rho > 2$: lower ray ($R^2/F \leq 1$, since the start at 1 is approached from above), window ($1 < R^2/F$ below the peak), and Empty (at or above the peak);
- $\rho \leq 2$: lower ray ($R^2/F < 1$), and Empty ($R^2/F \geq 1$).

Proof. Absolute: $\sigma^2\Pi/\lambda = [(w^2 - 1)/\rho + (w - 1)^2]/(2w)$, whose derivative in w is $[(w^2 - 1)(1 + 1/\rho) + 2/\rho]/(2w^2) > 0$. Relative: the numerator of the derivative, $-(1 + \rho)w^2 + 2(\rho - 1)w + (\rho - 1)$, is a downward parabola with value $2(\rho - 2)$ at $w = 1$ and vertex at $(\rho - 1)/(1 + \rho) < 1$. The classes follow from Theorem 2.2. □

Proposition 6.26 (Σ-tracking Kalman gain, misspecified λ′). Let $G = -\lambda' + \sqrt{(\lambda'^2 + 2\Sigma\Pi)}$. Absolute: σ^2 is strictly increasing from 0, and the class is upper ray. Relative: σ^2/F starts at 1 and tends to 0. It has a single maximum if and only if $\lambda' < \lambda/2$, and is strictly decreasing otherwise. The classes are lower ray ($R^2/F \leq 1$), window ($1 < R^2/F$ below the peak), and Empty when $\lambda' < \lambda/2$. Otherwise they are lower ray ($R^2/F < 1$) and Empty ($R^2/F \geq 1$).

Proof. With $\Sigma = (G^2 + 2\lambda' G)/(2\Pi)$, absolute $\sigma^2 = G(G + \lambda')/(\Pi(\lambda + G))$, with derivative numerator $G^2 + 2\lambda G + \lambda\lambda' > 0$. Relative $\sigma^2/F = 2\lambda(G + \lambda')/((G + 2\lambda')(G + \lambda))$, with derivative numerator $2\lambda[-G^2 - 2\lambda' G + \lambda'(\lambda - 2\lambda')]$. That bracket is a downward parabola, with vertex at $G = -\lambda' < 0$ and value $\lambda'(\lambda - 2\lambda')$ at $G = 0$. □

Proposition 6.27 (optimal filter along Σ). Absolute: $\sigma\star^2$ is increasing from 0, and the class is upper ray. Relative: $\sigma\star^2/F = 2/(1 + \sqrt{(1 + \nu)})$ is decreasing, and the classes are lower ray and Empty.

Proposition 6.28 (signal sweeps). $\sigma\star^2$ is increasing in Σ, decreasing in λ at fixed Σ, and increasing in λ at fixed F, since $\sigma\star^2 = 2F/(1 + \sqrt{(1 + 2F\Pi/\lambda)})$. So the optimal filter never windows along these sweeps either, for an elementary reason rather than an information-ordering one.

On a moving floor the tolerance convention decides the topology. With absolute tolerances every family is monotone. With relative tolerances windows return under the same factor-of-two over-confidence condition as in Chapters 3 and 5: $\rho > 2$, or $\lambda' < \lambda/2$ (that is, $F'/F > 2$). Under either convention the optimal filter never windows along Σ.

Figure B.10 (Appendix B): the process-noise sweep for the Σ-independent gain, the over-trusting Σ-tracking gain (ρ = 10), and the optimal filter. Absolute error (left) and relative error (right).

Verification for §6.6.

- Lemma 6.23 and Propositions 6.24–6.27 were checked symbolically in Task 12 (B1–B5).
- Propositions 6.25 and 6.26 were also checked on 2,000 parameter sets each. For 6.25

the maximum's position was checked to 6×10^{-31}.

- Propositions 6.24–6.27 were covered by Task 13 audit rows 35–42.
- Proposition 6.28 is a set of one-line derivatives (T11.1).

6.7 Summary

Sweep	Optimal filter	Relay or natural gains	Result
τ, matched zero-frequency power	decreasing: lower ray, Empty	fixed relay decreasing: All, lower ray, Empty. No relay is optimal.	Lemma 6.1–Proposition 6.9
τ, matched variance	unimodal: window, Empty	fixed relay increasing: All, upper ray, Empty	Proposition 6.10, Counterexample 6.11, Proposition 6.12
Δt, posterior	decreasing: lower ray, Empty	D1, D3: window. D2: also (1, 0, 1).	Theorem 6.19, Numerical result 6.21
Δt, prior and averaged	increasing: All, upper ray, Empty	two rays; (0, 1, 1)	Theorem 6.19, Proposition 6.20, Numerical result 6.21
Σ, absolute tolerance	increasing: upper ray	every family monotone	Propositions 6.24–6.27
Σ, relative tolerance	decreasing: lower ray, Empty	window if and only if $\rho > 2$, or $\lambda' < \lambda/2$	Lemma 6.23, Propositions 6.25–6.26

The optimal filter is monotone along every sweep in this chapter except matched-variance τ, where it windows. Along Δt each metric is monotone, but the posterior error moves opposite to the prior and averaged errors. Chapter 7 connects both facts to information ordering. An ordering by causal garbling forbids optimal windows (Corollary 7.3), and makes every metric move in the same direction (Proposition 7.16). Neither family is ordered in that way.

7. Optimality and Information

Chapters 3–6 found the optimal filter's error monotone along most sweeps, with two exceptions. Along matched-variance τ it windows (Counterexample 6.11). Along the sampling interval its metrics move in opposite directions (Theorem 6.19). The normal form of Chapter 3 cannot explain either, because both lie outside its scope. This chapter supplies the second pillar of the thesis. When one observation process can be built causally from another, using only the other's past and present together with independent randomness, the optimal filter on the richer process can never do worse. Along a sweep ordered this way the optimal filter cannot window. The chapter states that ordering (§7.1). It gives an exact spectral test for additive Gaussian noise (§7.2). It then tests the ordering's reach. The ordering is sharp within a natural family of coloured noises (§7.3). It must be causal (§7.4). It accounts for the sampling results of Chapter 6 (§7.5).

7.1 Causal garbling and monotonicity

Setting: observation sweeps. The signal s is the same stationary Ornstein–Uhlenbeck process for every θ. Only the observation process Y_θ, jointly stationary with s, changes. The optimal filtering error is

$$P(\theta) = \mathbb{E}\Big[\big(s(t) - \mathbb{E}[\, s(t) \mid Y_\theta(u),\ u \le t\,]\big)^2\Big]$$

Definition 7.1 (causal garbling). Write $\theta_1 \preceq_{\mathrm{c}} \theta_2$ (θ_2 is causally at least as informative as θ_1) if there are a random element U, independent of $(\mathrm{s}, \mathrm{Y}_{\theta 2})$, and a process Y' such that, for every t, the path $\mathrm{Y}'(\mathrm{u} \le \mathrm{t})$ is measurable with respect to $\mathcal{G}_\mathrm{t} = \sigma(\mathrm{Y}_{\theta 2}(\mathrm{u} \le \mathrm{t})) \vee \sigma(\mathrm{U})$, and $(\mathrm{s}, \mathrm{Y}')$ has the same joint law as $(\mathrm{s}, \mathrm{Y}_{\theta 1})$.

In words: the worse observations can be manufactured from the better ones as they arrive, using only the past and present of the better process together with independent randomness.

Theorem 7.2 (monotonicity under causal garbling). If $\theta_1 \preceq_{\mathrm{c}} \theta_2$, then $\mathrm{P}(\theta_2) \le \mathrm{P}(\theta_1)$.

Proof. Equality of joint laws gives $\mathrm{P}(\theta_1) = \mathrm{E}[(\mathrm{s}(\mathrm{t}) - \mathrm{E}[\mathrm{s}(\mathrm{t}) \mid \mathrm{Y}'(\le \mathrm{t})])^2]$. Since $\sigma(\mathrm{Y}'(\le \mathrm{t})) \subseteq \mathcal{G}_\mathrm{t}$, conditioning on the larger σ-field cannot increase the mean-square error. Because U is independent of $(\mathrm{s}, \mathrm{Y}_{\theta 2})$, $\mathrm{E}[\mathrm{s}(\mathrm{t}) \mid \mathcal{G}_\mathrm{t}] = \mathrm{E}[\mathrm{s}(\mathrm{t}) \mid \mathrm{Y}_{\theta 2}(\le \mathrm{t})]$. □

The argument is pointwise in t, so it holds for cyclostationary observations too. It also holds with any square-integrable target measurable with respect to s in place of s(t), which Theorem 7.6 uses.

Theorem 7.2 is a filtering instance of a known principle. Monotonicity of optimal performance under garbling goes back to Blackwell's comparison of experiments [18, 19], and to Le Cam's theory [49, 50]. It extends to sequential experiments [20]. In stochastic control, where observation channels are themselves design variables, it appears in [21]. The comparison of information structures in control and teams is studied in [22, 23]. What is specific here is the causal form of the order on observation processes, and its consequence for the topology of filtering failure sets. The general principle has close precedents, so the claim is restricted

to this filtering setting.

Corollary 7.3 (no optimal windows along causal orderings). Suppose $\theta_1 < \theta_2$ implies $\theta_1 \preceq_c \theta_2$ on the whole domain. Then P is non-increasing, so every optimal failure set is an initial segment of the domain. If moreover P is continuous (and satisfies H2′), the optimal failure set is Empty, All, or a lower ray at every R^2. For the reverse ordering it is Empty, All, or an upper ray. Windows, two rays and mixed classes are impossible.

Proof. A superlevel set of a non-increasing function is an initial segment. For continuous P it is open, hence Empty, the whole domain, or (θ_, x). Plateaus are allowed by Theorem 2.4. □

Strict monotonicity is not needed.

Corollary 7.4 (diagnostic). Along a causally ordered sweep, a failure set with a window, two rays, or a mixed class certifies that the filter is not optimal. With Theorem 5.1, the precision-sweep picture is complete. The optimal filter sits on the envelope $\sigma\star^2$ (classes lower ray and Empty). Every curve strictly above it belongs to some miscalibrated gain.

Proof. The first sentence is Corollary 7.3. The precision sweep is ordered (Proposition 7.7(a) below), so by Corollary 7.3 the optimal classes along it are Empty, All, or a lower ray. All is excluded, because $\sigma\star^2 = 2\Sigma/(\lambda + \sqrt{(\lambda^2 + 2\Sigma\Pi)}) \to 0$ as $\Pi \to \infty$ (Proposition 3.5), so a positive R^2 cannot lie below $\sigma\star^2$ everywhere. That leaves the lower ray and Empty. By Theorem 5.1 every continuous curve on or above $\sigma\star^2$ is the variance of some continuous gain. So every curve strictly above it belongs to a gain other than the optimal one. □

So the bounded window of §1.2, and the Hill mixed class (Proposition 5.13), are signatures of the filter, not of the environment.

7.2 A spectral test

For additive noise the ordering can be read off the spectra.

Proposition 7.5 (spectral dominance). Let $y_\theta = s + \eta_\theta$, with η_θ stationary Gaussian, independent of s, with spectral density S_θ (white components written as constant densities).

1. If $S_{\theta 1} \geq S_{\theta 2}$ almost everywhere, then $\theta_1 \preceq_c \theta_2$.
2. Suppose $S_s > 0$. If $Y' = h * y_{\theta 2} + n$, with h causal and linear time-invariant and n independent, realises θ_1, then $h = \delta$ and $S_n = S_{\theta 1} - S_{\theta 2} \geq 0$.

Proof. (1) Let ξ be stationary Gaussian with density $S_{\theta 1} - S_{\theta 2}$, independent of everything, and set $Y' = y_{\theta 2} + \xi$. (2) The cross-spectrum of s with Y′ is $H \cdot S_s$, which must equal S_s, so $H \equiv 1$. Matching the spectra of Y′ then gives S_n. □

The next theorem shows that for additive Gaussian noise the spectral test is exact. Non-linear causal garblings, and even non-causal garblings of whole paths, add nothing.

Lemma 7.6a (two-sided ordering). Suppose Y′ is measurable with respect to $\sigma(Y_{\theta 2}) \vee \sigma(U)$, with U independent of $(s, Y_{\theta 2})$, and (s, Y') has the law of $(s, Y_{\theta 1})$. Then, for every square-integrable Z measurable with respect to s,

$$\mathbb{E}\big[(Z - \mathbb{E}[Z \mid Y'])^2\big] \;\geq\; \mathbb{E}\big[(Z - \mathbb{E}[Z \mid Y_{\theta_2}])^2\big]$$

Proof. $\sigma(Y') \subseteq \sigma(Y_{\theta 2}) \vee \sigma(U)$. Conditioning on the larger σ-field cannot increase the mean-

square error. Because U is independent of (s, $Y_{\theta 2}$), E[Z | $Y_{\theta 2}$, U] = E[Z | $Y_{\theta 2}$]. The left side is therefore at least the right side. □

The same argument, applied at each t with the causal σ-fields, gives the causal version used in the proof of Theorem 7.2.

Theorem 7.6 (the orderings coincide for Gaussian noise). In the setting of Proposition 7.5, with $S_s > 0$ almost everywhere, the following are equivalent.

- (i) $\theta_1 \preceq_c \theta_2$ by some causal garbling, possibly non-linear.
- (ii) $S_{\theta 1} \geq S_{\theta 2}$ almost everywhere.
- (iii) $\theta_1 \preceq_c \theta_2$ by adding independent stationary Gaussian noise.
- (iv) $Y_{\theta 1}$ is a garbling of $Y_{\theta 2}$ as whole paths, not necessarily causal (pathwise Blackwell ordering).

Proof. (ii) ⇒ (iii) is Proposition 7.5(1). (iii) ⇒ (i) is immediate. (i) ⇒ (iv): a causal garbling is in particular a whole-path garbling, since each $Y'(\leq t)$ is measurable with respect to the whole path of $Y_{\theta 2}$ and U.

(iv) ⇒ (ii). Take $Z = \int g(u)s(u)\,du$ with g smooth and compactly supported. Lemma 7.6a, applied with the whole-path σ-field, gives the two-sided inequality for every such Z: the smoothing error of Z given $Y_{\theta 1}$ is at least that given $Y_{\theta 2}$. Because (s, Y_θ) is jointly Gaussian, the two-sided conditional mean is linear. Its error is $\int |\hat{g}(\omega)|^2\, h_\theta(\omega)\, d\omega/2\pi$, with $h_\theta = S_s\, S_\theta/(S_s + S_\theta)$. So $\int |\hat{g}|^2(h_{\theta 1} - h_{\theta 2}) \geq 0$ for every such g. Taking $g(u) = \psi(u/L)\cos(\omega_0 u)$ with $L \to \infty$ concentrates $|\hat{g}|^2/L$ at $\pm\omega_0$ as an approximate identity. The cross terms vanish for $\omega_0 \neq 0$, and the spectra are even. So $h_{\theta 1} \geq h_{\theta 2}$ at every Lebesgue point. Since $x \mapsto S_s\, x/(S_s + x)$ is strictly increasing where $S_s > 0$ (its derivative is ${S_s}^2/(S_s + x)^2$), $S_{\theta 1} \geq S_{\theta 2}$ almost everywhere.

(v) ⇒ (ii) follows in the same way. Lemma 7.6a is applied at each t, with the causal σ-fields $\mathcal{G}_t$ in place of the whole-path σ-field. The inequality holds for every t. As $t \to \infty$ the σ-fields $\sigma(Y_\theta(\leq t))$ increase to $\sigma(Y_\theta(\mathbb{R}))$, so by Lévy's upward theorem both sides converge to the two-sided errors, and the inequality survives. The rest of the argument is as above. □

Theorem 7.6 extends the characterisation of linear normal experiments [24], see also [50], from static observations to stationary Gaussian processes ordered by their spectra. A partial converse to Blackwell ordering for information structures is known in the game setting [51]. In the searches reported in §1.6, no version of the theorem for processes was found.

Proposition 7.7 (which studied sweeps are ordered).

- (a) Precision: $S_\Pi = 1/\Pi$ decreases in Π, so the sweep is ordered. This recovers the monotone optimal filter of the starting study.
- (b) Matched zero-frequency τ: $S_\tau(\omega) = (1/\Pi)/(1 + \tau^2\omega^2)$ decreases in τ at every ω, so $\tau_1 \preceq_c \tau_2$ for $\tau_2 > \tau_1$. Information increases with τ. The garbling noise is the second-order process

$$S_\xi(\omega) = \frac{(\tau_2^2 - \tau_1^2)\,\omega^2}{\Pi\,(1+\tau_1^2\omega^2)(1+\tau_2^2\omega^2)} = |H_\xi(i\omega)|^2,$$

$$H_\xi(s) = \sqrt{\frac{\tau_2^2 - \tau_1^2}{\Pi}}\;\frac{s}{(1+\tau_1 s)(1+\tau_2 s)}$$

Theorem 6.6's monotonicity follows. Its strictness comes from the closed form.

- (c) Matched variance τ: $\mathrm{S}_\tau = 2\mathrm{v}\tau/(1+\tau^2\omega^2)$. Every pair crosses: the larger τ has more power at $\omega = 0$, the smaller more power as $\omega \to \infty$. So by Theorem 7.6 no two values are ordered by any garbling, linear or not.

Proof. Proposition 7.5(1) for (a) and (b). The factorisation of S_ξ is direct. Theorem 7.6 for (c). □

7.3 The power family: the condition is sharp

A one-parameter family of coloured noises contains both normalisations of Chapter 6. Let η be Ornstein–Uhlenbeck with

$$S_{\beta,\tau}(\omega) = \frac{K\tau^\beta}{1+\tau^2\omega^2}, \qquad \operatorname{Var}\eta = v(\tau) = \tfrac{K}{2}\,\tau^{\beta-1}, \qquad \tau \in (0,\infty),\ K>0,\ \beta \in \mathbb{R}$$

$\beta = 0$ is matched zero-frequency power ($\mathrm{K} = 1/\Pi$), and $\beta = 1$ is matched variance ($\mathrm{K} = 2\mathrm{v}$). The optimal error is the root of Proposition 6.10 with $\mathrm{v} = \mathrm{v}(\tau)$. Coloured measurement noise, and its treatment by measurement differencing, is discussed in §6.2, which also gives the references. In the searches reported in §1.6, no comparison across normalisations and no exponent threshold was found.

Theorem 7.8 (sharpness in the power family).

1. $\beta \le 0$: $\mathrm{S}_{\beta,\tau}$ decreases in τ at every ω. The sweep is causally ordered, with information increasing in τ, so P is non-increasing. Optimal classes: lower ray and Empty.
2. $\beta \ge 2$: $\mathrm{S}_{\beta,\tau}$ increases in τ at every ω. The sweep is ordered with information decreasing in τ, so P is non-decreasing. Optimal classes: upper ray and Empty.
3. $0 < \beta < 2$: every pair $\tau_1 \neq \tau_2$ has crossing spectra. $\mathrm{P} \to 0$ at both ends while $\mathrm{P} > 0$. So for every $0 < \mathrm{R}^2 < \max \mathrm{P}$ the optimal failure set is bounded away from both ends: a window. By Theorem 7.9 it is exactly one window.

Hence, within this family: causally ordered if and only if P is monotone, if and only if there is no optimal-filter window, if and only if $\beta \notin (0, 2)$.

Proof. $\partial \log \mathrm{S}/\partial\tau = \beta/\tau - 2\tau\omega^2/(1+\tau^2\omega^2)$, and the second term lies in $[0, 2/\tau)$. So the derivative is ≤ 0 at every ω if $\beta \le 0$. It is > 0 at every ω if $\beta \ge 2$. It changes sign with ω if $0 < \beta < 2$. Parts 1 and 2 follow from Proposition 7.5 and Corollary 7.3. For part 3, $\mathrm{P} \le 2\mathrm{C}/\sqrt{(4\mathrm{AC})} = \sqrt{(\mathrm{C}/\mathrm{A})} = \sqrt{(2\Sigma\mathrm{K})}\cdot\tau^{\beta/2}/|1-\lambda\tau|$. This tends to 0 as $\tau \to 0$ if and only if $\beta > 0$, and as $\tau \to \infty$ if and only if $\beta < 2$. □

Where that bound does not vanish, the limits of P are as follows. For $\beta < 0$ and $\tau \to 0$, $\mathrm{P} \to \mathrm{F}$. For $\beta = 0$ and $\tau \to 0$, $\mathrm{P} \to \sigma\star^2(1/\mathrm{K})$. For $\beta = 2$ and $\tau \to \infty$, $\mathrm{P} \to \sigma\star^2(\lambda^2/\mathrm{K})$, not F. For $\beta > 2$ and $\tau \to \infty$, $\mathrm{P} \to \mathrm{F}$, approached like $\tau^{2-\beta}$.

Theorem 7.9 (unimodality for $0 < \beta < 2$). For $0 < \beta < 2$, P has exactly one turning point, a maximum. So the optimal failure set has k = 1.

Proof. Let $\Phi(\tau, P) = AP^2 + BP - C$, so $\Phi_P = 2AP + B > 0$. With $A' = (\lambda^2\tau^2 - 1)/\tau^2$, $A'' = 2/\tau^3$, $v'/v = (\beta - 1)/\tau$, and $v''/v' = (\beta - 2)/\tau$:

$$\Phi_\tau = A'P^2 - 4v'(\Sigma - \lambda P), \qquad \Phi_{\tau\tau} = A''P^2 - 4v''(\Sigma - \lambda P)$$

At a critical point $P' = 0$, so $\Phi_\tau = 0$, and $P'' = -\Phi_{\tau\tau}/\Phi_P$. For $\beta \neq 1$, substituting $\Sigma - \lambda P = A'P^2/(4v')$ gives

$$\Phi_{\tau\tau} = P^2\Big(A'' - A'\,\frac{v''}{v'}\Big) = \frac{P^2\,(\beta + (2-\beta)\lambda^2\tau^2)}{\tau^3} > 0$$

so $P'' < 0$. For $\beta = 1$, $\Phi_\tau = A'P^2$ vanishes only at $\tau = 1/\lambda$, where $A'' > 0$ gives the same sign. Every critical point is therefore a strict local maximum. Two strict maxima would enclose an interior minimum, a critical point that is not a strict maximum, so there is at most one critical point. There is at least one, because $P > 0$ and $P \to 0$ at both ends (Theorem 7.8). □

Proposition 7.10 (duality). $P_{\beta,K}(\tau) = P_{2-\beta,\,K\lambda^{2-2\beta}}(1/(\lambda^2\tau))$.

Proof. A is invariant under $\tau \mapsto 1/(\lambda^2\tau)$. Also $(K'/2)x^{1-\beta}$, at $x = 1/(\lambda^2\tau)$, equals $(K/2)\tau^{\beta-1}$ when $K' = K\lambda^{2-2\beta}$. □

In this natural family of coloured noises the information-ordering condition is exactly sharp. The two normalisations of Chapter 6 sit on opposite sides of the boundary: $\beta = 0$ is ordered, and $\beta = 1$ is in the windowed band. Duality maps the band onto itself about $\beta = 1$.

Figure B.11 (Appendix B): the optimal error relative to its supremum over τ, across β and $\log_{10}(\lambda\tau)$, with the three regimes labelled.

Verification for §7.1–7.3.

- Theorem 7.2: Task 11 (T11.1), with a Monte Carlo realisation of a garbling (T11.2). Corollaries 7.3 and 7.4: Task 11.
- Proposition 7.5: spectral checks on 25,000 pairs (T11.1, T11.3).
- Theorem 7.6: written proof, a symbolic check of the monotone map, and an independent referee pass (Task 13, A8). Lemma 7.6a, which supplies the whole-path step of the proof, postdates the referee pass, and its argument has not been reviewed independently.
- Proposition 7.7: T11.1, T11.3.
- Theorem 7.8: T11.3 over 129 values of β; Task 13 audit rows 17–19. The class check pre-registered in T11.3 failed, because its limits and brackets at $\beta \geq 2$ were wrong ($P \to \sigma_\star^2(\lambda^2/K)$ at $\beta = 2$, not F). They were corrected in the diagnosis (attribution: criterion; Appendix D, Task 11). With the corrected limits, the re-check passed in Task 19 on all 129 values of β (H-11.3).
- Theorem 7.9: symbolic identities and 10,000 curves (Task 13, A4). Task 11 had recorded it as a numerical result (12,900 of 12,900 curves).
- Proposition 7.10: T11.1, T11.3, to 8×10^{-51}.

7.4 Delay: why the ordering must be causal

Proposition 7.11 (latency). Let $y_d(t) = s(t - d) + \eta(t)$, with η white noise of intensity $1/\Pi$ and $d \geq 0$. For every d, the pair $(s(\cdot - d), y_d)$ has the same law as (s, y_0). So the experiments for different delays differ only by a time shift of the signal. The optimal filtering error is

$$P(d) = F - e^{-2\lambda d}(F - \sigma_\star^2(\Pi))$$

which is strictly increasing in d. Along the latency sweep $d(\theta) = \theta e^{-\theta}$, the delay rises for $\theta < 1$ and falls for $\theta > 1$. So the optimal filter has a window in θ for every $R^2 \in (\sigma\star^2, P(1/e))$. The causal order ($d_1 < d_2$ implies $d_2 \preceq_c d_1$) reverses at $\theta = 1$, where the window sits. The classes along θ are All ($R^2 \leq \sigma\star^2$), window ($\sigma\star^2 < R^2 < P(1/e)$) and Empty ($R^2 \geq P(1/e)$).

Proof. By the Ornstein–Uhlenbeck property, $s(t) = e^{-\lambda d}s(t - d)$ plus an independent innovation of variance $F(1 - e^{-2\lambda d})$. The delayed signal $s(t - d)$ is filtered with error $\sigma\star^2$ in the same way as s(t) at delay zero. So $P(d(\theta)) > \sigma\star^2$ for every $\theta > 0$. The limit at both ends is $\sigma\star^2$, and the maximum is at $\theta = 1$. □

In the worked example ($\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$), $P(0) = 0.0099505$ and $P(1/e) = 0.0099763$.

That delay degrades filtering is not new. The error of a delayed estimate follows from the standard prediction step of the Kalman filter [29, 40], and out-of-sequence and delayed measurements are handled in tracking [52, 53]. What the proposition adds is a statement about orders on experiments. In the searches reported in §1.6, no statement of this point was found. The delayed experiments differ only by a time shift, yet their optimal errors differ, so the order on experiments must be causal.

Counterexample 7.12 (pathwise order is not enough). In the latency family, the experiments are equivalent up to a time shift of the signal, and the optimal filtering error is not monotone in θ. So an order on whole paths, even one that allows a time shift, does not determine the monotonicity of the filtering error. Definition 7.1 must be causal.

The causal order also cannot be reversed into a characterisation of monotonicity.

Proposition 7.13 (ordering is sufficient, not necessary). Along matched-variance τ restricted to $(1/\lambda, \infty)$, P is strictly decreasing (Proposition 6.10). Yet no two values are causally ordered by any garbling (Proposition 7.7(c)). So a monotone optimal error does not require an information ordering.

Figure B.12 (Appendix B): the optimal filtering error along the latency sweep $d(\theta) = \theta e^{-\theta}$, with its window and the reversal at $\theta = 1$.

7.5 Sampling

Chapter 6 left two facts about the sampling interval unexplained. The posterior error and the averaged error of the point-sampled model move in opposite directions (Theorem 6.19). Its relays produce mixed classes. Integrate-and-dump sampling (M-I), in which each sample z_k is the average of the observation over the k-th interval, behaves differently, and the ordering explains why.

Corollary 7.14 (integrate-and-dump, nested grids). If $\Delta t_2 = m\Delta t_1$ for an integer

m and the grids are aligned, each coarse M-I sample is the average of the m fine samples completed by the time it is taken. That is a causal garbling. Hence the optimal error, pointwise in t and time-averaged, satisfies $P(\Delta t_2) \geq P(\Delta t_1)$ along every chain Δt_0, $m\Delta t_0$, $m^2\Delta t_0$, …

Proof. Theorem 7.2, pointwise in t, since the observations are cyclostationary. Then averaged over the period. □

On the continuum the ordering argument does not reach non-nested pairs, but the closed form does.

Theorem 7.15 (integrate-and-dump on the continuum). Both M-I optimal metrics, (a) the error at sample times and (b) the time average, are strictly increasing in Δt on the whole continuum, including non-nested pairs.

Proof. By scaling, $P_a = F \cdot p(x, \nu)$ with $x = \lambda\Delta t$ and $\nu = 2\Sigma\Pi/\lambda^2$. In reduced units ($\lambda = \Sigma = 1$, $\Pi = \nu/2$, $E = e^{-x}$), the integrate-and-dump model has process covariances $q_{11} = 1 - E^2$, $q_{12} = 2[(1 - E) - (1 - E^2)/2]$, and $q_{22} = 2[x - 2(1 - E) + (1 - E^2)/2]$ for the state and its interval average. With $h = (1 - E)/x$, $r_e = q_{22}/x^2 + 1/(\Pi x)$, and $g = q_{12}/x$, the posterior p is the positive root of

$$a_2 p^2 + a_1 p + a_0 = 0,$$

$$a_2 = h^2, \quad a_1 = r_e(1 - E^2) - q_{11}h^2 + 2Ehg, \quad a_0 = -(q_{11}r_e - g^2)$$

These coefficients satisfy $a_0 = -a_1$ identically, and $a_2/a_1 = \nu(1 - E)/(2K)$, with $K = \nu[x(1 + E) - 2(1 - E)] + x(1 + E)$. Hence

$$p = \frac{2}{1 + \sqrt{1 + 2\nu/k(x)}}, \qquad k(x) = \frac{K}{1 - E} = (1 + \nu)\, x \coth(x/2) - 2\nu$$

Since $d/dx[x \coth(x/2)] = (\sinh x - x)/(2 \sinh^2(x/2)) > 0$, k increases strictly from $k(0^+) = 2$ to ∞. So p increases strictly from $2/(1 + \sqrt{(1 + \nu)}) = \sigma\star^2/F$ to 1. Thus (a) is strictly increasing, from $\sigma\star^2$ (approached from above) to F (approached from below). For (b), $P_b = P_a \cdot \varphi + F(1 - \varphi)$ with $\varphi = (1 - E^2)/(2x)$. Its derivative $\varphi' = -E^2(e^{2x} - 1 - 2x)/(2x^2) < 0$. So $dP_b/dx = P_a'\varphi + (P_a - F)\varphi' > 0$. □

A second route, through the resultant of the cleared quadratic and its x-derivative, gives the same conclusion for (a). It was pre-registered in Task 13 (A3). The pre-registered script was inconclusive, and the route was completed post hoc in exact arithmetic, passing 22 of 22 checks (Appendix D, Task 13).

Proposition 7.16 (M-P is not a causal-garbling family). Along Δt, the point-sampled model is not ordered in either direction. For no pair $\Delta t_1 < \Delta t_2$ does a causal garbling exist in either direction.

Proof. The periodic error curve of §6.3 is defined for each Δt on one time axis. Posterior, prior and averaged errors are the minimum, the supremum and the time mean of that curve over its period. Theorem 7.2 applies pointwise in t to two processes on the same time axis. So an ordering in either direction would force all three metrics to move the same way between Δt_1 and Δt_2. But the posterior error falls while the prior and averaged errors rise (Theorem 6.19). □

So the two sampling models sit on opposite sides of the ordering. Integrate-and-dump discards information as the interval grows, and it is ordered. Point sampling with matched precision trades per-sample precision against sample density, and it is not ordered. Under integrate-and-dump every optimal metric worsens with Δt, so the optimal filter never windows there either. An interior optimal sampling interval then needs a cost outside the filtering error, such as the network delays of [48].

7.6 Summary: the answer to the optimality question

Along any causally ordered sweep, an optimal filter's failure set is Empty, All, or a single ray at the less-informative end. Every other class certifies a miscalibrated filter (Corollaries 7.3–7.4). For additive Gaussian noise the ordering is decided by the spectra alone (Theorem 7.6). Within the power-normalised family the condition is sharp (Theorem 7.8). In general it is sufficient but not necessary (Proposition 7.13). The garbling must be causal (Counterexample 7.12).

Sweep	Ordered?	Optimal filter	Result
Precision Π	yes, information increasing	lower ray, Empty	Proposition 7.7(a), Theorem 3.4
τ at matched zero-frequency power ($\beta = 0$)	yes, information increasing	lower ray, Empty	Proposition 7.7(b), Theorem 6.6
Power family, $\beta \leq 0$ or $\beta \geq 2$	yes	single ray, Empty	Theorem 7.8
τ at matched variance ($\beta = 1$); power family $0 < \beta < 2$	no: spectra cross	exactly one window	Proposition 7.7(c), Theorems 7.8–7.9
Latency $d(\theta) = \theta e^{-\theta}$	causal order reverses at $\theta = 1$	All, window, Empty	Proposition 7.11, Counterexample 7.12
Δt, integrate-and-dump	yes on nested grids; monotone on the continuum	All, upper ray, Empty	Corollary 7.14, Theorem 7.15
Δt, point sampling	no	posterior decreasing; prior and averaged increasing	Proposition 7.16, Theorem 6.19
Σ and λ (signal sweeps)	not an observation sweep	monotone, for an elementary reason	Proposition 6.28

Verification for §7.4–7.6.

- Theorem 7.2: Task 11 (T11.1), with a Monte Carlo realisation of a garbling (T11.2).
- Proposition 7.11 and Counterexample 7.12: T11.4, with a Monte Carlo check ($|z| \leq 1.91$). Task 13 audit row 20.
- Proposition 7.13: T11.5, 1,000 of 1,000 sets.
- Corollary 7.14: no violation in 9,000 pairs (T11.6).
- Theorem 7.15: closed form and limits (11 of 11), the resultant route (pre-registered, completed post hoc in exact arithmetic, 22 of 22), and 10,000 parameter sets at 100 digits with no violation (Task 13, A3). Also T11.6 and T12 (A9), 400,000 points each.
- Proposition 7.16: 1,000 of 1,000 pairs (Δt, $2\Delta t$) (T12, A10).

8. Verification

Every result in chapters 2–7 ends with a verification line naming the task and the check that tested it. This chapter describes how those checks were run, states the two numerical results that belong to the method rather than to the filters, and sets out what the checks found wrong. Appendix D gives the full record, task by task.

8.1 Protocol

Every verification task followed the same four rules.

1. **Pre-registration.** Each check and its pass criterion were written into a pre-registration file and committed before anything that produces a result was run.
2. **One recorded run.** The checks ran once, on one recorded commit. Any earlier runs made during development are disclosed in the task's report.
3. **Failures stay failures.** A failed check is reported as failed, with a diagnosis attributing it to the criterion, the input, the code, the document under test or a prediction. No criterion or artefact is changed after the recorded run. Where a check needed correcting, the corrected version was pre-registered again in a later task.
4. **Independent routes.** Where possible, a result is checked by two routes that share no code: a symbolic identity and a numerical evaluation, a closed form and a Riccati solution, root-finding and dense sampling. Values that depend on a cancellation are recomputed in extended precision (60 or 100 digits).

Tasks 1–7 verified the starting study, Tasks 8–13 the classification programme, and Tasks 14–18 the numbering, the bibliography, the figures and the text of this thesis. Task 19 re-ran earlier failures and checked the final text. Across Tasks 1–19 the reports' check tables hold 182 checks with a verdict, and 37 of them failed. Nearly all were pre-registered; a few rows record post-hoc completions or re-runs, such as Task 13's exact completion of the A3 route. Task 2 had three checks, and all three passed. Tasks 4 and 5 were exploratory by pre-registration: they have no pass criteria and add nothing to either count. Task 20 brought the repository into line with this text: register titles, a register entry for Lemma 7.6a, references [54]–[60], figure labels B.5 and B.6, and Appendix D's line for the Theorem 7.15 route. Its six checks are recorded separately and are not in the totals above. Five passed; the sixth failed on reference formatting, and this text follows the repository's format.

8.2 Classification by two routes

Each class is computed by two routes. Route (i) locates the turning points and reads the class from the critical values by the rule of Theorem 2.2. Route (ii) locates the crossings of R^2 directly, by sign tests at sample points and root-finding between them, with an equality tolerance at the critical values. The routes must agree, and every disagreement is diagnosed. Neither route relies on a fixed grid, and a grid alone is not enough.

Numerical result 8.1 (narrow windows need root-finding). Windows of relative

width 10^{-8} were found by the root-finding classifier in 20 of 20 curves, 55 windows in all. A logarithmic grid of 10^4 points found none of them.

Verification: Task 10 (T10.5).

The routes disagreed persistently in one place: row 12 of the Task 13 audit, described in §4.1. There route (ii) read Empty where route (i) read All, at tolerances set equal to the stored endpoint limit L_ of increasing curves. At the exact limit of an increasing curve the class is All, by Theorem 2.2(2), as route (i) gives. Route (ii) was asked to resolve a difference finer than the precision the stored limit carries. The original check failed and was attributed to code. Its re-check in Task 19 failed and was attributed to its criterion (§8.4).

8.3 Blind searches

The class tables of chapters 4–6 state that certain classes never occur. Those statements were tested by searches that did not use the tables.

Numerical result 8.2 (blind searches find no unpredicted class). Randomised searches, with tolerances concentrated near every critical tolerance, found no class outside the predicted sets: 10^6 classifications over the precision, design and τ sweeps; 4.4×10^6 over the Δt and Σ sweeps; and the audit of every row of the synthesis matrix.

Verification: Task 10 (T10.7), Task 12 (Part C), Task 13 (C2).

A blind search cannot prove that a class is impossible; the proofs do that. A search can catch a class table that disagrees with the curves, and the Task 13 audit did so once (§8.4).

8.4 What the checks found

The 37 failures fall into five kinds by attribution. Appendix D records them in 38 lines, because one Task 13 check (C2) failed in two rows, rows 8 and 12. Some lines name two kinds. Counting every kind a line names, the attributions are criterion 19, code 14, document 13, input 2 and prediction 1. Each failure is listed in Appendix D with its diagnosis.

Attribution	What went wrong	Examples
Criterion	the pre-registered pass rule was wrong or too tight	peak positions to 3 figures from a 0.4% grid (T6); limits at $\beta \geq 2$ (T11.3); a zero tolerance too coarse for the prior's dip (T12); SymPy routes with no handling of $\sqrt{(x^2)}$ (T10), or of an undecided sign and an unrewritten sinh (T13); a re-check tolerance finer than the stored limit's precision (T19); an old key left in recorded files (T19); double-precision flags (T9.5)
Code	the checking route could not do the computation	SymPy could not decide a sign or denest $\sqrt{(x^2)}$ (T16); a sample point below its equality tolerance (T13); a dry-run path resolver (T14); `sympy.limit` on the D2 coefficient (T12); precision loss at a cancellation (T13, T17); a coverage parser and a script docstring (T19)

Attribution	What went wrong	Examples
Document	the result as written was wrong	a class (1, 0, 1) stated for sums of Bayesian stages, which cannot occur (T13); a truncated worked value, in four places (T13); in an earlier version of this text, an incomplete proof, a miscounted number of test values, a mislabelled range and a sentence that claimed more than its source (T17); a citation range, the referee description and three reworded lines of Appendix D (T18); the route descriptions of §8.2, an example in §8.4 and the Hill condition of §9.2 (T19)
Input	the material given to the check was incomplete	two bibliography entries attached to no result, and DOIs unreachable from the session (T14); a source text unreachable from the session (T19)
Prediction	a number supplied in advance was wrong	the quick-check values in the Task 12 prompt (T12)

Every document failure was corrected, and none led to a withdrawn result. The most consequential was the class stated for sums of Bayesian stages. It was wrong, and the failure led to Proposition 5.14: a gain whose G/Π decreases, as it does for any sum of Bayesian stages, never gives the mixed class. The other document failures were a truncated worked value (T13) and, in this thesis's own text: two §2.5 table rows that left out hypotheses (T16); an incomplete proof of Corollary 7.4, a miscount, a mislabelled range and one sentence that overstated a source (T17); a citation range in §1.6, the description of the referee pass in §8.5 and three reworded lines of Appendix D (T18); and the descriptions of the two routes in §8.2, one example in §8.4 and the Hill condition in §9.2, which held for n = 2 only (T19). Task 16's legacy-label failure was attributed partly to the document, for a quotation in Appendix D. Separately, four passages of the starting-study manuscript are corrected by errata, each checked against the manuscript's page (Tasks 14 and 15).

Task 19 re-ran, under corrected criteria pre-registered in advance, every earlier failure that had been diagnosed but not re-checked. Of the 24 re-checks, 23 passed, each testing the claim at the precision the thesis prints. The one that failed is the route disagreement of §8.2.

8.5 What verification does not cover

- **Numerical results.** Numerical results 5.15, 6.21, 8.1 and 8.2 are evidence, not proof. So are the class lists of the sampled relays (rows 24–32 of the synthesis matrix): their excess sets and limits are proved, but their full class sets are not.
- **The conjecture.** Conjecture 6.22, that sampled relays have at most two turning points, has no proof.
- **Scope of the checks.** A check tests a statement within the model of §3.1 and its extensions. It does not test whether that model fits a particular application.
- **Independence.** The checks were written and run within the project. One independent referee pass was made inside the project: a separate agent read the whole consolidated document of the classification results and raised 14 points, all of which were fixed (Task 13). No result has been reviewed outside the project.

- **Lemma 7.6a.** Lemma 7.6a, and the whole-path step it supplies to the proof of Theorem 7.6, postdate Task 19 and the referee pass, and neither examined them. Task 20 entered the lemma in the register without checking it.

9. Discussion and Limits

9.1 What the thesis shows

A filter that must keep its error within a tolerance fails on a set of operating conditions, and that set has a shape. Chapter 2 showed that, along one parameter, the shape is fixed by a short list of numbers: the endpoint limits and turning-point values of the variance curve, together with two approach directions. The rest of the thesis computes those numbers for scalar stationary linear filters and reads off the shapes. Two results organise them.

The first is the normal form (chapter 3). For the white-noise relay every stationary variance lies on one surface, $\sigma^2/\mathrm{F} = \mathcal{V}(\nu, \mathrm{q})$, evaluated along a path. The floor crossing is the line $\mathrm{q} = 2$, the optimal filter is the lower envelope, and a sweep or a gain family only chooses the path. Along precision this gives a complete answer (chapter 5): the model itself rules out only a variance below the optimum, and every further restriction comes from the gain family.

The second is information ordering (chapter 7). When one observation process can be built causally from another, the optimal filter on the richer process does no worse, so along an ordered sweep the optimal filter never windows. For additive stationary Gaussian noise the ordering is decided by the spectra (Theorem 7.6). The spectral condition is sharp within the power-normalised family (Theorem 7.8) but not necessary in general (Proposition 7.13), and the garbling must be causal (Counterexample 7.12).

Together the two give a diagnostic. Along an ordered sweep, a window, two rays or a mixed class shows that the filter is miscalibrated; it does not come from the environment (Corollary 7.4).

9.2 How large the effects are

The shapes are exact, but their sizes span many orders of magnitude, and that decides whether they are seen in practice.

- **Windows need over-weighting by a factor of two.** The window conditions share one form: $\kappa > 2$ for the Bayesian weight, $\mathrm{F}' > 2\mathrm{F}$ for the misspecified Kalman gain, $\mathrm{g} > 2\mathrm{F}$ for the capped gain, and $\rho > 2$ or $\lambda' < \lambda/2$ on a moving floor. Below that threshold these relays never window. The Hill gain differs. Its window class needs $\sup \mathrm{q} > 2$, which is $\mathrm{c}/(\mathrm{F}\Pi_0) > 4$ for $\mathrm{n} = 2$ and $\mathrm{c}/(\mathrm{F}\Pi_0) > 2/\mathrm{h}(\mathrm{n})$ in general (§3.3). Below that threshold it can still produce a window component inside the mixed class (Proposition 5.13).
- **Peaks can be large.** At over-weighting ratio 100 the peak error is 9.3F for the Bayesian weight and 17.6F for the misspecified Kalman gain (Propositions 5.8 and 5.9).
- **Some bands are thin.** The Hill mixed band has median width 5×10^{-4} F (Proposition 5.13), and windows of relative width 10^{-8} occur (Numerical result 8.1).
- **Some dips are negligible.** The prior error of an over-weighted sampled relay dips

below the floor by a relative depth that shrinks like $e^{-2\lambda\Delta t}$ with the position of the minimum: from about 95% of F when it lies at $\lambda\Delta t < 1$ to below 10^{-300} for $\lambda\Delta t$ above about 380 (§6.5). The far-out dips exist mathematically but carry no practical weight. In the worked example the prior dip is about 10^{-90}, and the averaged error's dip is 0.125%.

A specification check based on a few operating points, or on a grid, can therefore report a filter as compliant when its failure set contains a thin window or a narrow band. The grid of Numerical result 8.1 missed every window of relative width 10^{-8}. Classifying the failure set by its critical values, with root-finding, avoids that.

9.3 Limits of scope

- **Scalar, stationary, linear.** Every result concerns a scalar state, stationary error and a linear filter. Transients, time-varying gains, non-linear filters and multivariate states are outside the thesis.
- **The signal model.** The signal is an Ornstein–Uhlenbeck process throughout. The framework of chapter 2 applies to any variance curve; the closed forms do not.
- **Gaussian noise for the spectral test.** Theorem 7.6, which makes the spectral test exact, needs additive stationary Gaussian noise. Theorem 7.2 does not.
- **Conventions.** Several statements depend on conventions stated where they are used: Σ is half the diffusion intensity (§3.1); each sample carries sampling variance $v_s = 1/(\Pi\Delta t)$ (§6.3); the tolerance is absolute, or relative (R^2/F) on a moving floor (§6.6); and the averaged error $\bar{P}$ is the primary metric along Δt. A different convention can change a result. In particular, Theorem 6.19 depends on the scaling of the sampling variance.
- **One parameter at a time.** Every sweep varies one parameter. In two or more parameters a failure set is a region rather than a union of intervals, and Theorem 2.2 does not describe it.

9.4 Open problems

1. **Turning points of sampled relays.** Prove Conjecture 6.22: D1–D3 have at most two turning points along Δt, for the posterior, prior and averaged errors. The census found no exception, but the fixed-gain error mixes two exponential rates with $1/\Delta t$, which the polynomial methods used elsewhere do not handle.
2. **Two-stage gains.** Bound the number of turning points of sums of natural stages (Numerical result 5.15). In 2,000 random sets per family, at most 3 were seen for Bayesian stages and at most 4 for Hill stages. Corollary 5.4 shows that constructed gains can have any number of window components; whether sums of natural stages can is open.
3. **A necessary condition for monotone optimal error.** The ordering condition is sharp in the power family but not necessary in general (Proposition 7.13). A simple necessary condition, for instance a one-sided spectral-crossing condition, is not known.
4. **Two-state and multivariate filters.** The fixed α–β filter has two states and does not fit the scalar model. For a multivariate state the failure set also depends on how

the tolerance is applied: to each component, to the trace, or to another norm. No multivariate case has been classified, coupled or decoupled.

5. **A gain that tracks correlation time.** Every gain studied along τ is set by precision alone. How a gain that adapts to τ changes the classes has not been studied.
6. **Applications.** No application is worked out here. A natural test is a filter with a published specification, for example a navigation filter: its failure set could be computed along a design or environment parameter and compared with the specification's own check.

10. Conclusion

Chapter 1 asked: under what conditions does the failure set of a stationary linear filter change shape as an environment or design parameter varies, and which shapes are impossible for an optimal filter? This chapter answers both parts at the scope the proofs support, lists the contributions, and closes the thesis.

10.1 The answer

When the shape changes. Along any sweep whose variance curve is continuous with finitely many turning points, the class of the failure set changes only when the tolerance crosses or equals a critical value: an endpoint limit or a turning-point value. At an endpoint limit the approach direction decides (Theorem 2.2). A family parameter changes which classes are available by changing that list itself, through onset conditions such as $\kappa = 2$ and through pair birth (§2.4–2.5). For the scalar stationary linear filters studied here, the lists are computed for each sweep in Table 1.5 and for the natural gain families used with those sweeps (chapters 4–6). Along precision, the model itself excludes only a variance below the optimum. Below the floor every failure set contains a lower ray, and every further restriction comes from the gain family (Theorem 5.1, Corollaries 5.2–5.3).

Which shapes are impossible for an optimal filter. Along a causally ordered sweep, the optimal filter's failure set is Empty, All or a single ray at the less informative end. A window, two rays and a mixed class are impossible (Corollary 7.3). For additive stationary Gaussian noise, whether a sweep is ordered is decided by the noise spectra (Theorem 7.6). Without that ordering the optimal filter can window: along correlation time at matched variance (Counterexample 6.11), within the power-normalised family exactly when $0 < \beta < 2$ (Theorem 7.8), and under sensor delay (Proposition 7.11, Counterexample 7.12).

What this means for a filter in use. Along an ordered sweep, any other class certifies that the filter is not optimal (Corollary 7.4). The bounded window that started this research (§1.2) is such a certificate: it comes from the gain, not from the environment.

Where the answer stops. The answer holds for a scalar state, stationary error, a linear filter and an Ornstein–Uhlenbeck signal, one parameter at a time (§9.3). The classification is complete, in the sense of §1.3, for the optimal filter on every sweep in Table 1.5 and for each relay whose class set is proved. It is not complete for the sampled relays along Δt, whose full class sets rest on a census (Numerical result 6.21) and on Conjecture 6.22, or for sums of two gain stages (Numerical result 5.15).

10.2 Contributions

The novelty statements below carry the search scope set out in §1.6: searches run on 11, 12 and 14 October 2026 of the English-language scholarly web, arXiv, publisher pages, bibliographic catalogues, Google Patents and the zbMATH abstract and review corpus. A statement that no prior work was found means that none was found in those sources.

1. **A classification of failure-set topology for stationary linear filters.** The thesis defines the failure set $\{\theta : \sigma^2(\theta) > \mathrm{R}^2\}$ of a sweep and its class, and gives a class table for each sweep in Table 1.5, including the forced lower ray below the floor (Corollary 5.2). No prior work that defines and classifies this set was found in those searches. Theorem 2.2 is the fixed-level reading of zero-dimensional persistence [30, 31]; what is added is that reading, with equality cases and approach directions, used as a classification tool.
2. **The expressivity of gain families.** Along precision, every continuous curve on or above the optimal envelope is reached by a continuous gain, given explicitly (Theorem 5.1). This is the scalar filtering analogue of covariance assignment [14, 15, 41–44]; the contribution is the scalar form, with an explicit gain, and its consequences for classes. Natural single-stage gains have at most one window component (Propositions 5.8–5.11), while constructed gains can have any number (Corollary 5.4). The Hill mixed class (Theorem 5.12, Propositions 5.13–5.14) appears to be new in those searches. The Hill-type gain has the form used in divisive-normalisation models [16, 17].
3. **Optimality.** Theorem 7.2 is a filtering instance of known results: monotonicity of optimal performance under garbling goes back to Blackwell [18, 19] and extends to sequential experiments and observation channels [20, 21], with related work on information structures [22, 23]. What is specific here is the causal garbling of observation processes and its consequence for failure sets (Corollaries 7.3–7.4). Theorem 7.6 extends the static characterisation for linear normal experiments [24] to stationary Gaussian processes ordered by their spectra; no process version was found in those searches. The power-family threshold (Theorem 7.8), the matched-variance optimal window (Counterexample 6.11) and the latency counterexample (Counterexample 7.12) appear to be new in those searches.
4. **Smaller contributions.** The normal form $\sigma^2/\mathrm{F} = \mathcal{V}(\nu, \mathrm{q})$ (Theorem 3.2), where dimensionless indices are precedented by Kalata's tracking index [25]. The explicit statement of the excess threshold $\mathrm{G} = 2\mathrm{F}\Pi$ (Corollary 3.3), an elementary consequence of the stationary variance formula, recorded because later results use it. The observation that, under matched-precision sampling, the posterior error falls with $\Delta\mathrm{t}$ while the prior and averaged errors rise (Theorem 6.19).

The fixed-gain variance, the Kalman–Bucy steady state [1], the discrete variance factor [26, 60], the coloured-noise methods [27, 28] and the delay prediction covariance [29] are background, and none is claimed.

10.3 Closing

The thesis replaces a question asked at one operating point, how large is the error, with a question about a set: for which conditions does the error exceed the specification, and what shape does that set have? For scalar stationary linear filters, a short list of numbers read from the variance curve decides the shape. The gain family limits which lists a filter can reach, and for the optimal filter the information order of its observations limits them further. Where that order holds, the optimal filter shows only the simplest shapes, so a window, two rays or a mixed class along such a sweep is evidence about the filter, not about the environment.

Appendix A — Register of results

Every numbered result with its thesis number, permanent repository key and the tasks that verified it (task identifiers as in Appendix D). Rendered from the repository register (`register/results.yaml`, freeze v1, with the Task 20 changes); 83 entries. Numbers are thesis numbers: one counter per chapter, shared by all result types. Earlier labels are listed in `register/crosswalk.md`.

No.	Type	Result	Key	Verified by	Status
2.1	Proposition	Topological realisability	`topological-realizability`	T8.5	proved
2.2	Theorem	Critical-tolerance theorem	`critical-tolerance`	T8.1, T8.3, T9–T13 classifiers	proved
2.3	Definition	Critical-value signature and stratification	`signature`	—	definition
2.4	Theorem	Plateaus (C2′)	`plateaus`	T10.5	proved
2.5	Corollary	Equal-height maxima	`equal-height-maxima`	T10.5	proved
2.6	Corollary	Degenerate stationary points	`degenerate-points`	T10.5	proved
2.7	Proposition	Pair birth	`pair-birth`	T10.5 (with diagnosis)	proved
2.8	Corollary	Reparameterisation (orientation swaps ℓ and h)	`reparameterisation`	T15 C11	proved
2.9	Proposition	Structural stability at a fixed tolerance	`structural-stability`	T15 C12	proved
3.1	Proposition	Stationary variance; constant-gain filter equivalence	`stationary-variance`	T1, T2	proved
3.2	Theorem	Normal form $\sigma^2/\mathrm{F} = \mathcal{V}(\nu, \mathrm{q})$	`normal-form`	T14 G	proved
3.3	Corollary	Excess identity $\sigma^2 > \mathrm{F} \Leftrightarrow \mathrm{q} > 2$	`excess-identity`	T1, T8.1	proved
3.4	Theorem	Window theorem ($\sup \mathrm{q} > 2$); optimal monotone along Π	`window-theorem`	T1, T3, T4, T13 C2	proved
3.5	Proposition	Optimal gain and envelope $\mathcal{V}_\star = \mathrm{q}_\star$	`optimal-envelope`	T1, T7	proved
3.6	Lemma	Single minimum in the gain	`single-minimum-gain`	T8.1	proved
3.7	Lemma	Elasticity form (path derivative)	`elasticity`	T10.1	proved

No.	Type	Result	Key	Verified by	Status
4.1	Theorem	Monotone gain sweeps cannot window	`monotone-gain-sweeps`	T8.4, T13 C2	proved
4.2	Proposition	Prior precision: endpoints and extrema	`prior-endpoints`	T8.1, T8.2	proved
4.3	Proposition	Prior precision: classification	`prior-classification`	T8.3, T13 C2	proved
4.4	Proposition	Coupling: endpoints and extrema	`coupling-endpoints`	T8.1, T8.2	proved
4.5	Proposition	Coupling: classification	`coupling-classification`	T8.3, T13 C2	proved
5.1	Theorem	Gain realisability	`realizability`	T10.1, T10.2	proved
5.2	Corollary	Forced lower ray	`forced-lower-ray`	T10.1 (factor-radicand route), T10.2	proved
5.3	Corollary	Exact class set along precision	`exact-class-set`	T10.2	proved
5.4	Corollary	Unbounded windows for regular gains	`unbounded-windows`	T10.2 (supplementary)	proved
5.5	Corollary	What regularity pins	`regularity-pins`	T10 (derivation)	proved
5.6	Lemma	Excess components	`excess-components`	T8.1 (Corollary 3.3, excess identity)	proved
5.7	Lemma	Bounded gains approach H from above	`bounded-gain-limit`	T10.3, T10.7	proved
5.8	Proposition	Bayesian-weight gain along precision (onset $\kappa > 2$; four regimes)	`bayesian-weight`	T1, T3, T7, T10.3, T10.7, T13 C2	proved
5.9	Proposition	Misspecified Kalman gain	`misspecified-kalman`	T10.1, T10.3, T13 C2	proved
5.10	Proposition	Capped gain	`capped-gain`	T10.3, T10.5, T13 C2	proved
5.11	Proposition	Precision-blind gain	`precision-blind`	T4, T10.3, T13 C2	proved
5.12	Theorem	Hill gain for every real $n > 1$	`hill-turning-points`	T4, T7, T10.1, T10.3 ($n = 2$), T13 A5	proved
5.13	Proposition	Hill classification and the mixed class	`hill-classification`	T10.4, T13 C2	proved
5.14	Proposition	No mixed class when G/Π decreases	`no-mixed-decreasing-ratio`	T13 C2 (row-8 diagnosis, t13_audit_classes_diag2.py)	proved

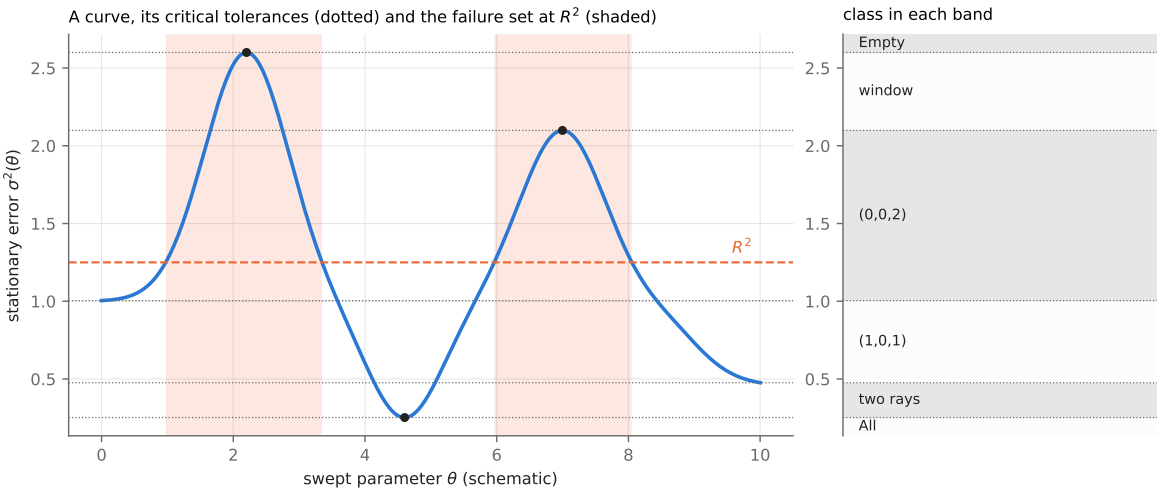
No.	Type	Result	Key	Verified by	Status
5.15	Numerical result	Two-stage gains	`two-stage-gains`	T10.3, T13 C2	numerical
6.1	Lemma	Relay variance at matched power (with coloured excess condition)	`tau-relay-variance`	T4, T6, T9.1	proved
6.2	Proposition	Relay monotone in τ	`tau-relay-monotone`	T9.1, T9.4	proved
6.3	Proposition	Relay classification along τ	`tau-relay-classes`	T9.4, T13 C2	proved
6.4	Proposition	Optimal error along τ (closed form)	`tau-optimal-error`	T9.1, T9.2 (Wiener route; ARE route via diagnosis)	proved
6.5	Corollary	Normalised optimal error P/F	`tau-optimal-normal-form`	T14 G	proved
6.6	Theorem	Optimal error decreases in τ	`tau-optimal-monotone`	T9.1, T9.5	proved
6.7	Proposition	Optimal classification along τ	`tau-optimal-classes`	T9.4, T13 C2	proved
6.8	Proposition	Best constant-gain relay	`tau-best-constant-gain`	T9.5 (with mp diagnosis)	proved
6.9	Proposition	Strict gap $P < V$	`tau-strict-gap`	T13 A7	proved
6.10	Proposition	Optimal filter at matched variance	`matched-variance-optimal`	T10.6, T13 C2	proved
6.11	Counterexample	The optimal filter can window	`optimal-can-window`	T10.6	proved
6.12	Proposition	Relay at matched variance	`matched-variance-relay`	T10.6, T13 C2	proved
6.13	Proposition	Discrete excess condition $L/(2-L) > FI\Pi_d$ for $0 \le L < 2$	`discrete-excess`	T4, T7	proved
6.14	Lemma	Shared excess set (M-P)	`shared-excess-set`	T12	proved
6.15	Proposition	D1 tanh lemma	`d1-excess`	T12	proved
6.16	Proposition	D2 excess set ($0 < c < 2$)	`d2-excess`	T12	proved
6.17	Proposition	D3 excess set; onset $\rho > 2 + \nu$	`d3-excess`	T12	proved
6.18	Proposition	Endpoint limits along Δt	`dt-endpoint-limits`	T12 (with series-route diagnosis)	proved
6.19	Theorem	Optimal M-P along Δt: posterior $\downarrow$; prior $\uparrow$; averaged $\uparrow$	`dt-optimal-mp`	T12 (a), T13 A1, A2 (b), (c)	proved
6.20	Proposition	Two rays from an over-weighted sampled relay	`dt-two-rays`	T12 (with diagnosis)	proved
6.21	Numerical result	Census of sampled relays	`dt-relay-census`	T12, T13 C2	numerical

No.	Type	Result	Key	Verified by	Status
6.22	Conjecture	At most two turning points (sampled relays)	`dt-two-turning-points`	T12, T13 A6, C2	conjecture
6.23	Lemma	Σ scaling	`sigma-scaling`	T12	proved
6.24	Proposition	Σ-independent gain	`sigma-independent-gain`	T12, T13 C2	proved
6.25	Proposition	Σ-tracking gain with over-trust ρ	`sigma-overtrust`	T12, T13 C2	proved
6.26	Proposition	Σ-tracking gain with misspecified λ'	`sigma-misspecified-lambda`	T12, T13 C2	proved
6.27	Proposition	Optimal filter along Σ	`sigma-optimal`	T12, T13 C2	proved
6.28	Proposition	Signal sweeps (Σ and λ): optimal monotone	`signal-sweeps`	T11.1	proved
7.1	Definition	Causal garbling	`causal-garbling`	—	definition
7.2	Theorem	Monotonicity under causal garbling	`causal-monotonicity`	T11.1, T11.2	proved
7.3	Corollary	No optimal windows along causal orderings	`no-optimal-windows`	T11	proved
7.4	Corollary	Diagnostic	`diagnostic`	T11	proved
7.5	Proposition	Spectral dominance	`spectral-dominance`	T11.1, T11.3	proved
7.6	Theorem	Causal; pathwise and spectral orderings coincide (Gaussian)	`gaussian-orderings-coincide`	T13 A8	proved
7.6a	Lemma	Two-sided ordering (step (iv) $\Rightarrow$ (ii) of Theorem 7.6)	`whole-path-garbling`	—	proved; not independently checked
7.7	Proposition	Which studied sweeps are ordered	`sweeps-ordered`	T11.1, T11.3	proved
7.8	Theorem	Sharpness in the power family	`power-family-sharpness`	T11.3 (with diagnosis), T13 C2	proved
7.9	Theorem	Unimodality for $0 < \beta < 2$	`power-family-unimodal`	T13 A4, T11.3 (numerical)	proved
7.10	Proposition	Duality	`power-family-duality`	T11.1, T11.3	proved
7.11	Proposition	Latency	`latency`	T11.4, T13 C2	proved
7.12	Counterexample	Pathwise ordering is not enough	`pathwise-not-enough`	T11.4	proved
7.13	Proposition	Ordering sufficient; not necessary	`ordering-not-necessary`	T11.5	proved
7.14	Corollary	Integrate-and-dump; nested grids	`mi-nested`	T11.6	proved
7.15	Theorem	Integrate-and-dump on the continuum	`mi-continuum`	T13 A3, T11.6, T12	proved

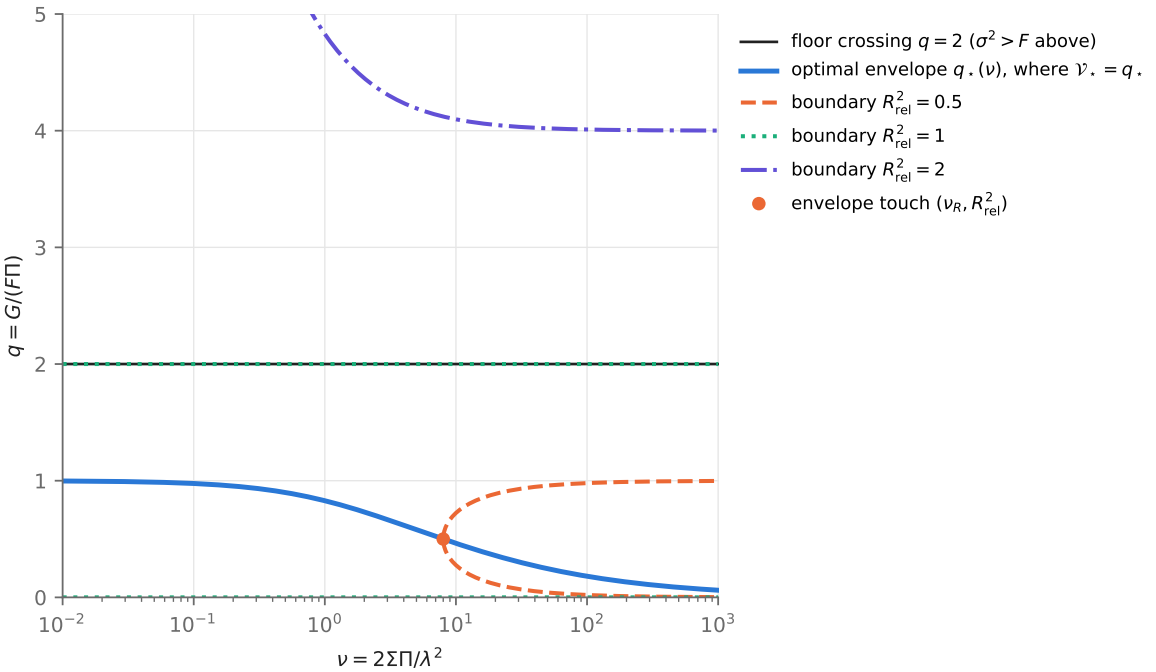
No.	Type	Result	Key	Verified by	Status
7.16	Proposition	M-P is not a causal-garbling family	`mp-not-ordered`	T12	proved
8.1	Numerical result	Narrow windows need root-finding	`narrow-windows`	T10.5	numerical
8.2	Numerical result	Blind searches find no unpredicted class	`blind-searches`	T10.7, T12 Part C, T13 C2	numerical

Appendix B — Figures

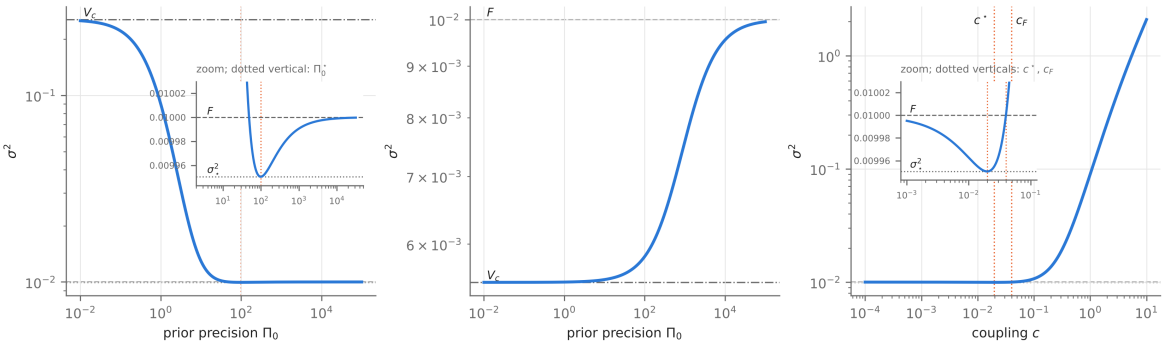
The images carry no numbers. The text refers to each one as Figure B.n, so renumbering a result never forces a redraw. Figures are numbered in order of first citation. The captions describe what each regenerated image plots (Task 15); the images are produced by `t15/make_figs_thesis.py` (Appendix E).



B.1 — A schematic curve $\sigma^2(\theta)$ for $0 \leq \theta \leq 10$ (three Gaussian bumps and a logistic step; not the variance of any filter), with its endpoint limits and turning-point values dotted as critical tolerances, its turning points marked, and the failure set at $R^2 = 1.25$ shaded. Right: the class in each band between consecutive critical tolerances.

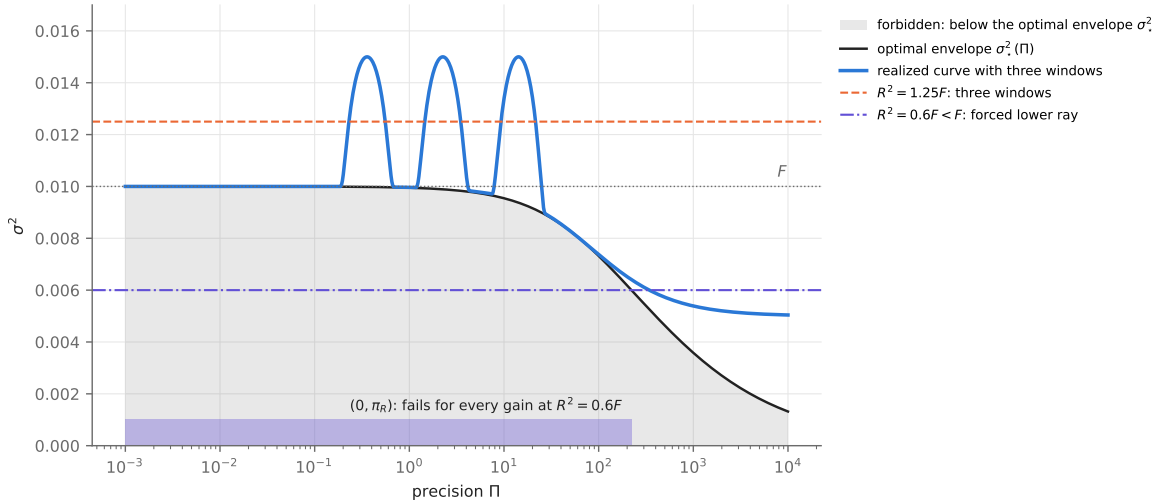


B.2 — The (ν, q) plane, with $\nu = 2F\Pi/\lambda$ from 10^{-2} to 10^{3} (log axis) and $q = G/(F\Pi)$ from 0 to 5: the floor crossing $q = 2$, the optimal envelope $q_\star(\nu) = 2/(1 + \sqrt{(1 + \nu)})$, and the boundaries $\mathcal{V}(\nu, q) = R^2/F$ for $R^2/F = 0.5$, 1 and 2. The $R^2/F = 1$ boundary is the pair of lines $q = 0$ and $q = 2$. The $R^2/F = 0.5$ boundary touches the envelope at $\nu_R = 8$ (marked).

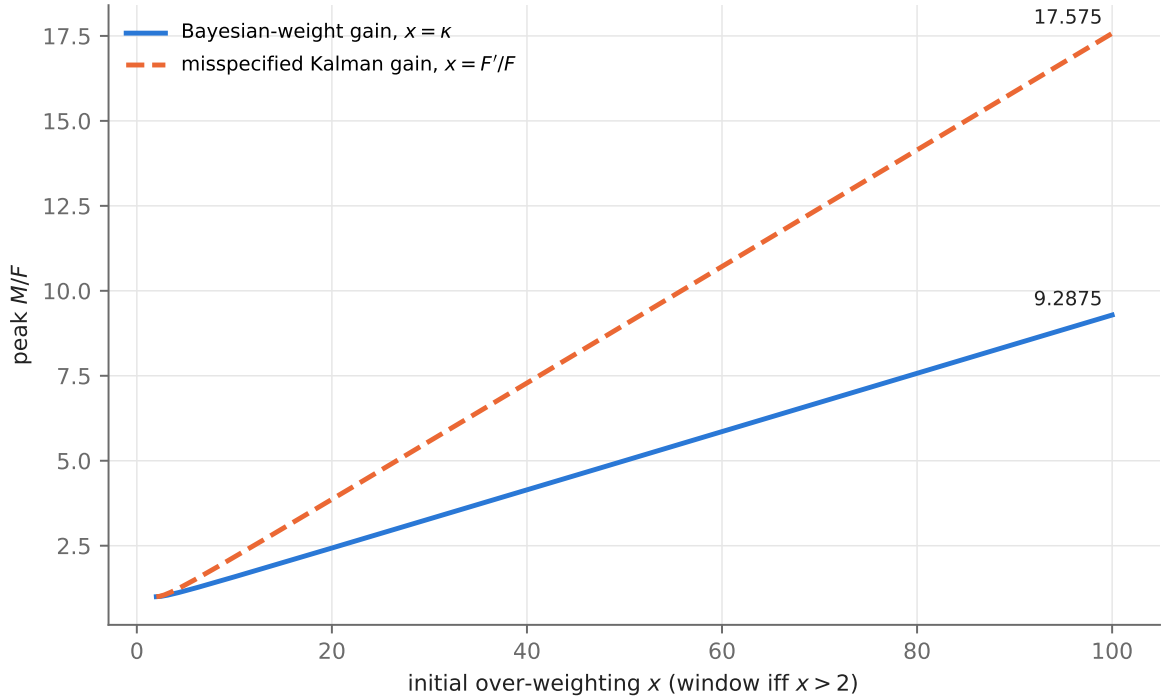


(a) prior sweep, $\Pi = 1$ (case P3) (b) prior sweep, $\Pi = 500$ (case P1) (c) coupling sweep, $\Pi = \Pi_0 = 1$

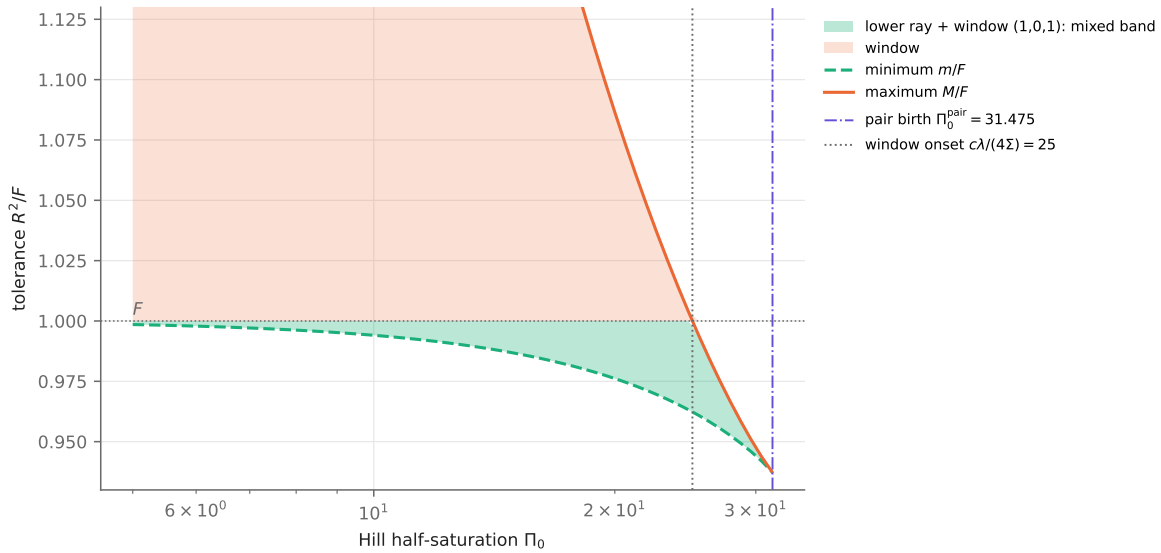
B.3 — σ^2 (log scale) at $\lambda = c = 1$, $\Sigma = 0.01$ along three design sweeps: (a) Π_0 from 10^{-2} to 10^{5} at $\Pi = 1$ (case P3, U-shaped; F, V_c, $\sigma\star^2$ and $\Pi_0\star$ marked); (b) the same sweep at $\Pi = 500$ (case P1, monotone; F and V_c marked); (c) c from 10^{-4} to 10 at $\Pi = \Pi_0 = 1$ (F, $\sigma\star^2$, c⋆ and c_F marked). Insets zoom on the minima in (a) and (c).



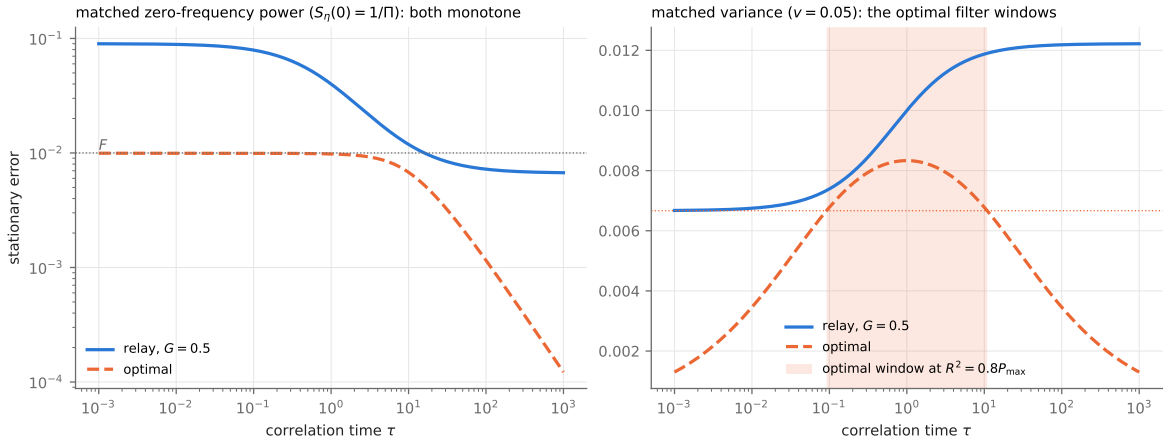
B.4 — σ^2 against Π (log axis) at $\lambda = 1$, $\Sigma = 0.01$ (F = 0.01): the optimal envelope $\sigma\star^2(\Pi)$, with the unreachable region below it shaded; a Bayesian-weight base curve ($\kappa = 1.5$) carrying three compactly supported bumps of peak 1.5F, placed before the envelope touch point $\Pi_c = 56.20$; the tolerance 1.25F, which gives three windows; and the tolerance 0.6F, with its forced lower ray $(0, \pi_R)$, $\pi_R = 222.2$.



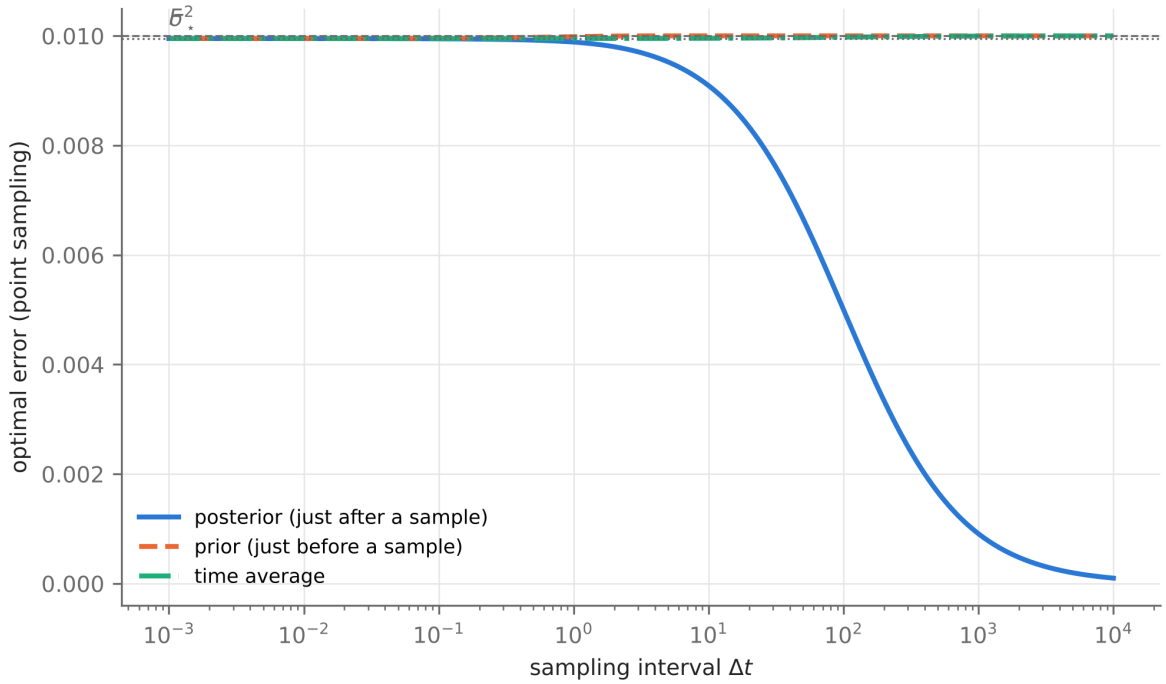
B.5 — Peak variance M/F against the initial over-weighting x, for the Bayesian-weight gain ($x = \kappa$) and the misspecified Kalman gain ($x = F'/F$), at $\lambda = \lambda' = c = 1$, $\Sigma = 0.01$. Both rise from 1 at $x = 2$. At $x = 100$ they reach 9.2875 and 17.575.



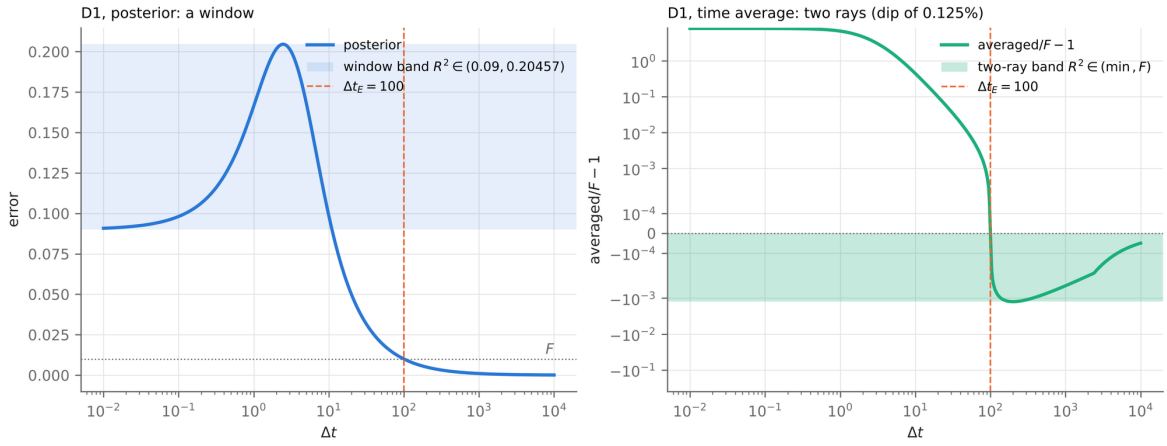
B.6 — Hill gain, n = 2, at $\lambda = c = 1$, $\Sigma = 0.01$, over the half-saturation Π_0 (log axis): the turning-point values m/F (dashed) and M/F (solid). The band $m \le R^2 < \min(M, F)$ is the mixed class, lower ray plus window; the band $F \le R^2 < M$ is a window. Vertical lines mark the pair birth $\Pi_0 = 31.475$ and the window onset $c\lambda/(4\Sigma) = 25$.



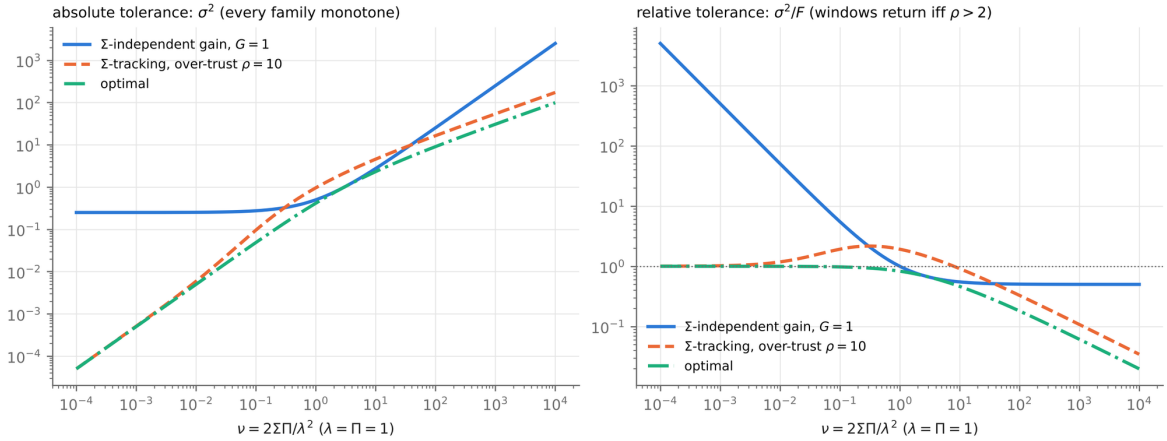
B.7 — Stationary error against the correlation time τ (log axes) at $\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$, for the relay with G = 0.5 and for the optimal filter. Left, matched zero-frequency power: both decrease. Right, matched variance v = 0.05: the relay increases, and the optimal error is unimodal, with maximum $P_{max} = 0.008333$ at $\tau = 1/\lambda$. The optimal filter's failure set at $R^2 = 0.8\ P_{max}$ is shaded.



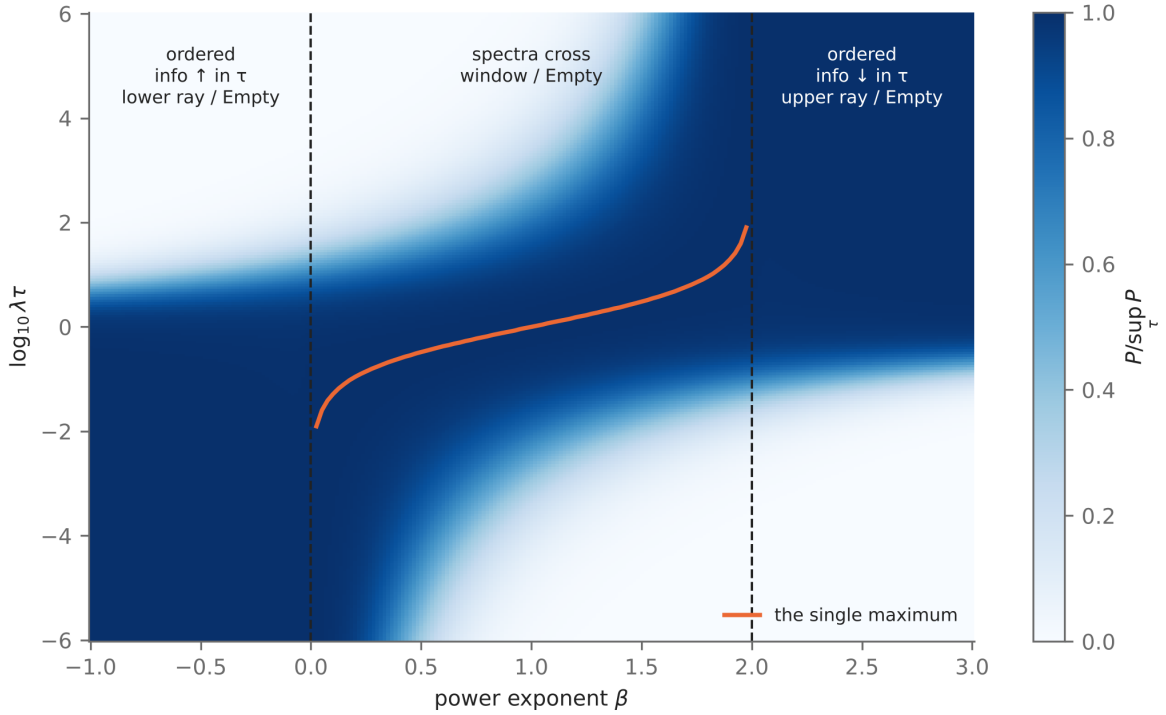
B.8 — Optimal error against the sampling interval Δt (log axis) in the point-sampled model at $\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$, with per-sample sampling variance $v_s = 1/(\Pi\Delta t)$. The posterior error P^+ falls from $\sigma\star^2$ towards 0, while the prior error P^- and the averaged error $\bar{P}$ rise from $\sigma\star^2$ to F.



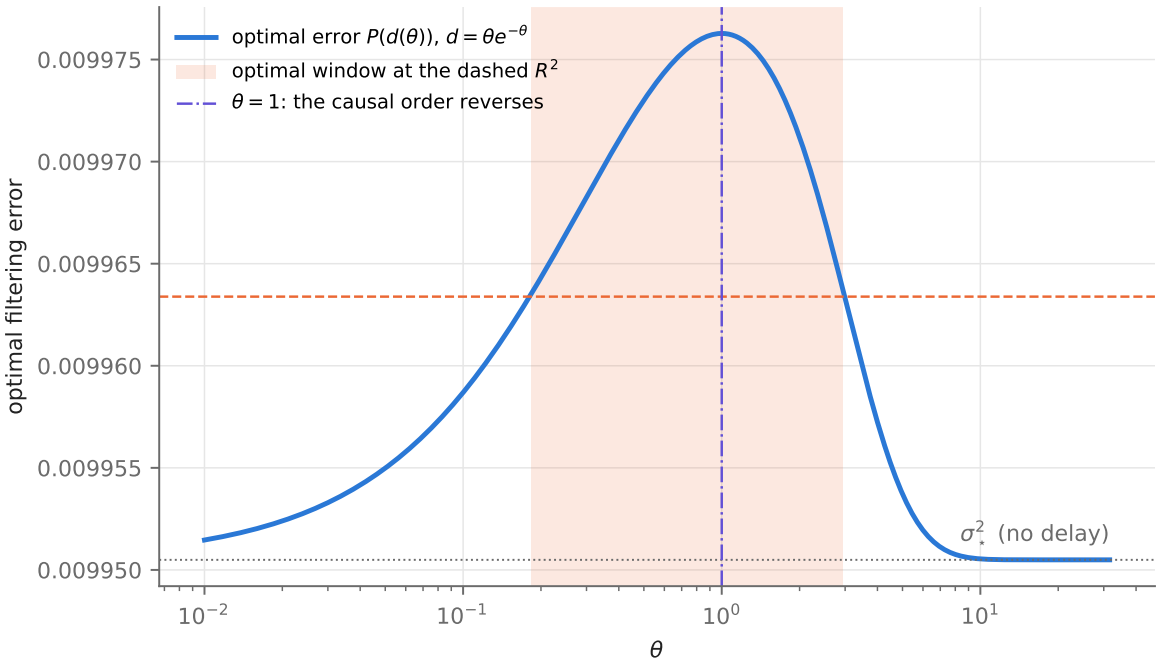
B.9 — The sampled relay D1 (gain $1 - e^{-G\Delta t}$, $G = 0.5$, $\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$) along Δt (log axis). Left: the posterior error P^+, with the window band $0.09 < R^2 < 0.20457$ shaded and the excess crossing $\Delta t_E = 100$ marked. Right: the averaged error, plotted as $\bar{P}/F - 1$ on a symmetric-log axis, with the two-ray band between its minimum and F shaded.



B.10 — The process-noise sweep at $\lambda = \Pi = 1$ against $\nu = 2F\Pi/\lambda$ (log axes), for the Σ-independent gain ($G = 1$), the Σ-tracking gain that over-trusts precision by $\rho = 10$, and the optimal filter. Left: absolute error. Right: σ^2/F, where the over-trusting gain rises above 1 to a peak of about 2.17 and then falls.



B.11 — The optimal error relative to its supremum over τ, P/sup P, for the power-normalised family $S = K\tau^\beta/(1 + \tau^2\omega^2)$ at $\lambda = 1$, $\Sigma = 0.01$, $K = 1$, over the exponent β (horizontal) and $\log_{10}(\lambda\tau)$ (vertical). The single maximum for $0 < \beta < 2$ is drawn, and the three regimes are labelled: ordered with information increasing in τ; spectra cross; ordered with information decreasing in τ.



B.12 — Optimal filtering error P(d(θ)) for observations delayed by $d(\theta) = \theta e^{-\theta}$, at $\lambda = 1$, $\Sigma = 0.01$, $\Pi = 1$, against θ (log axis). It rises from $\sigma\star^2 = 0.0099505$ to $P(1/e) = 0.0099763$ at $\theta = 1$, then falls back. The failure set at $R^2 = 0.0099634$ is shaded, and $\theta = 1$, where the causal order reverses, is marked.

Appendix C — References

References [1]–[53] are numbered in order of first citation. A number, once assigned, does not change, so the seven references first cited in §1.6 after the list was fixed carry the next free numbers, [54]–[60].

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Appendix D — Verification record, Tasks 1–19

For each task: the pre-registration and results files, the pre-registered checks that passed and failed, and every failure with the attribution its report gives (criterion, input, code, document or prediction). Outcomes are copied from the reports, not re-judged, and every failure stays recorded.

Task 1. verification/PREREG.md · verification/REPORT.md. 12 checks with a verdict; 12 pass, 0 fail.

Task 2. verification/PREREG.md · verification/REPORT.md. 3 checks with a verdict; 3 pass, 0 fail.

Task 3. verification/PREREG.md · verification/REPORT.md. 8 checks with a verdict; 8 pass, 0 fail.

Task 4. verification/PREREG.md · verification/REPORT.md. Exploratory by pre-registration (outcomes reported either way; no pass/fail criteria).

Task 5. verification/PREREG.md · verification/REPORT.md. The report itself (`verification/REPORT.md`); no pass/fail criteria.

Task 6. verification/PREREG_t6.md · verification/REPORT.md. 6 checks with a verdict; 5 pass, 1 fail.

- Failure — C5 (peak locations to 3 s.f. against Task 4b): criterion — Task 4b located peaks on a grid of about 0.4 % steps; heights all match, three locations differ in the third figure.

Task 7. verification/PREREG_t7.md · verification/REPORT.md. 15 checks with a verdict; 15 pass, 0 fail.

Task 8. classification/PREREG_t8.md · classification/REPORT.md. 5 checks with a verdict; 5 pass, 0 fail.

Task 9. classification/PREREG_t9.md · classification/t9/RESULTS_t9.md. 7 checks with a verdict; 5 pass, 2 fail.

- Failure — T9.2 ARE route: criterion — the regularised ARE converges as $\sqrt{\varepsilon}$ with a large prefactor; mp ARE and the closed-form Wiener filter match the document.
- Failure — T9.5 monotonicity and ordering: criterion — float64 resolution; 0 of 285 and 0 of 13 flags are real at higher precision.

Task 10. classification/PREREG_t10.md · classification/t10/RESULTS_t10.md. 10 checks with a verdict; 6 pass, 4 fail.

- Failure — T10.1 symbolic (2 items): criterion / route — no $\sqrt{}$(square) handling in the pre-registered SymPy route; identities hold.
- Failure — T10.2 A1.3 as specified: plan's construction (criterion as pre-registered); the supplementary construction passes.
- Failure — T10.5 pair birth: criterion.
- Failure — T10.6 matched-variance τ (zero-limit shape check): criterion.

Task 11. classification/PREREG_t11.md · classification/t11/RESULTS_t11.md. 11 checks with a verdict; 10 pass, 1 fail.

- Failure — T11.3 classes: criterion — the pre-registered limits/brackets at $\beta \geq 2$.

Task 12. classification/PREREG_t12.md · classification/t12/RESULTS_t12.md. 13 checks with a verdict; 10 pass, 3 fail.

- Failure — Symbolic A5 (D2 post at $\Delta t \to \infty$): code — the SymPy check route (`sympy.limit`); the series route confirms.
- Failure — A7 minimum below F and two rays: criterion — the zero tolerance is too coarse for the prior dip.
- Failure — A8 qualitative regression: prediction — the prompt's quick-check numbers.

Task 13. classification/PREREG_t13.md · classification/t13/RESULTS_t13.md. 15 checks with a verdict; 10 pass, 5 fail.

- Failure — Part A symbolic (2 of 68 items): criterion / route — identities verified by a post-hoc rewrite route.
- Failure — A2 numerical (auxiliary cross-check, 277 sets): code — precision loss; all agree at higher precision.
- Failure — C1 numbers, first run: document (4, a truncated worked value) and code (2, parser); passes after the fixes.
- Failure — C2 classes, row 8: document — the stated class (1,0,1) cannot occur (proved; class set corrected).
- Failure — C2 classes, row 12: code — route (ii) sample point below its equality tolerance at an exact limit.
- Failure — D reproduction, Task 10: code — `run_t10.sh` omits `t10_realize_diag.py`; Task 12: criterion — masking of the provenance runtime list.
- Recorded — A3 (Theorem 7.15): the pre-registered resultant route, completed post hoc in exact arithmetic (`t13/t13_A3_proof.py`, 22 of 22 checks; the pre-registered attempt `t13_A3_attempt.py` reported 'unresolved'): the resultant of the cleared quadratic and its x-derivative is $4u^2x^2(u+1)^2(E-1)^4(E^2+2xE-1)^2$, and $E^2+2xE-1 = -2e^{-x}(\sinh x - x) < 0$ on $E = e^{-x}$, so dp/dx never vanishes.

Task 14. classification/PREREG_t14.md · classification/t14/RESULTS_t14.md. 11 checks with a verdict; 8 pass, 3 fail.

- Failure — C4 paper render reproduces v1: criterion — the register types the excess identity as a Corollary (accepted in Task 15).
- Failure — C9 bibliography: input / criterion — two entries unattached and DOIs unresolvable from the session; resolved by the Task 15 input (verified bib, attachments).
- Failure — C10 reproduction: code — the dry-run resolver; all 9 flagged paths resolve as their scripts build them.

Task 15. classification/PREREG_t15.md · classification/RESULTS_t15.md. 13 checks with a verdict; 11 pass, 2 fail.

- Failure — C4 [n] rendering: criterion / code — the pre-registered bracket pattern flags the interval '[0, 10]' in the Appendix B description of Figure B.1; it is not a citation, and no unresolved citation remains.
- Failure — C10 thesis phrase check: criterion — all 81 A–E matches are Task 15's figure

labels B.1–B.12, which coincide in form with the legacy labels the pattern targets; 0 'Paper 1', no legacy label remains.

Task 16. classification/PREREG_t16.md · classification/RESULTS_t16.md. 8 checks with a verdict; 5 pass, 3 fail.

- Failure — D1 new proof steps of the thesis text (2 of 9 items): code — SymPy could not decide a sign and could not denest a square root of a square; both items hold (post-hoc diagnosis).
- Failure — D2 statements against the repository: document / criterion — four statements differ in content: 3.1 and 3.5 gather conclusions the repository states in other results (verified in D1), and the §2.5 table rows for 2.5 and 2.7 left out hypotheses (full statements were then added to the thesis).
- Failure — D6 legacy labels: criterion / document — three hits, none a legacy label: two figure labels in Appendix E outside the exemption window, and a quotation in Appendix D.

Task 17. classification/PREREG_t17.md · classification/RESULTS_t17.md. 7 checks with a verdict; 3 pass, 4 fail.

- Failure — E1 new proof steps of the thesis text (2 of 9 items): code / document — the 50-digit test of the optimal prior $M < F$ (Proposition 6.13) cannot resolve it when $e^{-2\lambda\Delta t} < 10^{-50}$ (it holds exactly, $f(F) = F^2a^2 > 0$); Corollary 7.4's proof as written omits $\sigma\star^2 \to 0$ and Theorem 5.1 (the corollary holds).
- Failure — E3 numbers (1 of 88): document — "37 values of β" for Theorem 7.8, in the repository's label and the thesis text; Task 11 ran 129.
- Failure — E4 cross-references: document — "Prop. 6.10–6.12" spans Counterexample 6.11.
- Failure — E5 prior-work sentences: document — §7.3 makes [47] an analysis of coloured measurement noise; it studies time-correlated model error (2 sentences undetermined: no abstracts reachable from the session).

Task 18. classification/PREREG_t18.md · classification/RESULTS_t18.md. 7 checks with a verdict; 3 pass, 4 fail.

- Failure — F1 Task 17 fixes (2 of 6): code / document — the [47] search included Appendix D's quotation of the old finding (both chapter sentences now say model error); §1.6 still cites [21–23] for monotonicity under garbling (fixed in §7.1 only).
- Failure — F2 chapter 8 counts: code — the recount parser skipped Task 2's 'PASS (info)' row (143); read as pre-registered the total is 144, with 28 failed and Tasks 4 and 5 without criteria.
- Failure — F3 chapter 8 claims: document — §8.5 names a referee pass on Theorem 7.6 only; Task 13 C4's independent referee read the whole consolidated document (14 points, all fixed).
- Failure — F6 appendices: code / document — Appendix A differs only in math delimiters; the thesis's Appendix D rewords three Task 17 failure lines (tense, P^- for M, 'the proof was completed').

Task 19. classification/PREREG_t19.md · classification/RESULTS_t19.md. 31 checks with a verdict; 26 pass, 5 fail.

- Failure — G1 citations: input — Roberts (1959) unreachable from the session; that it states $r/(2 - r)$ cannot be determined from its text (secondary sources agree); the §6.3 identity, [29], [40] and both [21] sentences hold.
- Failure — G3 chapter 8: code / document — the coverage parser missed two Appendix D lines (both covered); §8.2 interchanges the route descriptions; §8.4 lists Task 13 under $\sqrt{(x^2)}$ handling, which Task 13's report does not support. Recount 151 / 32 and §8.5 match.
- Failure — G4 chapter 9: document — §9.2's $c/(F\Pi_0) > 4$ for the Hill gain holds for $n = 2$ only.
- Failure — G7 sync: criterion / code — the old key remains in ten recorded files of Tasks 14, 15 and 17 (not editable) and in Task 19's register script's docstring; 0 in live files; the other three clauses hold.
- Failure — H-13.C2-12 C2 row 12 re-check: criterion — 45 persisting cases; the 10^{-300} tolerance is finer than the stored limit's precision (about 10^{-25}), so the routes compare below what the critical value carries.

Appendix E — Reproduction

- Python 3.13.16 on Linux-6.18.44-fc-v114-x86_64-with-glibc2.39
- Packages (from the installed environment, `importlib.metadata`; the repository has no lockfile): numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0, matplotlib 3.11.2, PyYAML 6.0.1.
- Tasks 1–7 record their own environment in `verification/results/env.json`; Tasks 8–13 in each task's `results/env.json`.

All commands run from the repository root unless stated.

Task	What it regenerates	Command
T1–T7	starting-study checks, simulations, figures, `verification/REPORT.md`	`cd verification && bash run_all.sh`
T8	chapter 2–3 checks, figure, `classification/REPORT.md`	`cd classification && bash run_all.sh`
T9	τ-sweep checks (chapter 6), `t9/RESULTS_t9.md`	`cd classification && bash t9/run_t9.sh`
T10	gain-family and matched-variance checks, figures, `t10/RESULTS_t10.md`	`cd classification && bash t10/run_t10.sh` (then `python3 t10/t10_realize_diag.py`, which the run script omits; see Task 13)
T11	optimality checks (chapter 7), figures, `t11/RESULTS_t11.md`	`cd classification && bash t11/run_t11.sh`
T12	Δt and Σ sweeps, figures, `t12/RESULTS_t12.md`	`cd classification && bash t12/run_t12.sh`
T13	open items, audits, figures, `t13/RESULTS_t13.md`	`cd classification && bash t13/run_t13.sh`
T8–T13 at once	clean checkout, manifest	`bash classification/reproduce_all.sh WORKDIR 8 9 10 11 12 13`
T14	register, sources, renders, bibliography check	see the render chain below
T15	figures B.n, appendices, renders, checks	see the render chain below
T16	register fixes, renders, checks D1–D8	render chain below, then `python3 t16/t16_checks.py`
T17	thesis-only imports, renders, checks E1–E7	render chain below, then `python3 t17/t17_checks.py`
T18	record corrections, Theorem 2.2 import, archived thesis, checks F1–F7	render chain below, then `python3 t18/t18_checks.py`
T19	reference 26, Theorem 2.2 import, open problems 5–6, archived thesis, checks G1–G7 and re-checks H-*	render chain below, then `python3 t19/t19_checks.py`, `python3 t19/t19_routes.py` and `python3 t19/h13d_replay.py`

Task	What it regenerates	Command
T20	register titles, Lemma 7.6a, references [54]–[60], figure labels B.5–B.6, Appendix D note, checks C1–C6	render chain below, then `python3 t20/t20_checks.py`

The Task 14–20 render chain, from `classification/`:

```
# register from the T14 table (T14; overwrites T15 metadata, re-apply next line)
python3 t14/build_register.py
# T15 citations/status, references.yaml, figures.yaml
python3 t15/t15_register.py
# T16 legacy-label prefixes, crosswalk, result 5.6 wording, B.4 Pi_c
python3 t16/t16_register.py
# T19 reference 26: key roberts1959 in references.yaml and results.yaml
python3 t19/t19_register.py
# T20 titles, 8.1/8.2 swap, Lemma 7.6a, references 54-60, crosswalk
python3 t20/t20_register.py
# T17 thesis-only imports (placed by register/thesis_overrides.yaml)
python3 t17/make_thesis_only_t17.py
# T18 thesis-only Theorem 2.2 (parts (1)-(4))
python3 t18/make_thesis_only_t18.py
# T19 Theorem 2.2 with the proof's closing sentences (replaces T18's file)
python3 t19/make_thesis_only_t19.py
# figures B.1, B.3-B.12 (text-only edits) and B.2
python3 t15/make_figs_thesis.py
# Stages_0_4_Thesis.md, Roadmap_v3.md, figs_thesis/*.txt, Appendices A-C,
 ↪  reference/phase1/
python3 register/render.py --numbering thesis
# Stages_0_4_Paper2.md
python3 register/render.py --numbering paper
# Appendices D (Tasks 1-19) and E
python3 t15/make_appendix_de.py --with-t19
# Task 15 checks
python3 t15/t15_checks.py && python3 t15/t15_partG.py
```

The sources were derived once from the Task 14 versions by `t14/make_source.py`, `t14/make_roadmap_source.py` and `t15/make_sources_t15.py`; they are now edited directly and are not regenerated by this chain.